

Anatomy

World of Revision

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Instructions

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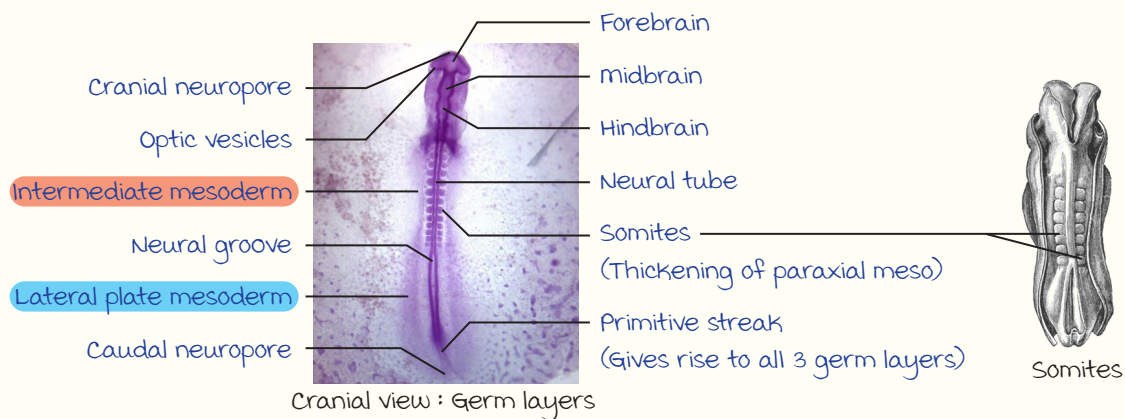
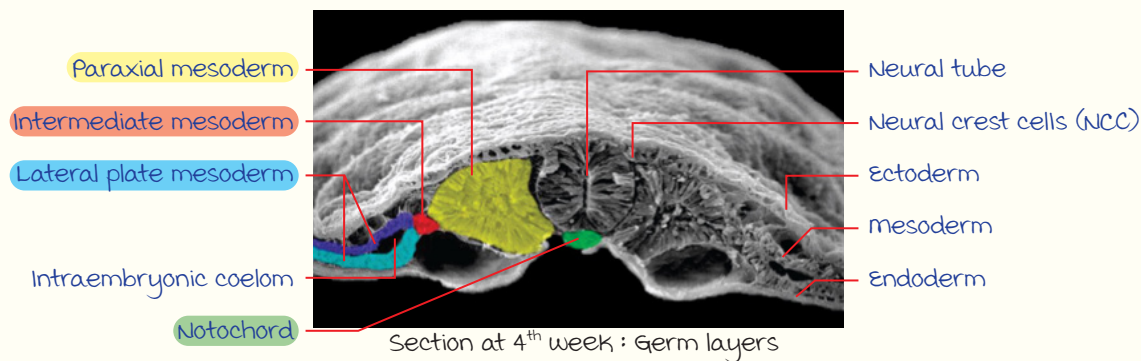
EMBRYOLOGY : PART 1

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Development of Germ Layers



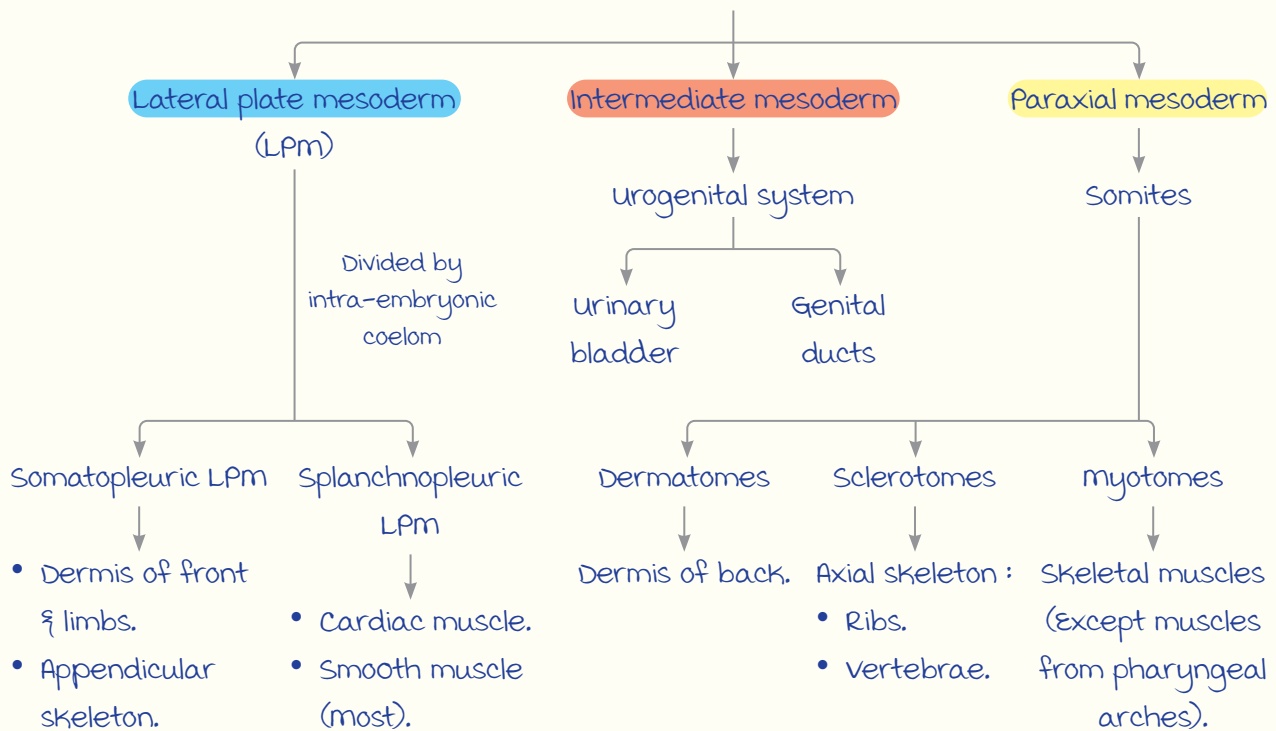
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Applied anatomy :

	Normal	Developmental anomaly
Primitive streak	Disappears	Persists → Sacrococcygeal teratoma
Cranial neuropore	Closes by : Day 25	Fails → Anencephaly
Caudal neuropore	Closes by : Day 28	Fails → Spina bifida

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mesoderm :Notochord :

- Derived from **primitive knot**.
- made of axial mesoderm.
- Remnants :
 - a. **Nucleus pulposus** (Central part of intervertebral disc).
 - b. Apical ligament of dens.
 - c. Tectorial membrane (Cranial continuation of posterior longitudinal ligament).

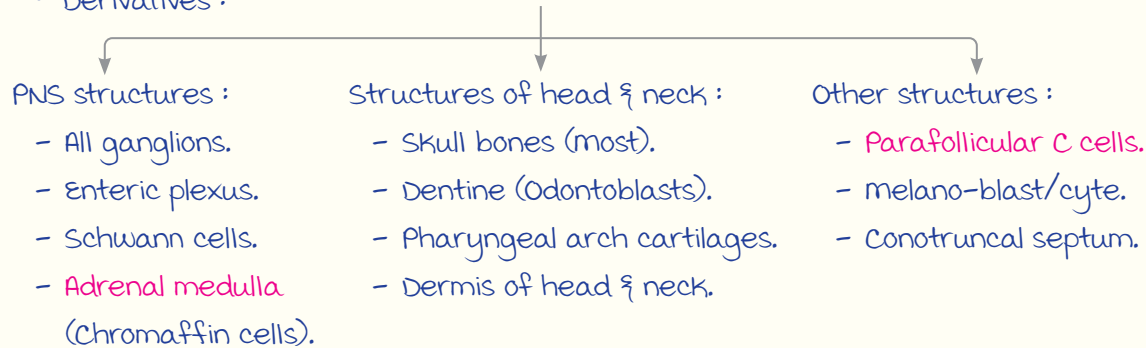
Ectoderm :Neural tube :

- **Notochord** $\xrightarrow{+}$ Ectoderm $\rightarrow$ Neuroectoderm $\rightarrow$ Neural tube.
- Derivatives : Forms **CNS structures**.
 - Brain & spinal cord.
 - Oligodendrocytes (myelination).
 - Astrocytes (Blood-brain barrier).
 - Ependymal cells (Lining of ventricles).
 - Retina & pigment.

Neural Crest Cells (NCC) :

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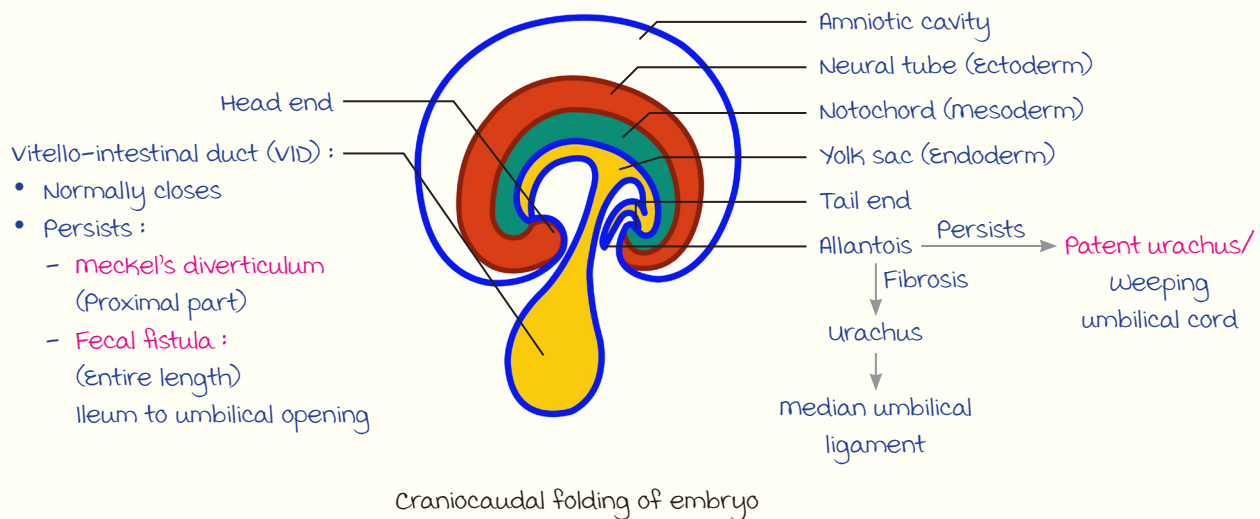
- Present at junction of neural tube & ectoderm.
- AKA 4th germ layer.
- Derivatives :

**Note :**

Adrenal cortex : Derivative of intermediate mesoderm.

Allantois and Vitellointestinal Duct

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Umbilical cord :

Contents :

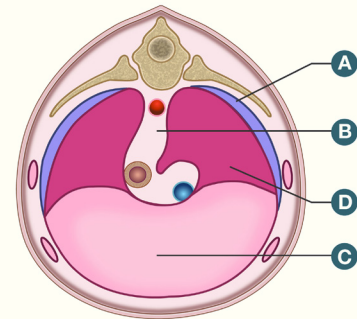
- Wharton's jelly.
- umbilical arteries (2).
- Left umbilical vein.
- **Allantois.**
- **VID.**

Development of Diaphragm

00:28:46

Derivatives :

	Structures	Derivatives
A	Body wall mesoderm/ cervical somites	muscles of diaphragm
B	Dorsal mesentery of esophagus	Crura of diaphragm
C	Septum transversum (Undivided Lpm)	Central tendon
D	Pleuroperitoneal membrane	Incorporated by muscles



Diaphragm development

Applied Anatomy :

Congenital diaphragmatic hernia/Bochdalek hernia:

- Pathogenesis : Absence of pleuroperitoneal membrane

↓
Bochdalek foramen

Intestinal herniation into thoracic cavity.

- Left > Right.
- Complication: Left lung hypoplasia.

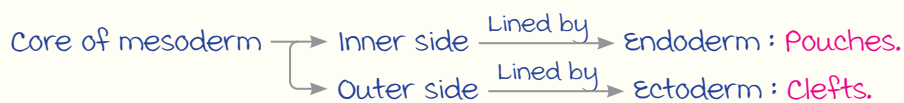
Eventration of diaphragm:

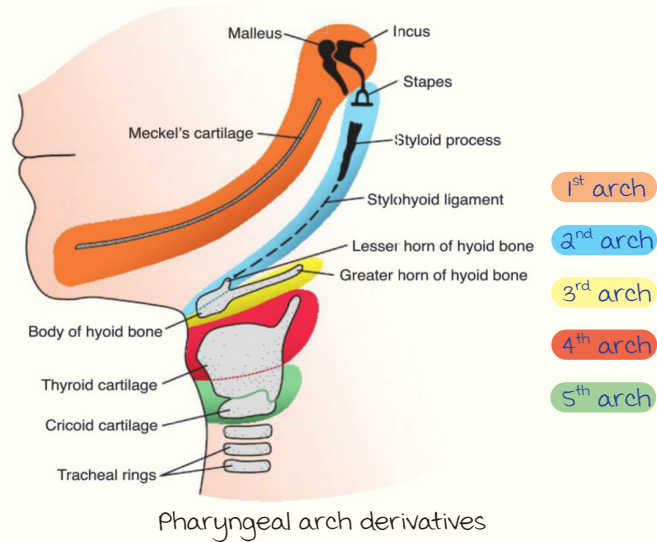
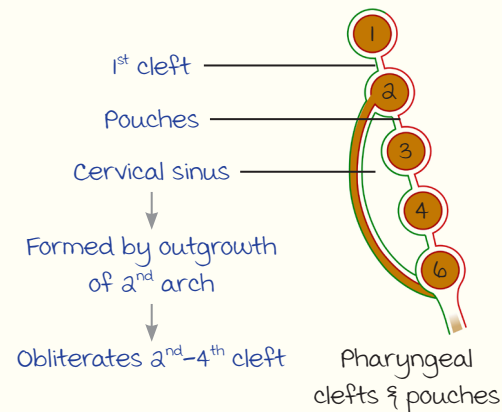
Pathogenesis: Failure of muscles to incorporate into pleuroperitoneal membrane

Intestinal contents in thorax :
Separated by a thin membrane (Not a true hernia).

Pharyngeal Arches

00:34:38



Derivatives :

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1st cleft :

- Only cleft to persist.
- Derivatives :
 - External auditory canal.
 - Outer layer of tympanic membrane (Tm).

5th pouch :

- Disappears.
- **Ultimobranchial body (Remnant) :**
NCC migrate here → Para-follicular C cells.

	Pouch derivatives	Arch derivatives
1	• Inner layer of Tm • Tympanic cavity • Auditory/eustachian tube • Mastoid antrum	• Meckel's cartilage : - malleus - Incus - Anterior ligament of malleus - Sphenomandibular ligament • mandible
2	Lining epithelium of palatine tonsil	• Reichert's cartilage : Stapes (except footplate & annular ligament : Derived from otic capsule) • Styloid process • Stylohyoid ligament • Lesser cornua of hyoid • upper part of hyoid body
3	• Thymus • Inferior parathyroids	• Greater cornua of hyoid • Lower part of hyoid body
4	Superior parathyroids	Laryngeal cartilages
6	-	

Applied Anatomy :

	Developmental anomalies
Pharyngeal cleft	Branchial cyst : Persistent cervical sinus
Pharyngeal pouches	DiGeorge syndrome : • Anomaly of 3rd > 4th pouch • 22q11 microdeletion syndrome • Absent thymus and hypoparathyroidism

EMBRYOLOGY : PART 2

Development of Heart

00:00:40

Heart Tube :

Develops from splanchnopleuric LPM (Cranial end of embryo).

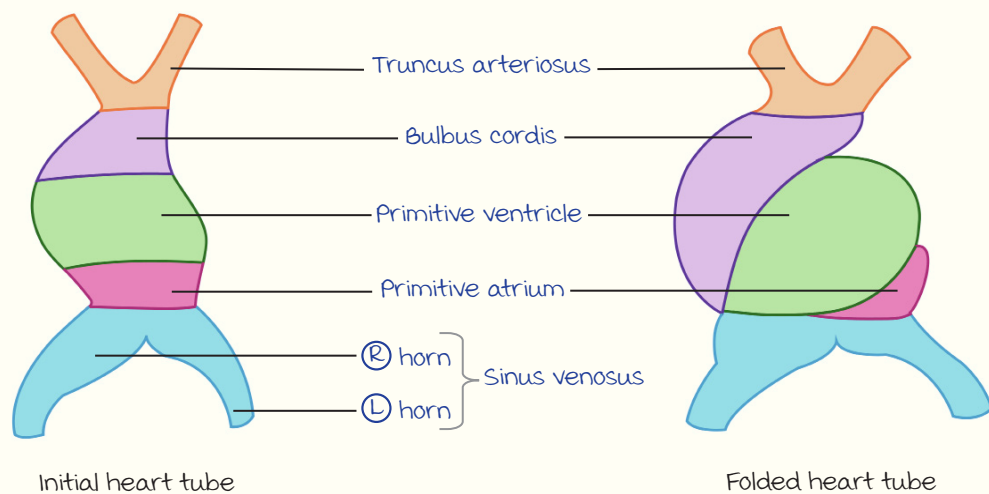
Parts of heart tube		Adult derivatives
Truncus arteriosus		Ascending aorta & pulmonary trunk
Bulbus cordis	Proximal	Trabeculated part of RV
	Distal/conus	Smooth part of both ventricles
Primitive ventricle		Trabeculated part of LV
Primitive atrium		Rough part of RA & LA
Sinus	Ⓡ	Smooth part of RA
venosus	Ⓛ	Coronary sinus

Divided by → Conotruncal septum

(NCC)

⊖

Patent truncus arteriosus.



Note :

Absorption of pulmonary veins → Smooth posterior wall of LA.

Arterial Development

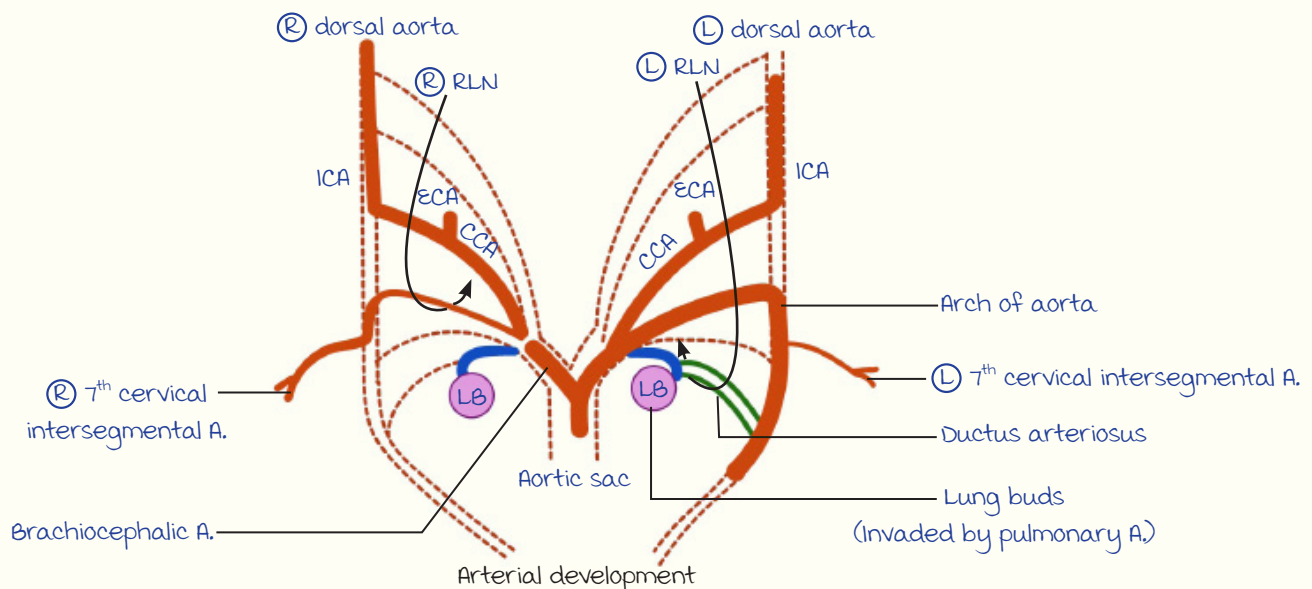
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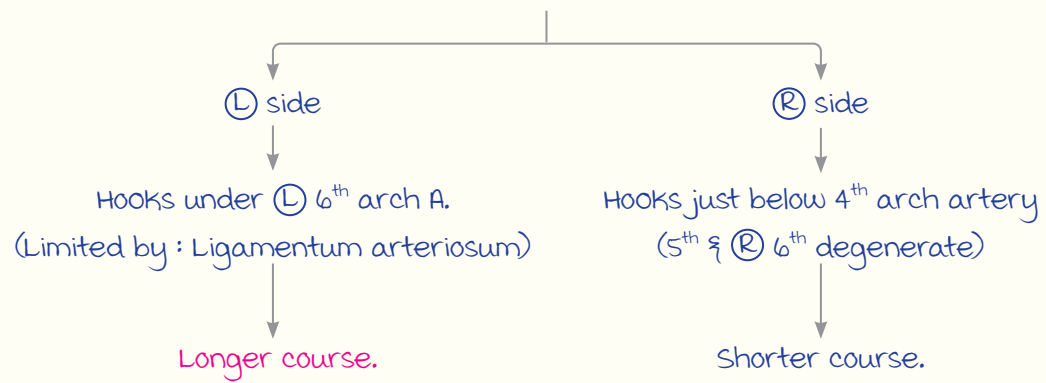
Aortic sac horns (ventral) → Pharyngeal arch arteries → Dorsal aortas → Segmental arteries.

Embryonic structures			Developed arteries
1 st arch artery			maxillary A.
2 nd arch artery			Hyoid & stapedial A. (Disappear after birth)
3 rd arch A.	Proximal part		CCA
	Distal part		ICA
	Bud		ECA
Ⓐ 4th arch artery - Aortic sac Ⓐ horn of aortic sac - Part of Ⓐ dorsal aorta			Arch of aorta
Ⓑ horn of aortic sac			Brachiocephalic A.
5 th arch artery			- (Degenerates)
6 th arch A.	Proximal		Pulmonary artery
	Distal	Ⓑ	- (Degenerates)
		Ⓐ	Ductus arteriosus — (Fibroses after birth)
Ⓑ 4th arch artery Ⓑ 7th cervical intersegmental A.			Ⓑ subclavian
Ⓐ 7 th cervical intersegmental A.			Ⓐ subclavian

Persists → Patent ductus arteriosus.

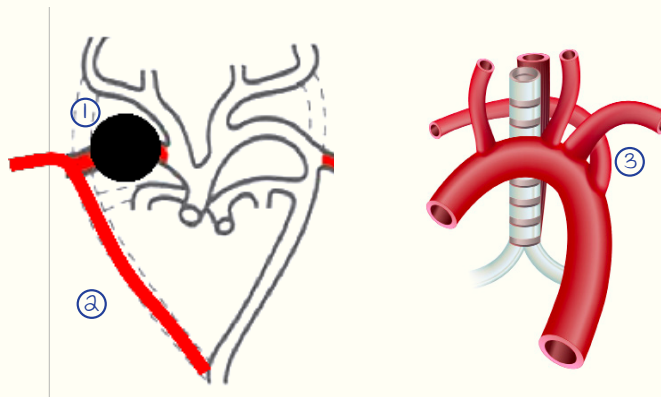


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RLN Course in Relation to Arch Arteries :Applied Anatomy :

Dysphagia lusoria

1. Obliteration of Ⓑ 4th arch A. → 2. Persistence of caudal part of Ⓑ dorsal aorta → 3. Aberrant Ⓑ subclavian artery → 4. Esophagus compressed → 5. Dysphagia.

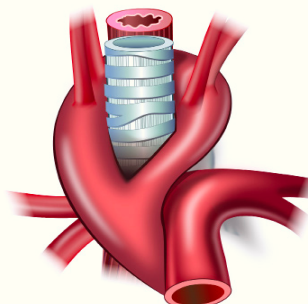


Double aortic arch :

Persistence of caudal part of Ⓑ dorsal aorta + 4th arch A.

Forms vascular ring around trachea & esophagus

Dyspnea & dysphagia.



Double aortic arch

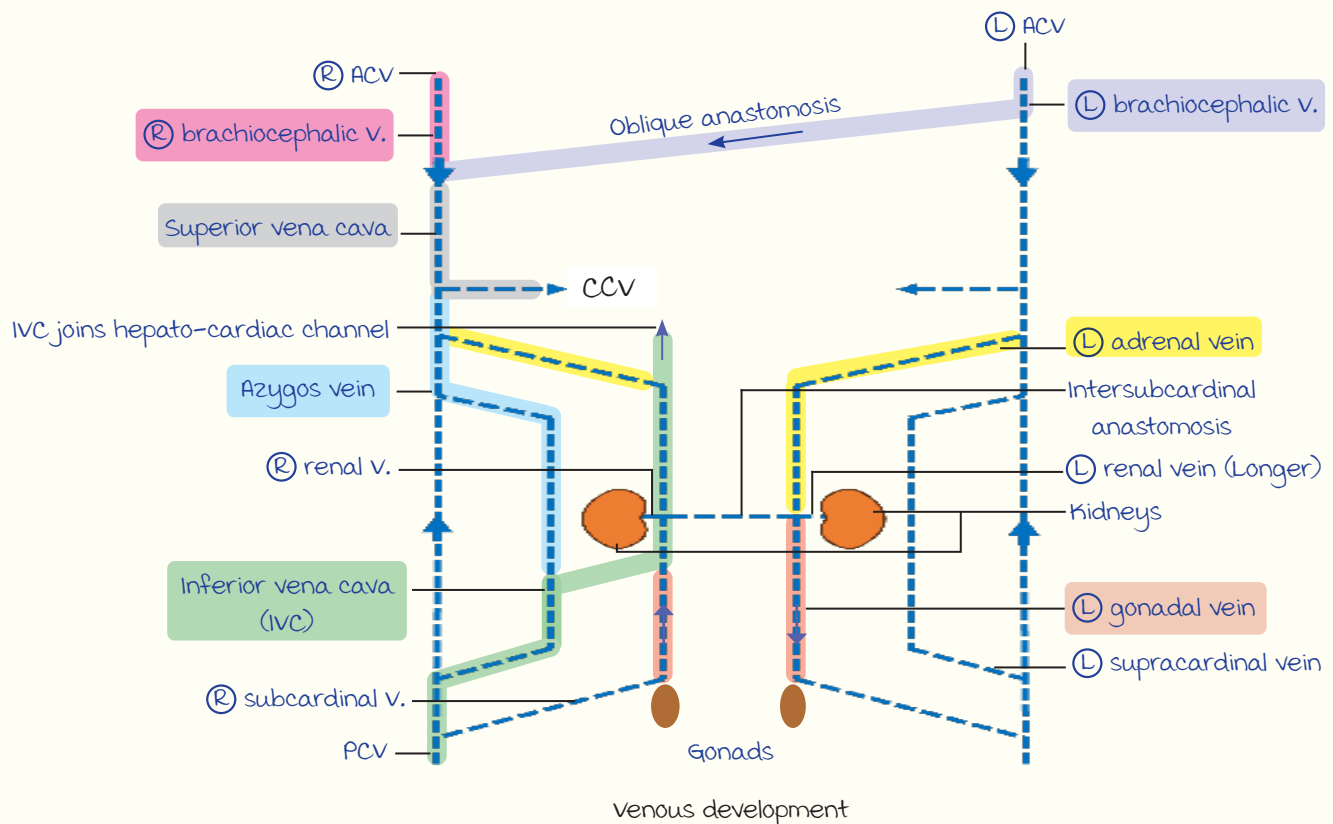
Venous Development

00:26:08

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Upper body → Anterior cardinal veins (ACV)
 Lower body → Posterior cardinal veins (PCV) } Common cardinal veins (CCV).

Embryonic structures	Developed veins
Ⓡ ACV	Ⓡ brachiocephalic vein
Ⓛ ACV + Oblique anastomosis	Ⓛ brachiocephalic vein
Ⓡ ACV + Ⓡ CCV	Superior vena cava
Ⓡ PCV + Ⓡ supracardinal vein + Ⓡ subcardinal vein	Inferior vena cava
Ⓡ mesonephric vein	Ⓡ renal vein
Ⓛ mesonephric vein + Intersubcardinal anastomosis	Ⓛ renal vein
Ⓡ supracardinal vein > Ⓡ PCV	Azygos vein
Caudal part of subcardinal vein	Gonadal vein
Cranial part of subcardinal vein	Adrenal vein



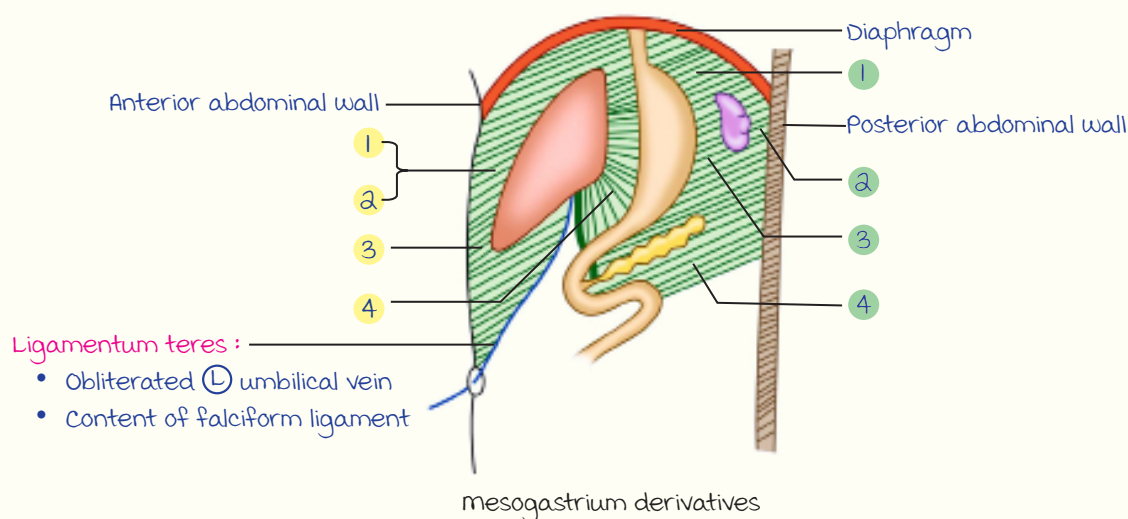
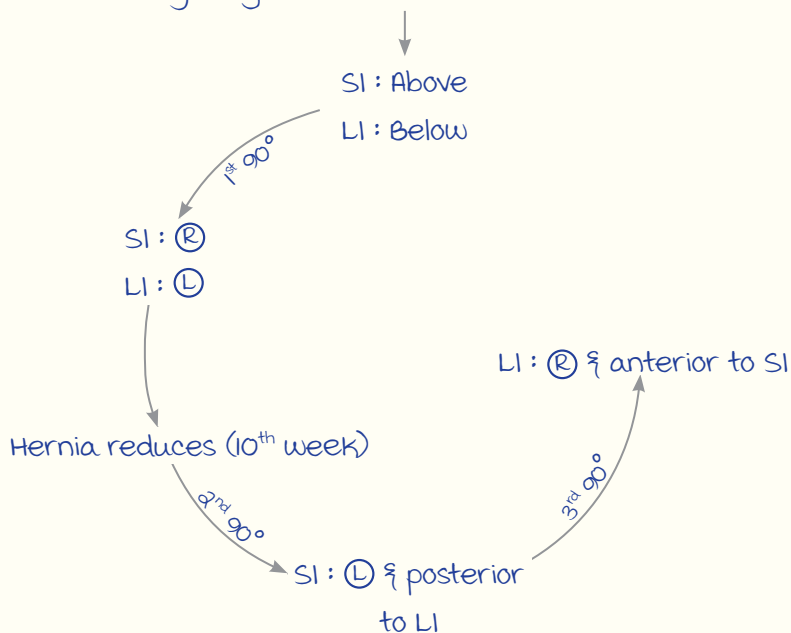
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Development of GIT

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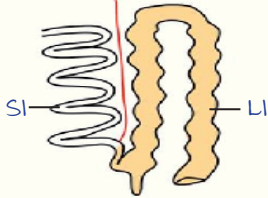
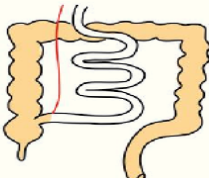
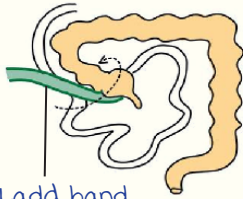
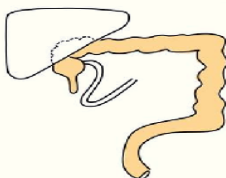
Mesogastrium & its Derivatives :

	Ventral mesogastrium	Dorsal mesogastrium
Derived from	Septum transversum	Lateral plate mesoderm
Relation to stomach	Front of developing stomach	Behind developing stomach
Derivatives	1 Triangular ligament 2 Coronary ligament 3 Falciform ligament 4 Lesser omentum	1 Gastrophrenic ligament 2 Lienorenal ligament 3 Gastrosplenic ligament 4 Greater omentum

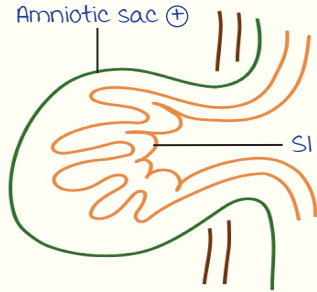
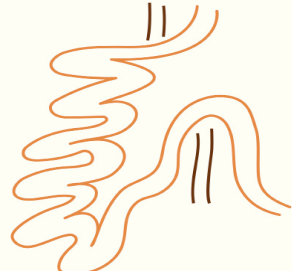
midgutNormal rotation : 270° ($90^\circ + 90^\circ + 90^\circ$) anti-clockwise.Physiological umbilical hernia (6th week)

Rotational anomalies :

----- Active space -----

	Defect in rotation	Features
Non-rotation 	1st rotation ↓ Arrest	• SI : Ⓜ • LI : Ⓛ
Reverse rotation 	1st rotation ↓ Reverse rotation (180°)	Transverse colon overlapped by SMA (Superior mesenteric artery) & duodenum
Mixed rotation 	Absent 3rd rotation	• Subpyloric caecum (midline) held by Ladd band : Compresses SI • Rx : Ladd band resection
Incomplete rotation 	Incomplete 3rd rotation	Subhepatic caecum : Appendicitis mimics cholecystitis

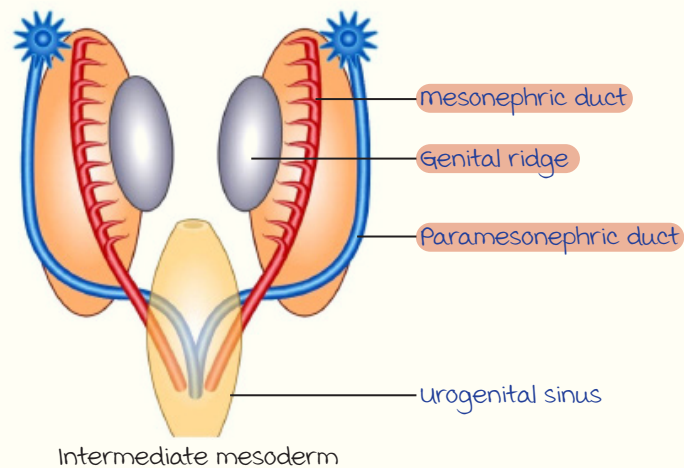
Omphalocele vs. gastroschisis :

	Omphalocele	Gastroschisis
Defects	Failure of reduction of physiological hernia	Reherniation of midgut from anterior abdominal wall defect
Features	Amniotic sac ⊕ 	Amniotic sac ⊖ 
Complications	A/w multiple anomalies ↓ ↑ mortality	Intestinal obstruction

----- Active space -----

Development of Urogenital System

00:57:10

Derived from **intermediate mesoderm**.**Derivatives of Intermediate mesoderm :**

	males	Females
mesonephric/ wolffian duct	- Posterior wall of proximal urethra - ureteric bud → Collecting system of kidneys - Trigone of bladder	
	- Spermatic pathway - Appendix of epididymis	Remnant : Gartner's duct (In broad ligament)
Paramesonephric/ mullerian duct	Remnants : - Appendix of testis - Prostatic utricle	- Fallopian tube - uterus - upper vaginal wall
Genital ridge	Testes	Ovaries

HISTOLOGY

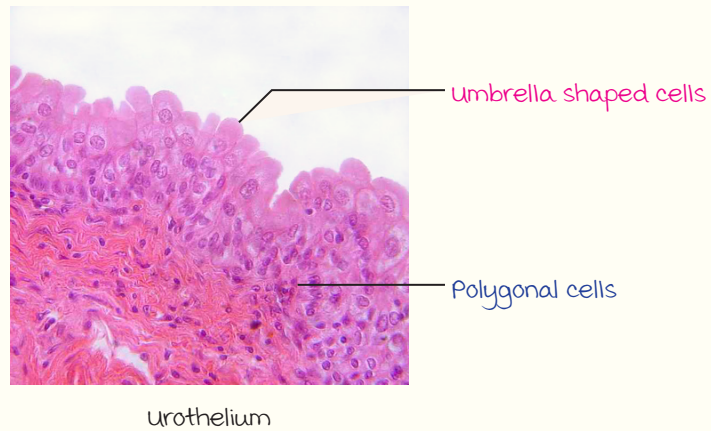
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Types of Epithelium

00:01:02

Epithelium		Examples
Simple squamous		• Endothelium • mesothelium • Alveoli
Simple cuboidal (Central, round nucleus)		• Renal tubules • Ducts of glands • Surface of ovary (Germinal epithelium)
Simple columnar (Basal, oval nucleus)	Non-ciliated	• Stomach • Small intestine • Large intestine • Gall bladder
	Ciliated	• Uterine tube • Uterus • Small bronchi
Pseudostratified columnar ciliated		Respiratory tract (Trachea, bronchi)
Stratified Squamous	Keratinized	Epidermis of skin
	Non-keratinized	• Oral cavity • Esophagus • Vagina • Anal canal
Stratified cuboidal		Large ducts of : • Salivary glands • Sweat glands • Pancreas
Stratified columnar		• Conjunctiva • male urethra • Large excretory ducts
Transitional (Urothelium) 		• urinary bladder (Trigone) • ureters • major & minor calyces • Renal pelvis • Proximal urethra

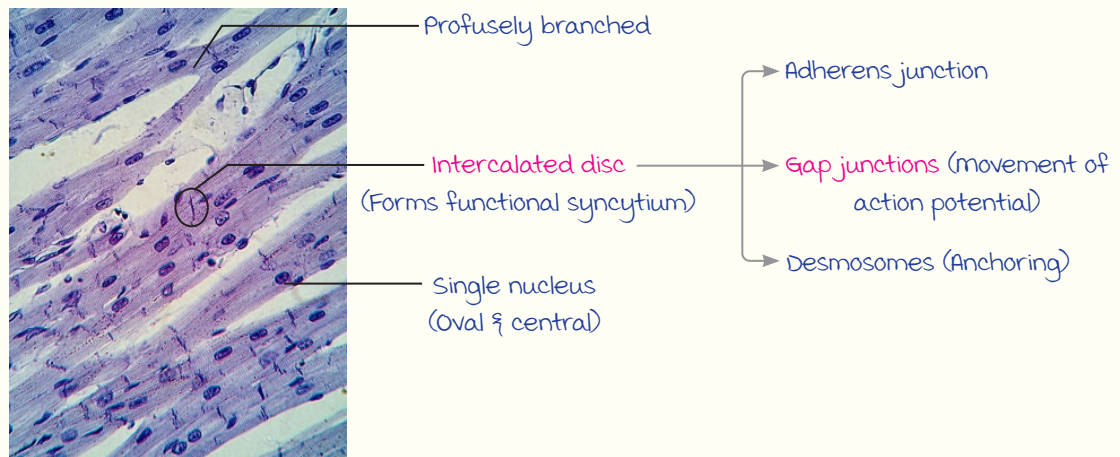
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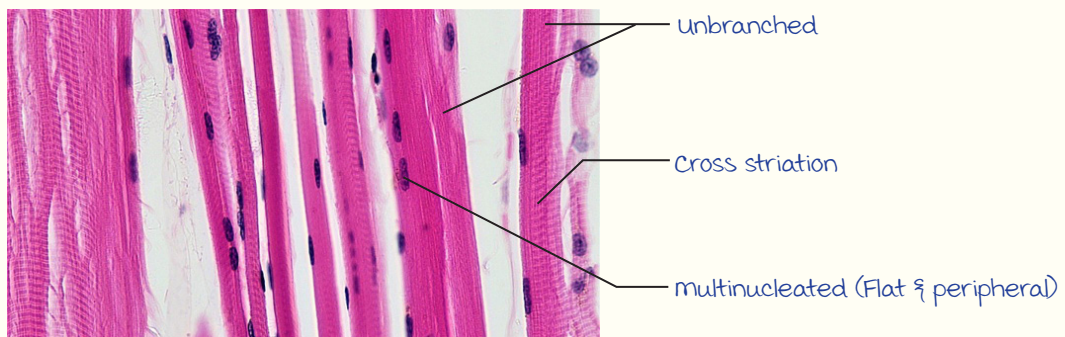
Muscles

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Cardiac muscle :



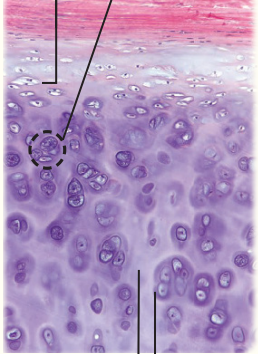
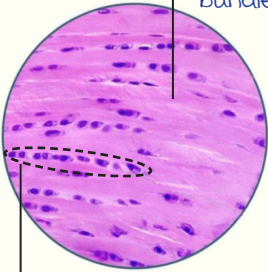
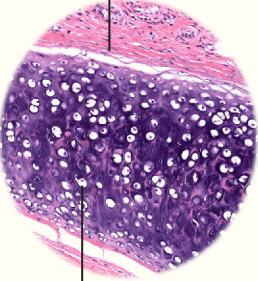
Skeletal muscle :



Cartilages

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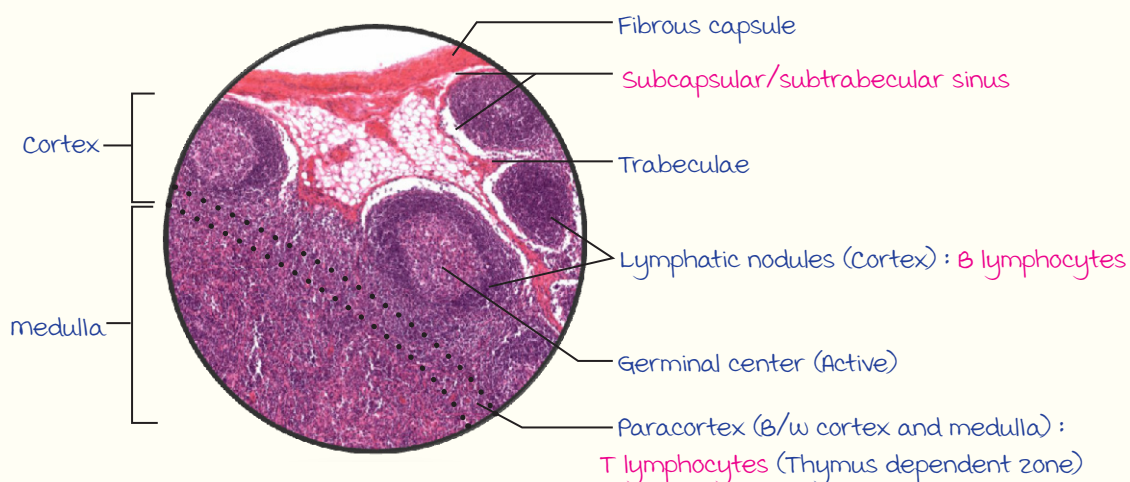
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	Cartilage		
	Hyaline	Fibro	Elastic
Arrangement of chondrocytes (In lacunae)	 Cell nest	 Linear (Rows)	 Elastic fibres Individual large chondrocytes
Histology	 Perichondrium Cell nest Inter-territorial matrix Territorial matrix	 Collagen bundles Linear chondrocytes	 Perichondrium Individual chondrocytes
Perichondrium	⊕	⊖	⊕
Example	Articular cartilage	Intervertebral disc	-

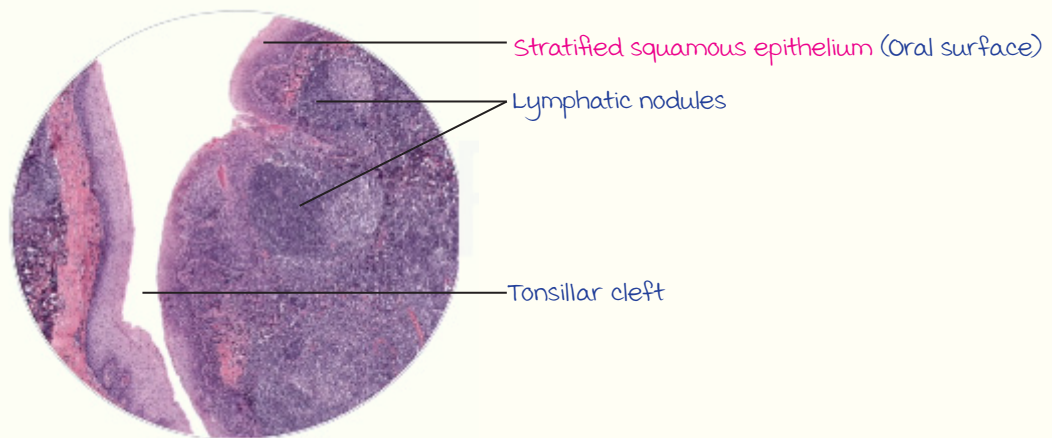
Lymphoid System

00:24:44

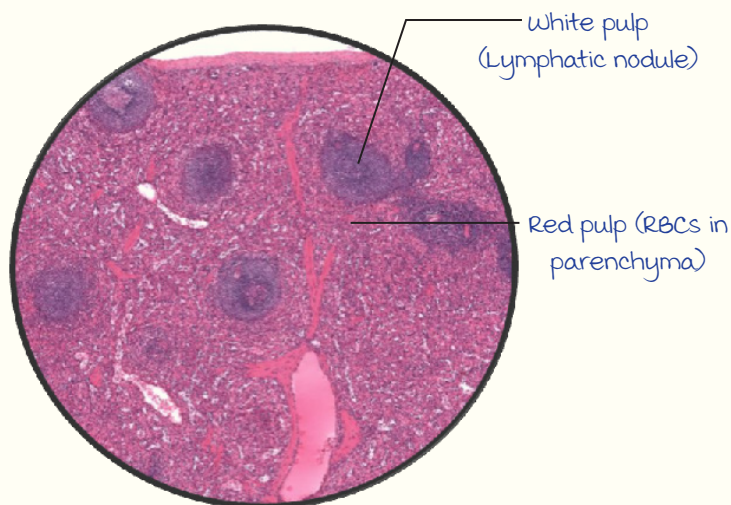
Lymph Node :



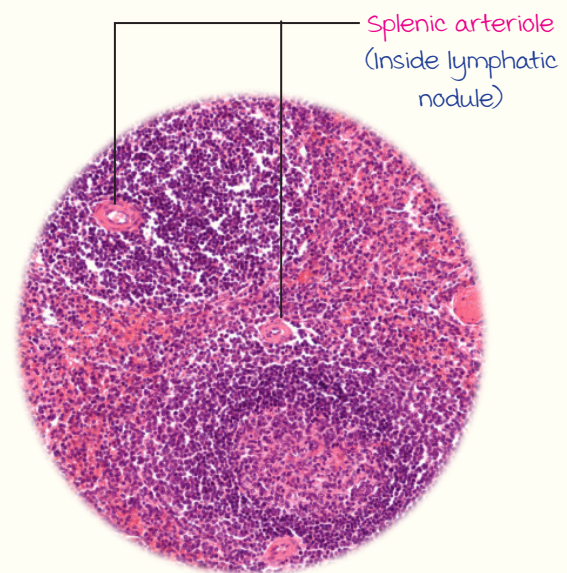
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Palatine Tonsil :Spleen :

No differentiated cortex & medulla.



Splenic tissue



Lymphatic nodule in spleen

Glands

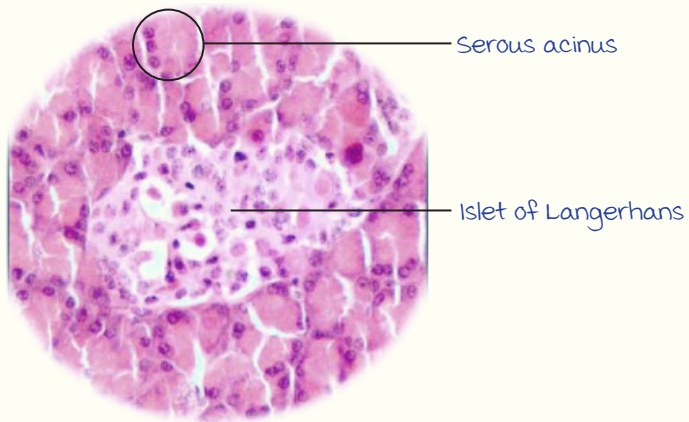
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	Serous acinus	Mucous acinus
Histology features	Pyramidal cells Round, central nucleus Rough endoplasmic reticulum	Columnar cells Flat, basal nucleus
Location	• Parotid gland • Pancreas	-

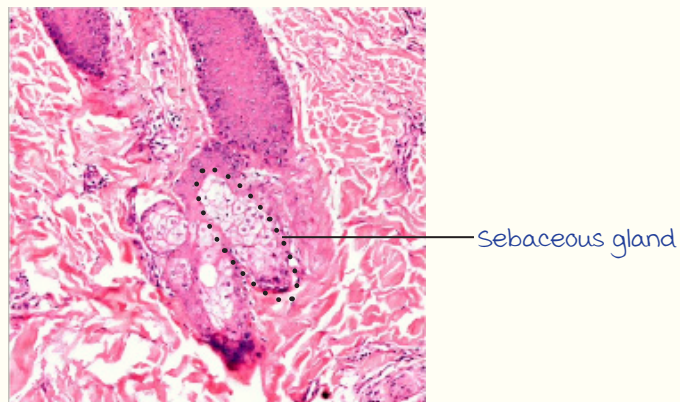
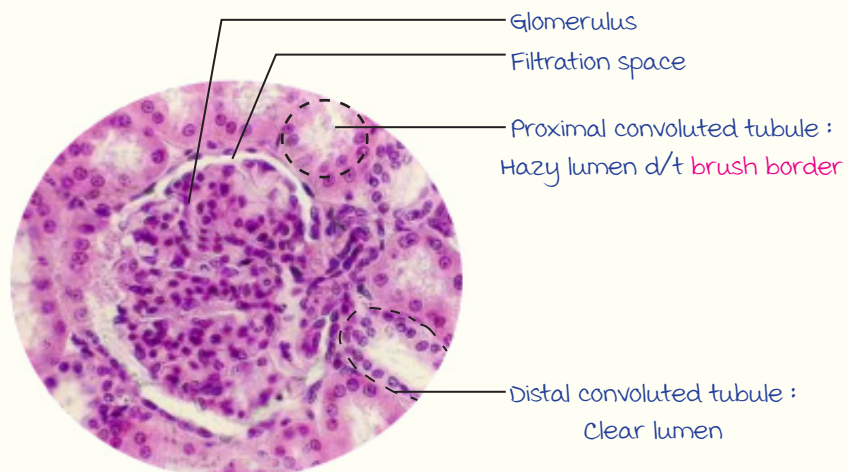
Pancreas :

----- Active space -----

Centroacinar cells : Cells of serous acini duct → Nuclei in lumen ⊕.

Sebaceous Gland :

- Polyhedral cells.
- Central round nucleus.
- Foamy cytoplasm.
- Holocrine secretion.

Note :Kidney :

----- Active space -----

HEAD, NECK AND FACE : PART 1

Cranial Nerve Nuclei and Columns

00:01:45

		Cranial nerve column							
		GSE		SVE	GVE	GVA	SVA	GSA : Trigeminal nucleus	SSA
		Nerve	At level of						
Brainstem	mid-brain	 III	Superior colliculus	-	EW (Edinger westphal)	-	-	 mesencephalic nucleus	
		 IV	Inferior colliculus						
	Pons	 VI	Facial colliculus	 V ₃  VII	 SSN  ISN	-	-	 Principal sensory nucleus	
	medulla	 XII	-	 IX X XI (NA) Nucleus ambiguus	 Dorsal nucleus of X	 VII IX X (NTS) Nucleus of tractus solitarius		 Spinal nucleus	
						 Cochlear nuclei  Vestibular nuclei			
Activity		Supplies :				Sensation :			
		Skeletal muscles derived from :			• Smooth muscles • Glands	General sensation (Pain, touch, temperature) from :		• Olfaction • Vision • Hearing & balance	
		Somites : • Extraocular muscles • Tongue	Pharyngeal arches	viscera		Body wall			

III

VII, IX, X

● : Parasympathetic CNS & columns they belong to

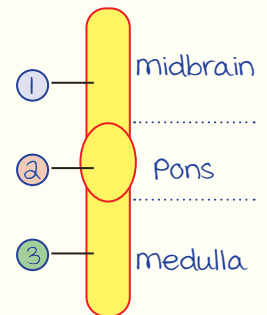
GSE = General somatic efferent
 SVE = Special visceral efferent
 GVE = General visceral efferent
 GVA = General visceral afferent
 SVA = Special visceral afferent

GSA = General somatic afferent
 SSA = Special somatic afferent
 SSN = Superior salivatory nucleus
 ISN = Inferior salivatory nucleus

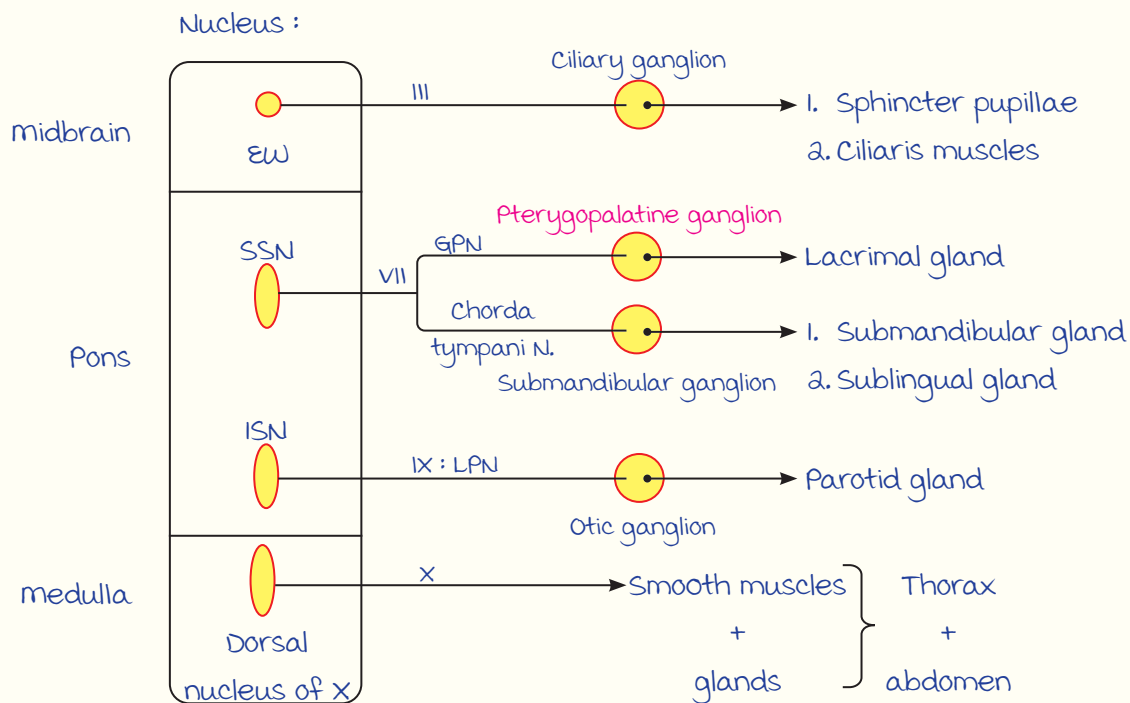
GSA  :

----- Active space -----

Nuclei parts	Sensation	Key points
mesencephalic	Proprioception (Only here)	- Only site for pseudo-unipolar neuron in CNS - Centre for jaw reflex
Principal/chief sensory	Touch & pressure (Primarily)	-
Spinal	Pain & temperature (Primarily)	Can also receive touch & pressure



Trigeminal nuclei

GVE  :

GPN : Greater Petrosal N.

LPN : Lesser Petrosal N.

Pterygopalatine ganglion : Largest parasympathetic ganglion.

----- Active space -----

Note :

- Parasympathetic outflow (Craniosacral)
 - Cranial : III, VII, IX, X.
 - Sacral : S₂, S₃, S₄.
- Sympathetic outflow (Thoracolumbar) : T₁ - L₂.

Cranial Nerves & Columns :

Cranial Nerve	Columns						
	GSE	SVE	GVE	GVA	SVA	GSA	SSA
I			No nuclei				
II							
III	⊕		⊕				
IV	⊕						
V		⊕				⊕	
VI	⊕						
VII		⊕	⊕	⊕	⊕	⊕	
VIII							⊕
IX		⊕	⊕	⊕	⊕	⊕	
X		⊕	⊕	⊕	⊕	⊕	
XI	⊕	⊕					
XII	⊕						

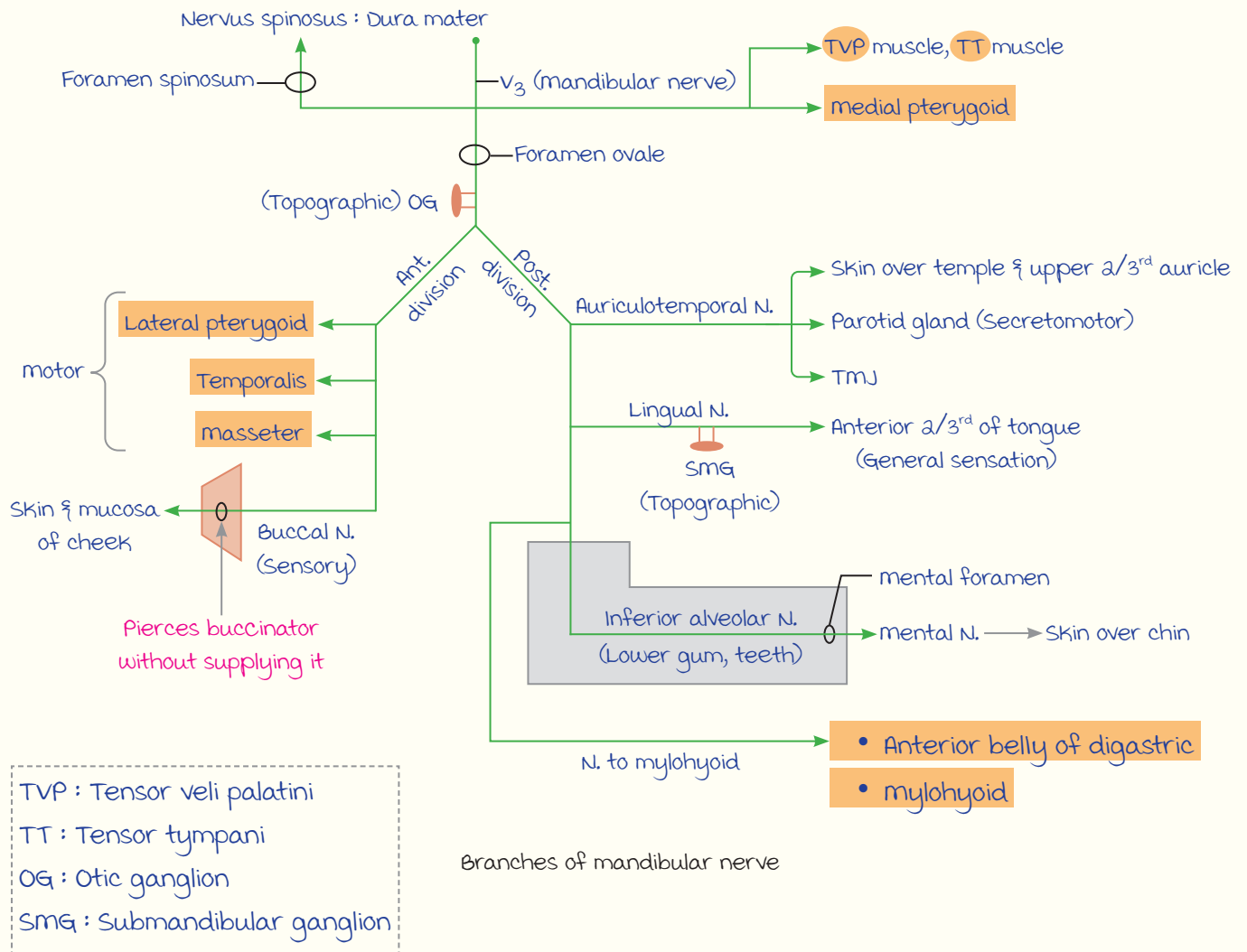
● : Spinal accessory N. ● : Cranial accessory N.

Mandibular Nerve

00:27:54

----- Active space -----

Nerve of 1st arch.



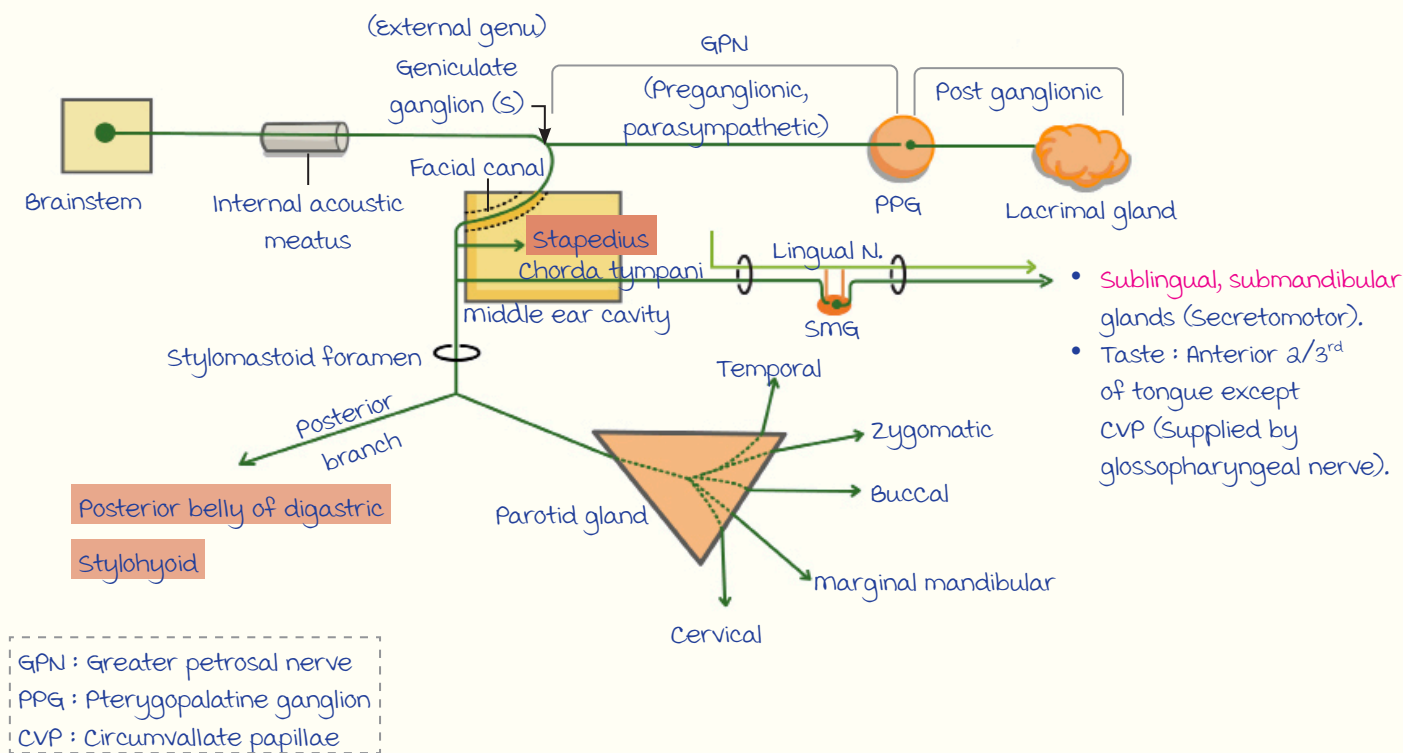
Note :

Buccinator muscle : Supplied by buccal branch of facial nerve.

----- Active space -----

Facial Nerve

00:36:20

Nerve of 2nd arch.**Branches of Facial Nerve :**

Branch	muscles supplied	Actions hindered in injury
Temporal	Frontalis	Wrinkling of forehead
Zygomatic	Orbicularis oculi	Tight closing of eyes
Buccal	Buccinator + upper lip muscles	Inflating cheeks, smiles
marginal mandibular	Lower lip muscles	Pouting
Cervical	Platysma	2 ^o muscle for lower lip action

Facial Nerve Injury :

	I/L facial paralysis	Loss of taste sensation from anterior 2/3 rd of tongue	Loss of salivation	Hyperacusis (Stapedius)	Loss of lacrimation
Stylomastoid foramen	⊕	⊖	⊖	⊖	⊖
Above chorda tympani	⊕	⊕	⊕	⊖	⊖
medial wall of middle ear	⊕	⊕	⊕	⊕	⊖
Geniculate ganglion	⊕	⊕	⊕	⊕	⊕

muscle & Nerve Supply : Summary

----- Active space -----

muscle	Nerve supply	Exception
muscle of tongue	CN XII	Palatoglossus → VAC (Vago-accessory complex)
muscle of palate	VAC	Tensor veli palatini → V ₃ (Trunk of mandibular nerve)
muscle of larynx	RLN	Cricothyroid → ELN
muscle of pharynx	VAC	- Stylopharyngeus → CN IX - Cricopharyngeus → RLN

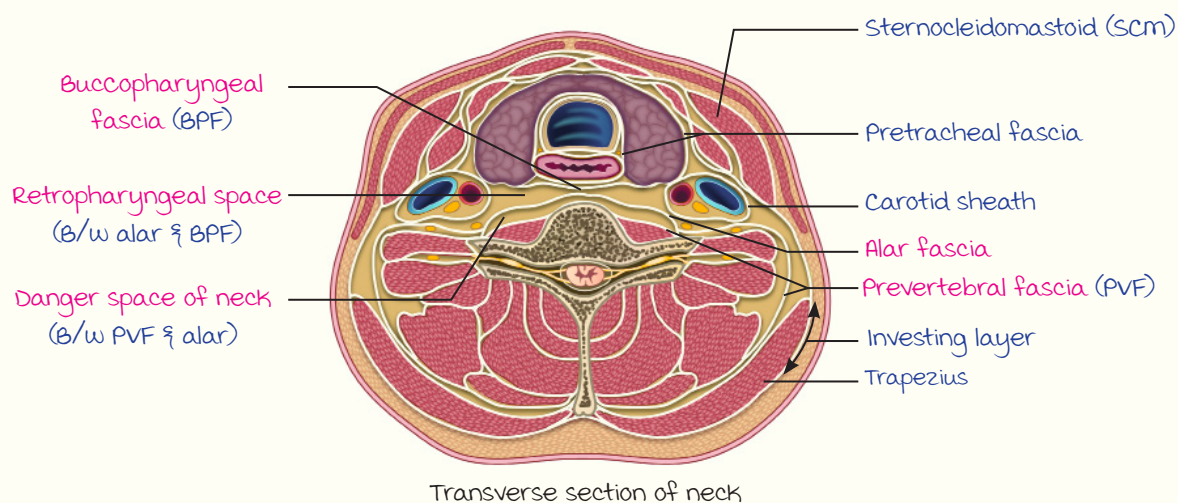
Rule of 17 :

Lesion : X + VII	Lesion : XII + V
Deviation : Contralateral deviation  - Right CN X lesion ↓ Deviation of uvula to the left  - Right CN VII lesion ↓ Deviation of angle of mouth to left	Deviation : Ipsilateral deviation  - Left CN XII lesion ↓ On protrusion, deviation of tongue to the left  - Left CN V lesion ↓ Deviation of jaw to the left

Deep Cervical Fascia

00:55:58

Fascia & Spaces :



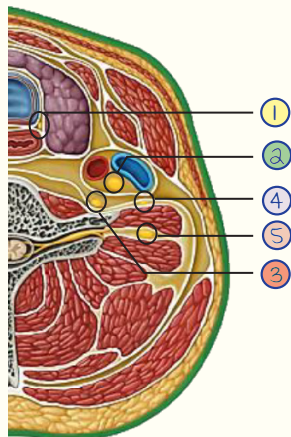
----- Active space -----

Fasciae	Significance
Investing layer	Covers SCM + trapezius
Prevertebral fascia	- Covers vertebral column & its muscles - Extends to form : Axillary sheath
Pretracheal fascia	Forms false capsule of thyroid gland
Carotid sheath	Extension of : Pretracheal + investing + prevertebral fascia
Buccopharyngeal fascia	Extension of pretracheal fascia
Alar fascia	- Anterior layer of PVF - merges with BPF

Danger space of neck :

- Extent : Open till diaphragm.
- Significance : Infection/abscess descends down to posterior mediastinum → Dyspnea, dysphagia.

Nerves :



Transverse section : Neck

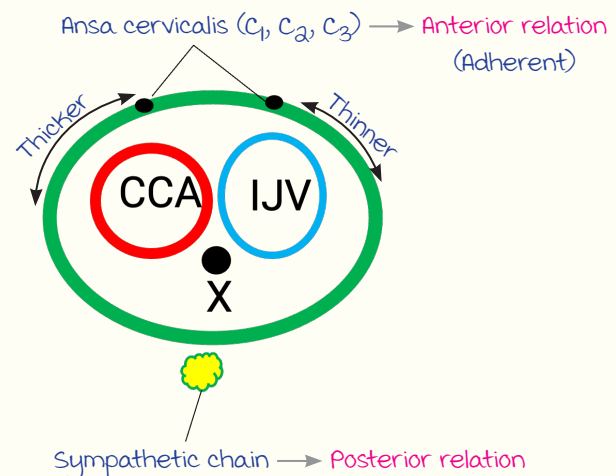
Name	Location
RLN (Recurrent laryngeal N.)	Tracheo-esophageal groove
CN X	Inside carotid sheath
Sympathetic chain	B/w carotid sheath & PVF
Phrenic N.	Anterior to scalenus anterior & posterior to PVF
Brachial plexus (Cervical nerves/roots)	B/w scalenus anterior & medius

Carotid Sheath :

In carotid triangle of neck.

Contents :

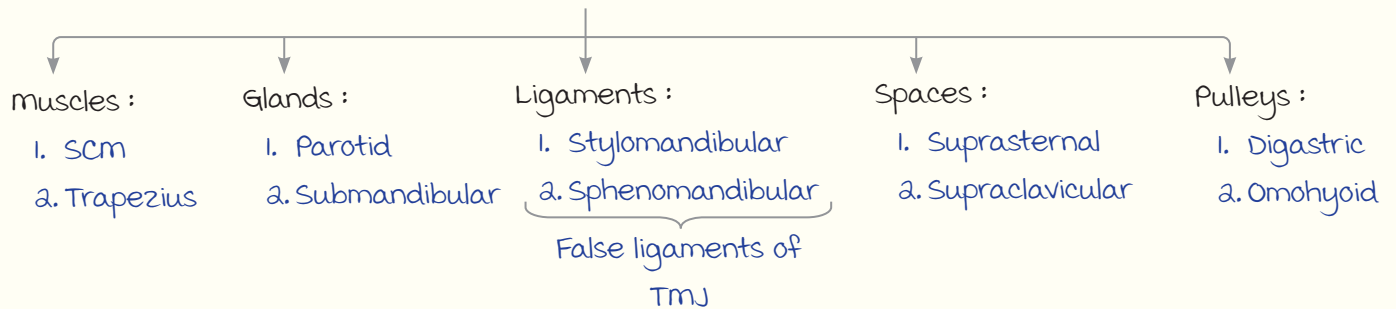
- Common carotid artery (CCA)
- Internal carotid artery (ICA)
- Internal jugular vein (IJV).
- CN X.
- CN IX, XI, XII : Traverse upper part.



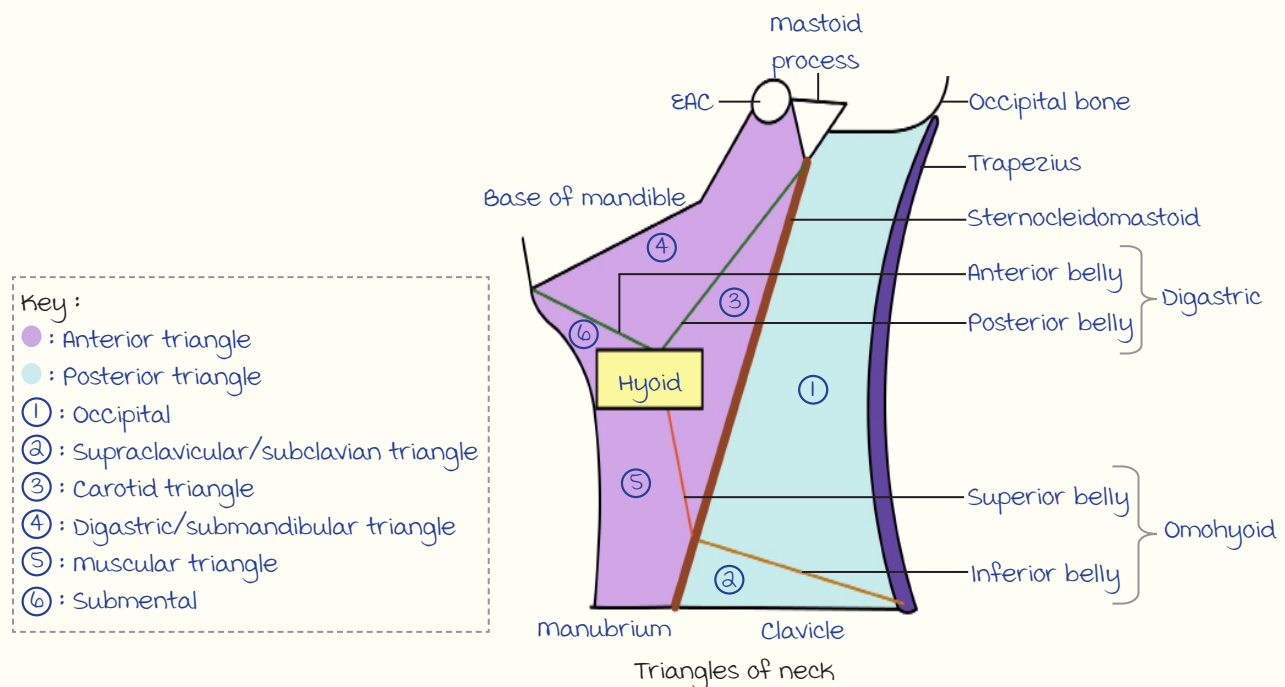
Investing Layer :

----- Active space -----

Rule of 2 :

Triangles of Neck

01:13:15



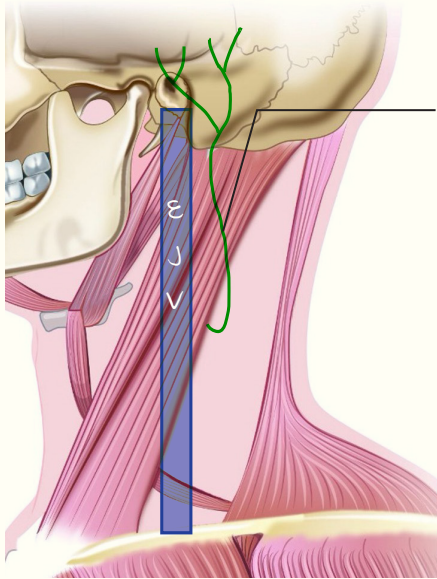
Contents of muscular triangle : All strap muscles except inferior belly of omohyoid.

----- Active space -----

Posterior Triangle of Neck :

Contents :

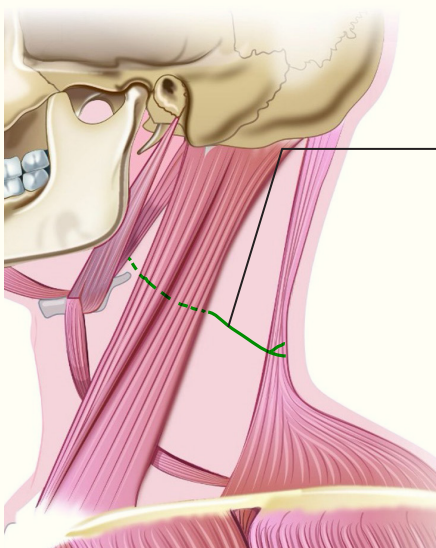
- Great auricular nerve (Roof).
- Spinal accessory nerve.

Great Auricular Nerve (GAN) :

- Root value : C₂-C₃.
- Runs with EJV.
- Supplies :
 - Ear lobule.
 - Skin at angle of mandible & parotid region.
- Frey's syndrome : Auriculotemporal nerve (ATN) communicates to GAN (m/c).
- Injured in parotid Sx → Sensory loss in shaving area.

Note :

ATN injury during parotid Sx → Sensory loss on upper 2/3rd of auricle & temple.

Spinal accessory nerve :

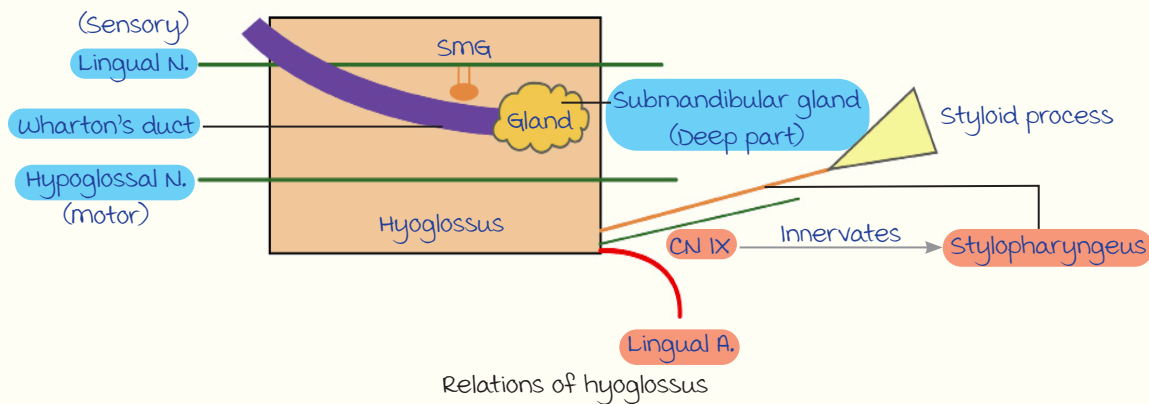
- Supplies SCM $\xrightarrow{\text{Followed by}}$ Trapezius.
- Runs b/w investing layer & prevertebral layer.
- Injury in posterior triangle → SCM spared.
- Trapezius affected → Difficulty in :
 - Shrugging of shoulder.
 - Retraction of scapula.
 - Overhead abduction.

Digastric Triangle :

----- Active space -----

Contents  :

- Hyoglossus
 - Lingual N.
 - Hypoglossal N.
 - Submandibular gland + Wharton's duct (Deep part)
- Superficial to hyoglossus + Deep to mylohyoid
- Stylopharyngeus
 - CN IX
 - Lingual artery
- Deep to hyoglossus



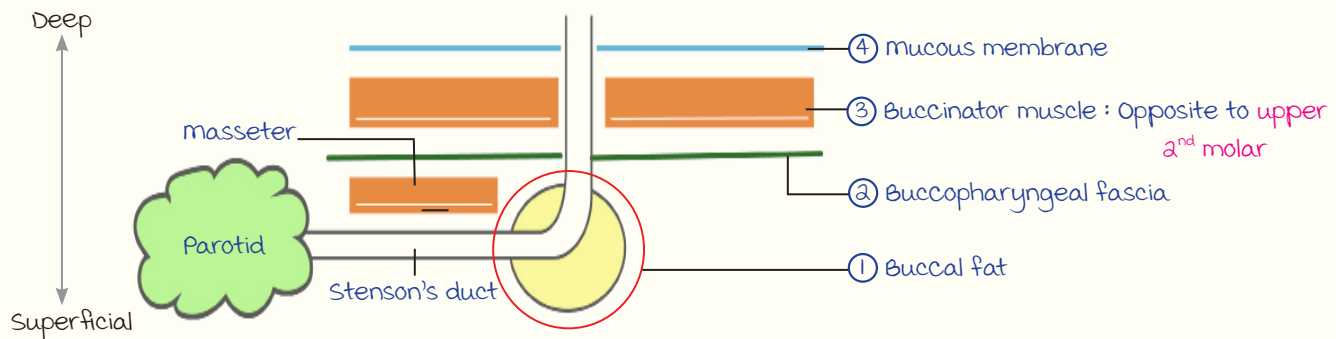
Applied anatomy :

- Wharton's duct Sx → Lingual N. injured (m/c) → Sensory loss on anterior 2/3rd of tongue.
- Submandibular gland Sx :
 - Hypoglossal N. injured → Tongue muscle weakness
 - Glossopharyngeal N. injured → Stylopharyngeus weakness/ Sensory loss on posterior 1/3rd of tongue.

----- Active space -----

Note :

- Structure pierced by Stenson's duct :



- Structures piercing buccinator :
 - a. Buccal branch of mandibular N.
 - b. Stenson's duct.
 - c. molar glands/glands of buccopharyngeal fascia.

HEAD, NECK AND FACE : PART 2

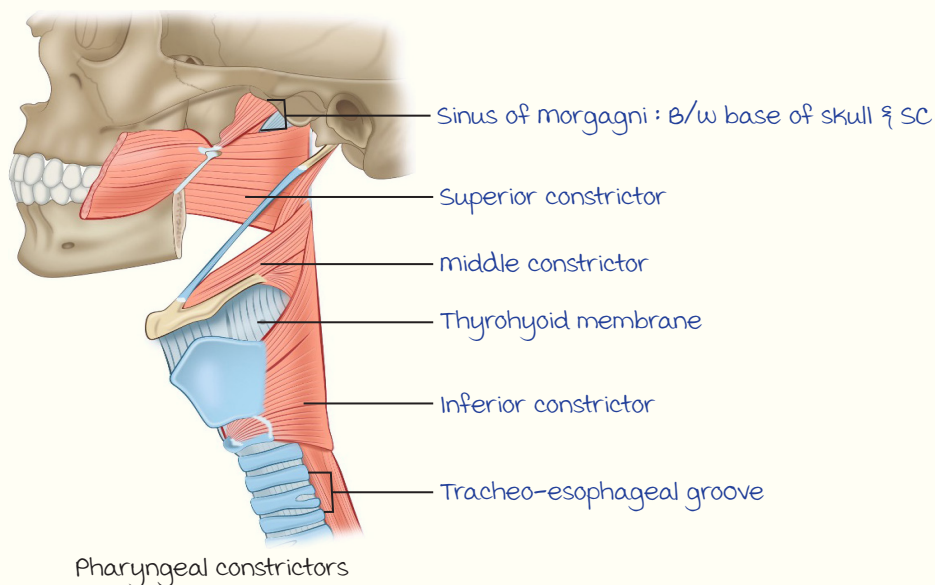
----- Active space -----

Pharyngeal Constrictors

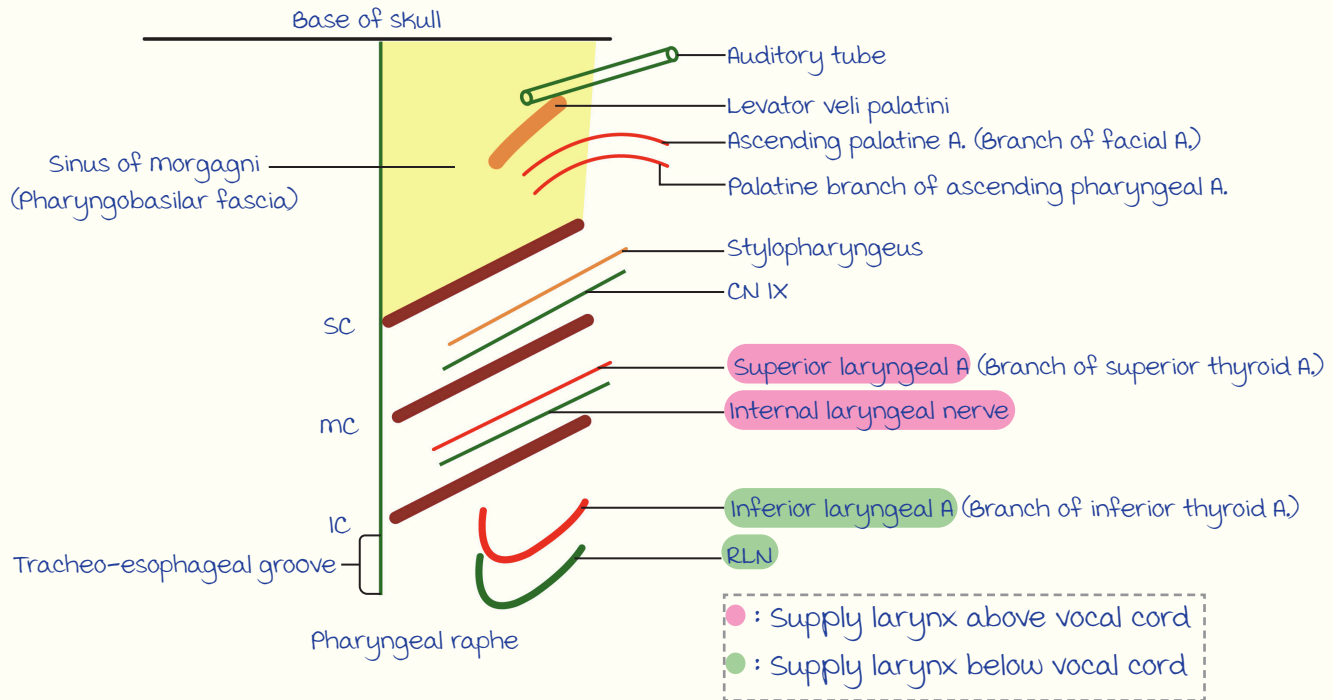
00:00:45

Origin & Insertion :

muscles		Origin	Insertion
Superior constrictor (SC)		Pterygomandibular raphe	Pharyngeal raphe : modification of pharyngobasilar fascia
middle constrictor (mc)		Greater cornua of hyoid + Stylohyoid ligament	
Inferior constrictor (IC)	Thyropharyngeus	Thyroid cartilage (Oblique line)	
	Cricopharyngeus	Cricoid cartilage	



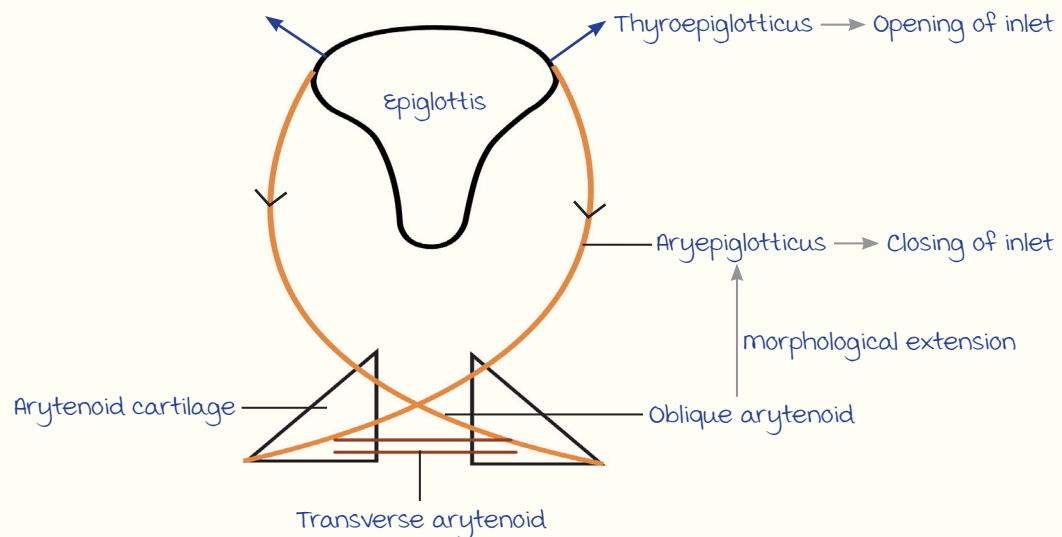
----- Active space ----- Structure Passing b/w Pharyngeal Constrictors :



Laryngeal Muscles

00:10:07

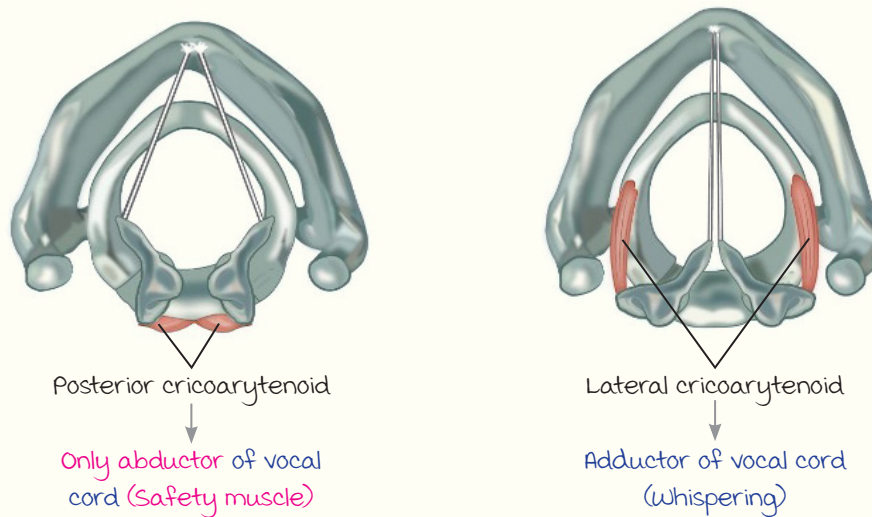
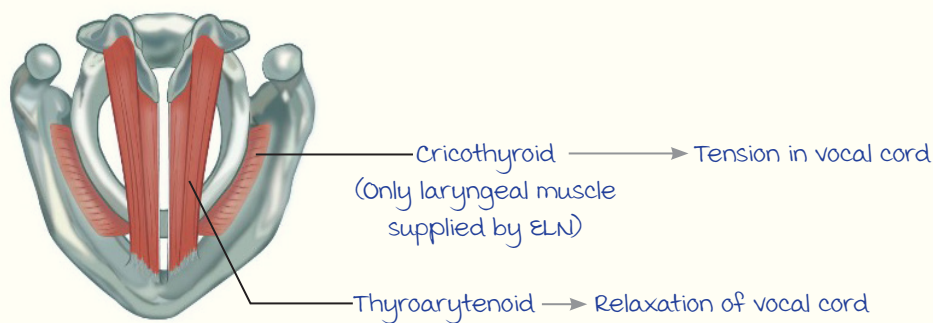
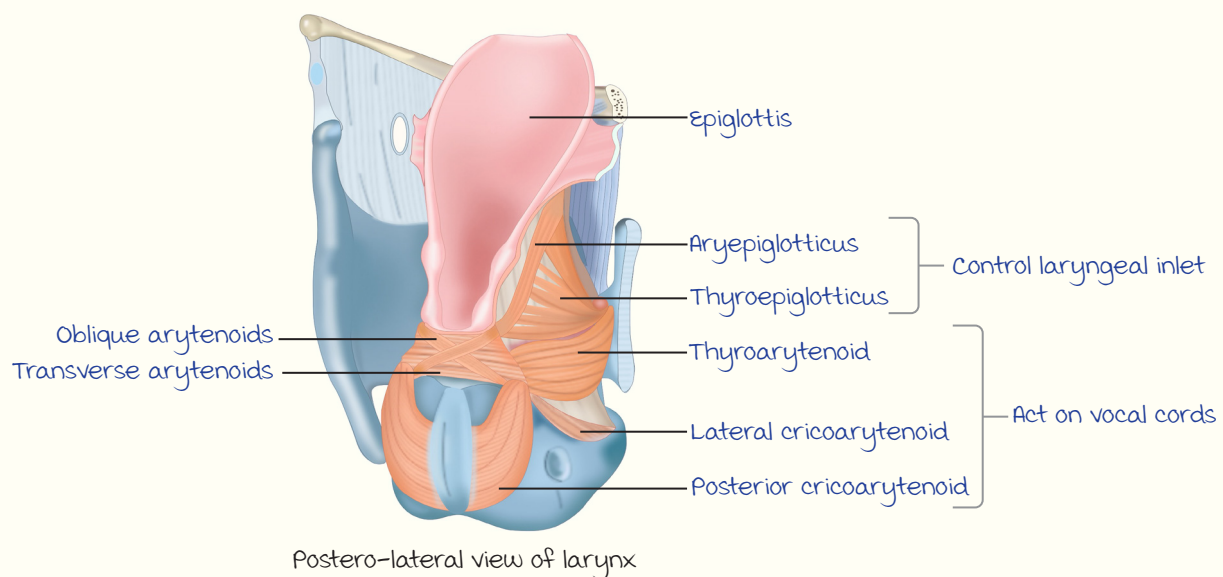
muscle Controlling Laryngeal Inlet :



Muscles Controlling Rima Glottidis :

----- Active space -----

Rima glottidis : Space b/w vocal cords.

Muscles Controlling Tension in Vocal Cords :Summary & Identification :

----- Active space -----

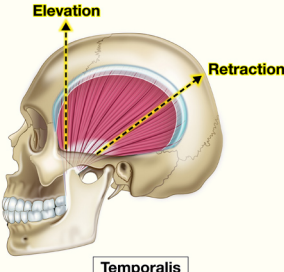
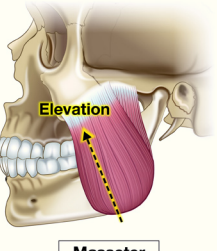
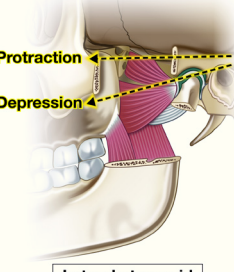
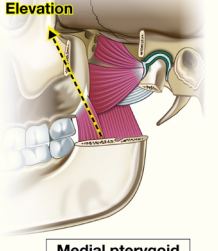
TMJ & Muscles of Mastication

00:21:47

TMJ :

- Condylar joint.
- Lateral pterygoid (LP) tendon extends to form Articular disc of TMJ.
- m/c dislocation : Anterior dislocation d/t excessive mouth opening (Yawning).

muscles of mastication :

	Temporalis	masseter	Lateral pterygoid	medial pterygoid
Nerve supply	mandibular N. : Anterior division			mandibular N. : Trunk
Origin	Temporal fossa	Zygomatic bone + zygomatic arch	LP plate	
			Lateral surface	medial surface
Insertion	mandible : • Coronoid process • Ant. border of ramus	Ramus of mandible : Outer surface	Neck of mandible : + TMJ capsule	Ramus of mandible : Inner surface
Action	• Ant. fibres : Elevation > • Post. fibres : Retraction (Unique)	Elevation (Teeth clenching)	• Depression (Unique) • Protraction	• Elevation > • Protraction
			u/L : l/L contraction → Deviates jaw c/L	
Image				

Innervation of Dura Mater

00:32:09

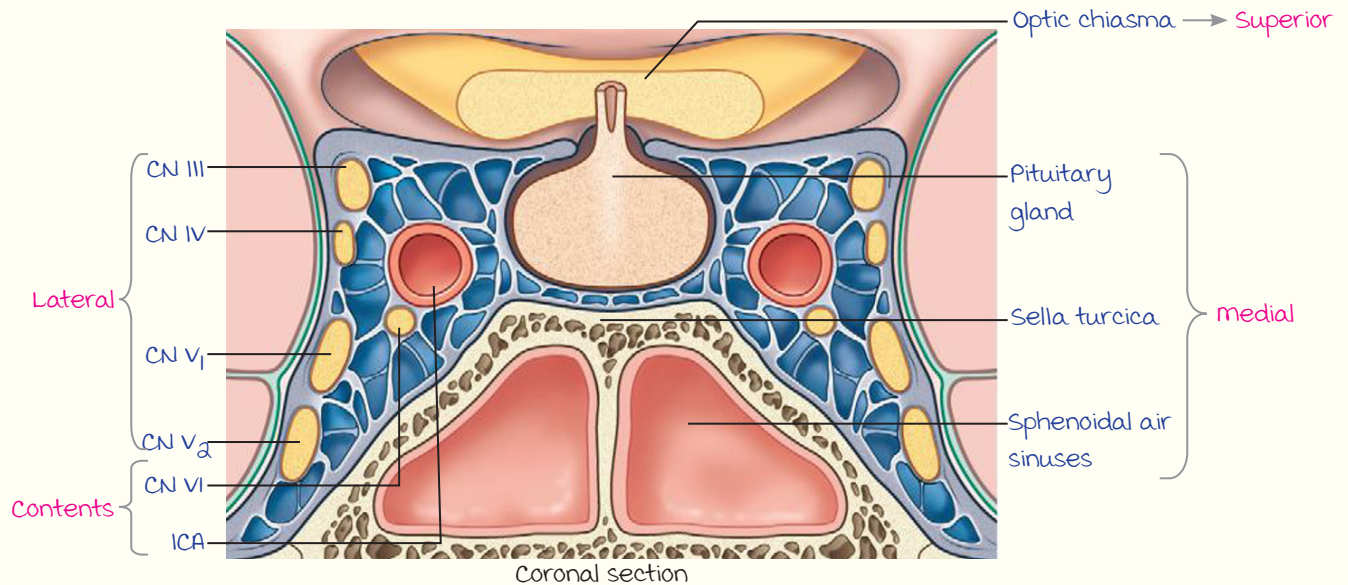


Cavernous Sinus (CS)

00:36:29

----- Active space -----

Relations :



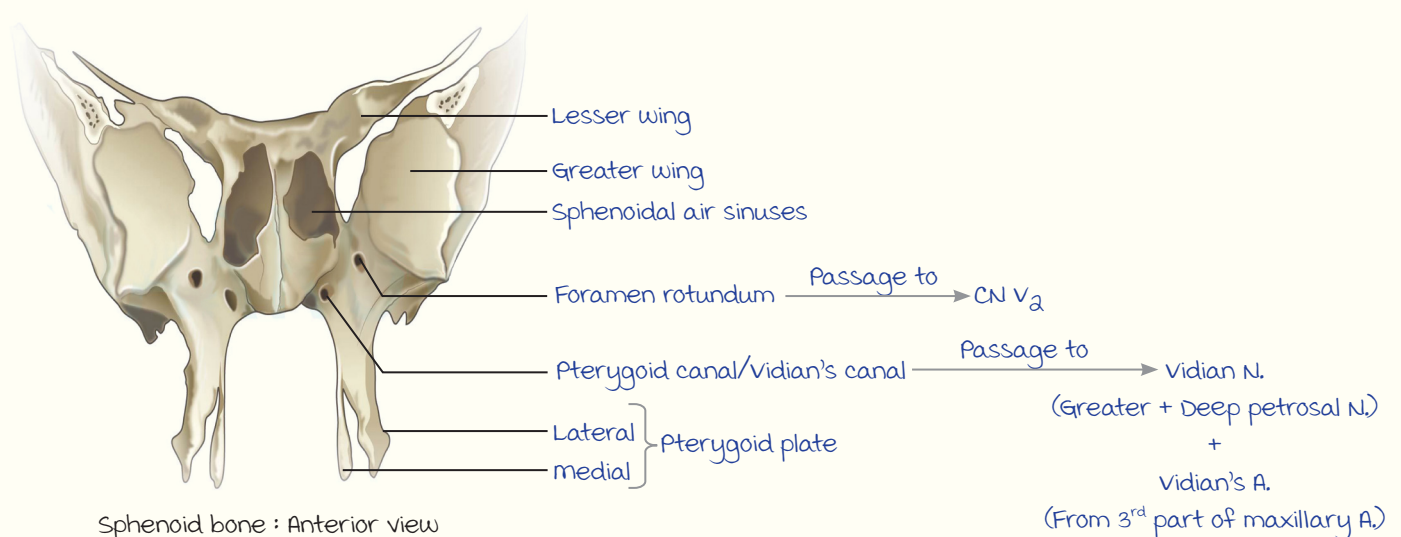
Applied Anatomy :

1. Cavernous sinus thrombosis : **CN VI** m/c affected.
2. Carotid-cavernous fistula :
 - Rupture of ICA aneurysm inside CS → Carotid-cavernous fistula.
 - ICA pulsation → Transmission via cavernous sinus → **Pulsatile proptosis**.

Head & Neck : Osteology

00:40:27

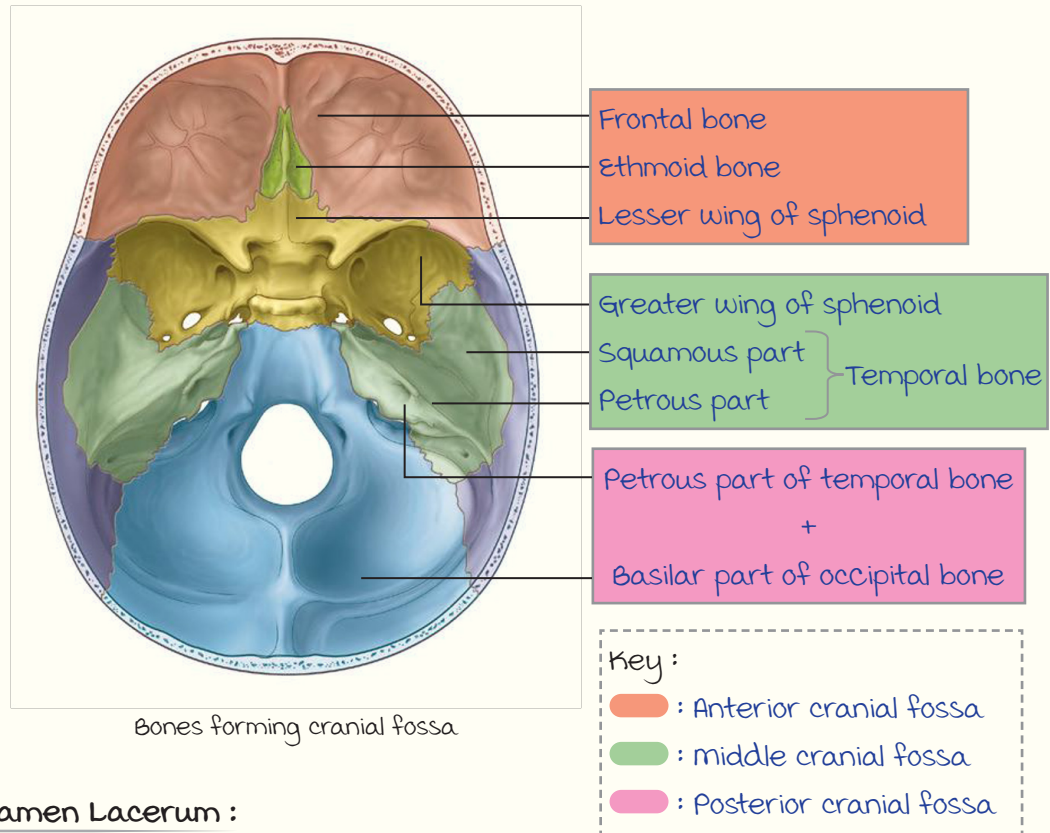
Sphenoid Bone :



----- Active space -----

Structures seen from posterior view :

- Entire greater wing.
- Foramen : Ovale, spinosum, rotundum.

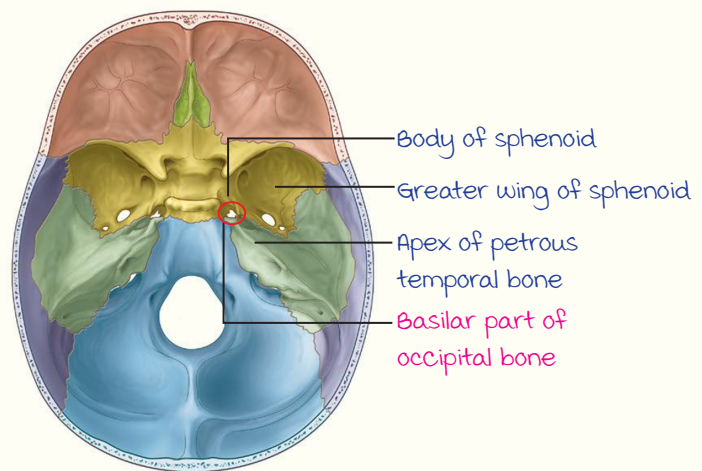
Bones Forming Cranial Fossa :Foramen Lacerum :

Structures traversing :

- meningeal branch of ascending pharyngeal A.
- Emissary veins.

Structures close to/partially traversing :

- ICA + sympathetic plexus.
- DPN (Deep petrosal N.).
- GPN (Greater petrosal N.).

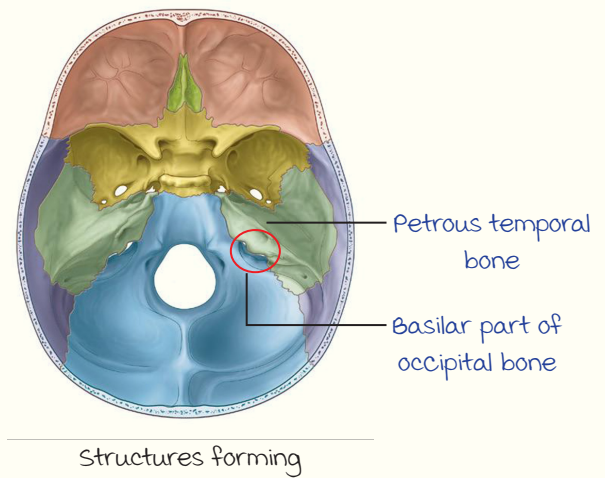
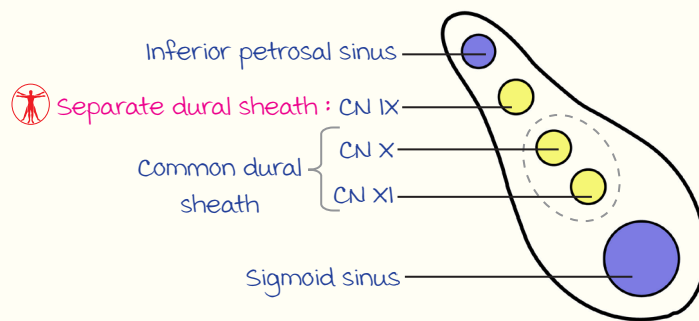


Structures contributing

Jugular Foramen :

----- Active space -----

Contents :

Identification of Cranial Nerves With Dura mater Intact :

Key :

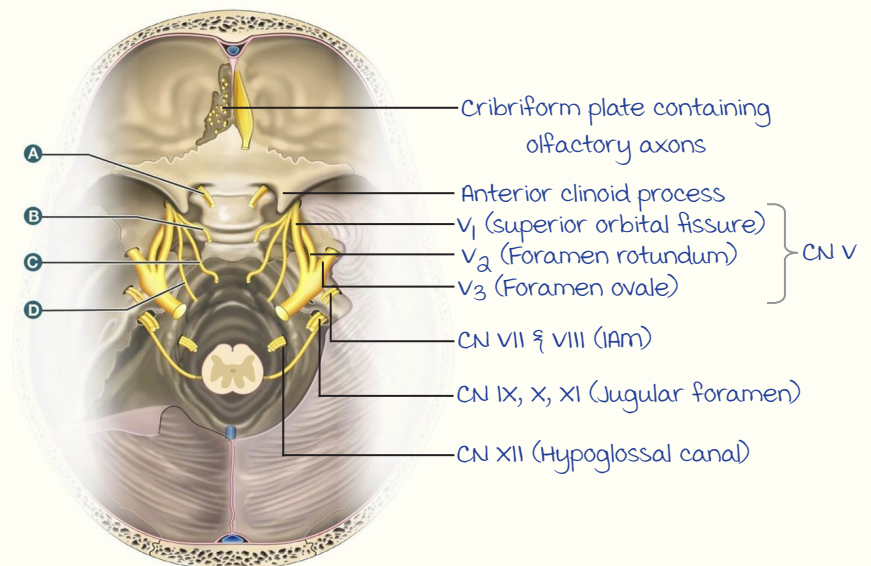
A : Optic N. in optic canal
(medial to anterior clinoid process)

B : CN III

C : CN VI

- Longest intradural course.
- Runs in cavernous sinus.

D : CN IV



Cranial nerves exiting cranial cavity

Lateral wall of cavernous sinus :

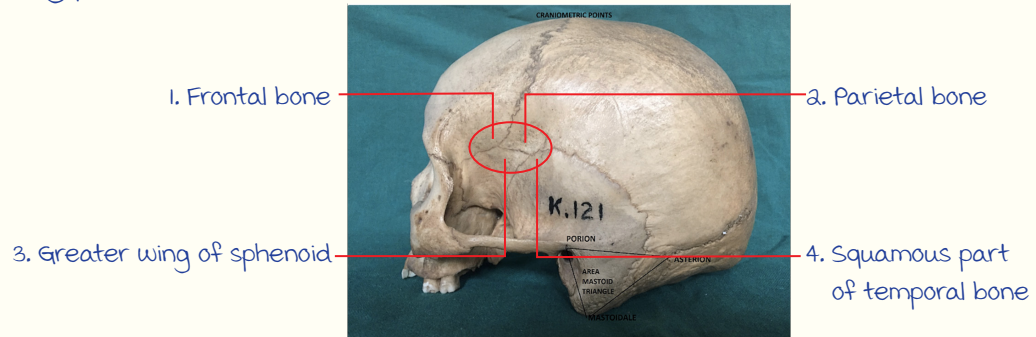
- CN III
- CN VI
- CN V₁
- CN V₂

----- Active space -----

Pterion :

Formation :

meeting point of 4 bones :



Superficial relations :

- Superficial temporal vessels.
- Auriculotemporal N.

Deep relations :

- Anterior division of middle meningeal A.
- middle cerebral vessels. Running in
- Sylvian fissure. under
- Insula. under
- Broca's motor speech area (On the (L)).
- Lesser wing of sphenoid.

NEUROANATOMY : PART 1

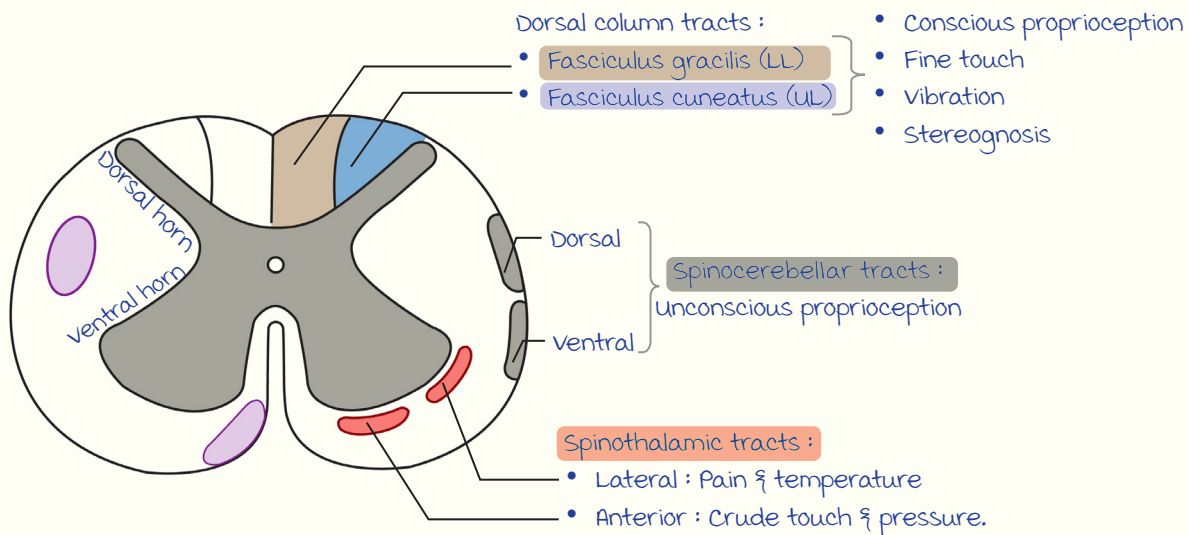
----- Active space -----

Salient external features of spinal cord :

- Extent : Lower border of L₁.
- Extent of dura & arachnoid : S₂.
- Conus → Coccygeal vertebrae : Filum terminale.

Ascending Tracts

00:01:19

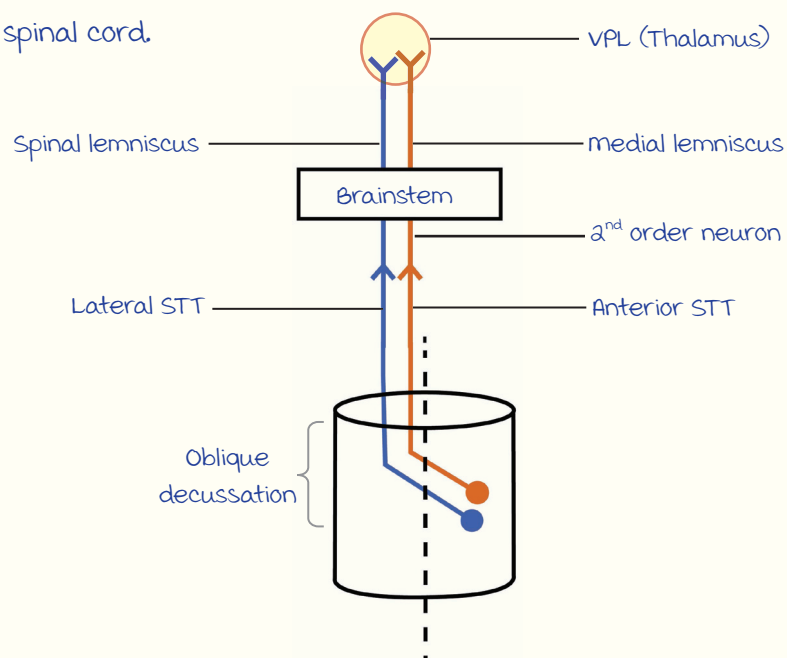


Spinothalamic Tract :

Decussation : Obliquely in spinal cord.

Note :

- VPL : Limbs.
- VPM : Face/mouth.

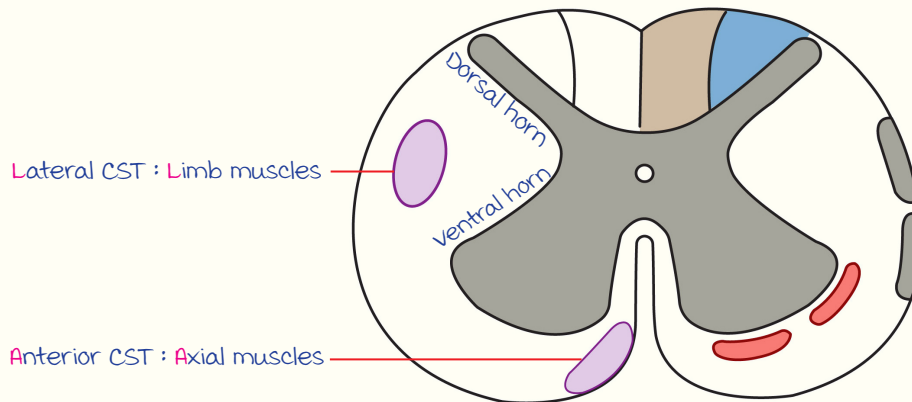


Descending Tract

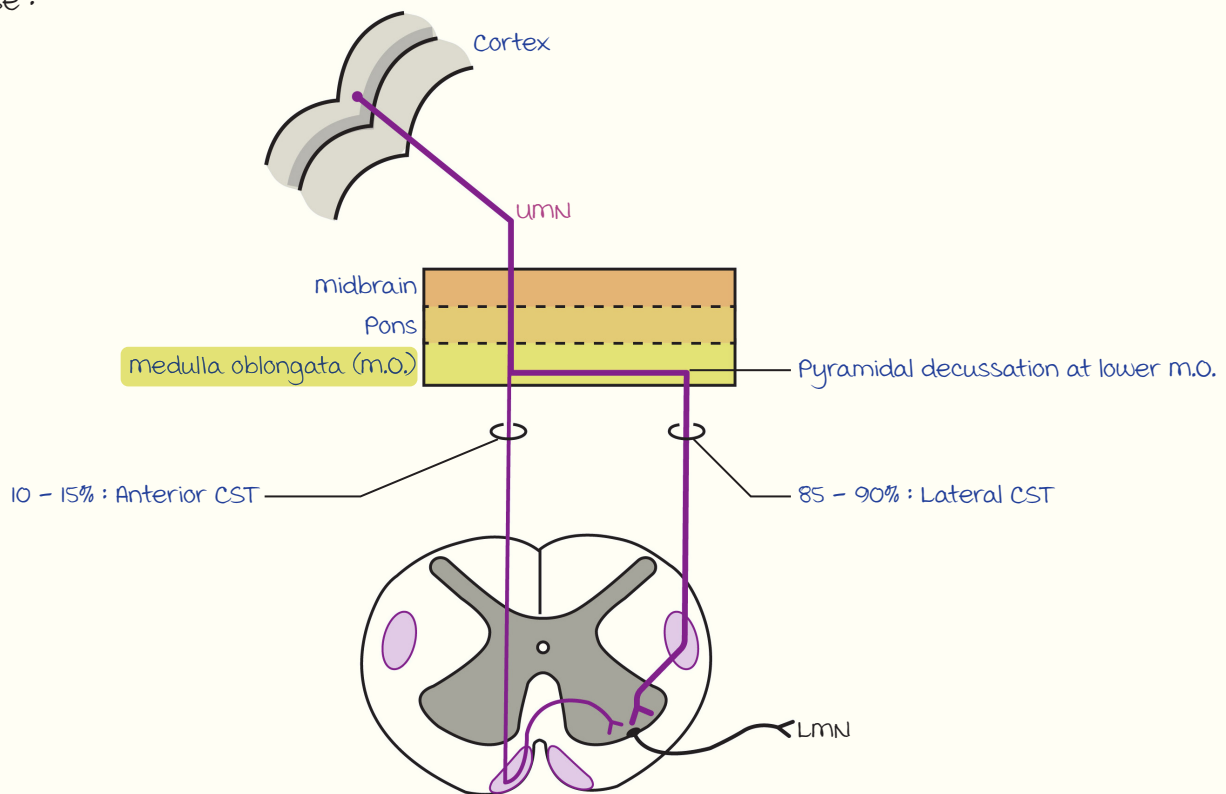
00:19:00

----- Active space -----

Corticospinal (CST)/Pyramidal Tracts :



Course :



Brown-Sequard Syndrome

00:25:30

Hemisection of spinal cord.

Tract	Features
Spinothalamic	- At level of lesion : I/L loss - Below level of lesion : C/L loss
Dorsal column	At & below level of lesion : I/L loss
Spinocerebellar	
Corticospinal (Lateral)	- At the level : I/L flaccid paralysis (LMN lesion) - Below the level : I/L spastic paralysis (UMN lesion)

----- Active space -----

Ventral Aspect of Brainstem

00:37:49

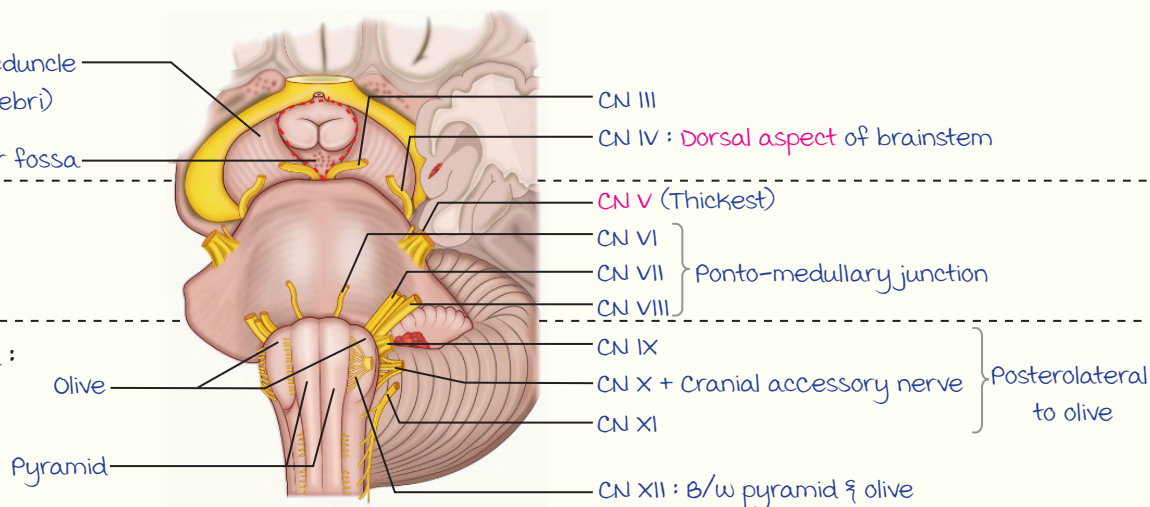
midbrain :

Cerebral peduncle
(Crus cerebri)

Interpeduncular fossa

Pons :

medulla oblongata :

**Interpeduncular Fossa :**

Boundaries :

Optic chiasma

Optic tract

Cerebral peduncle

Contents :

Pituitary gland infundibulum & Tuber cinereum

mammillary bodies

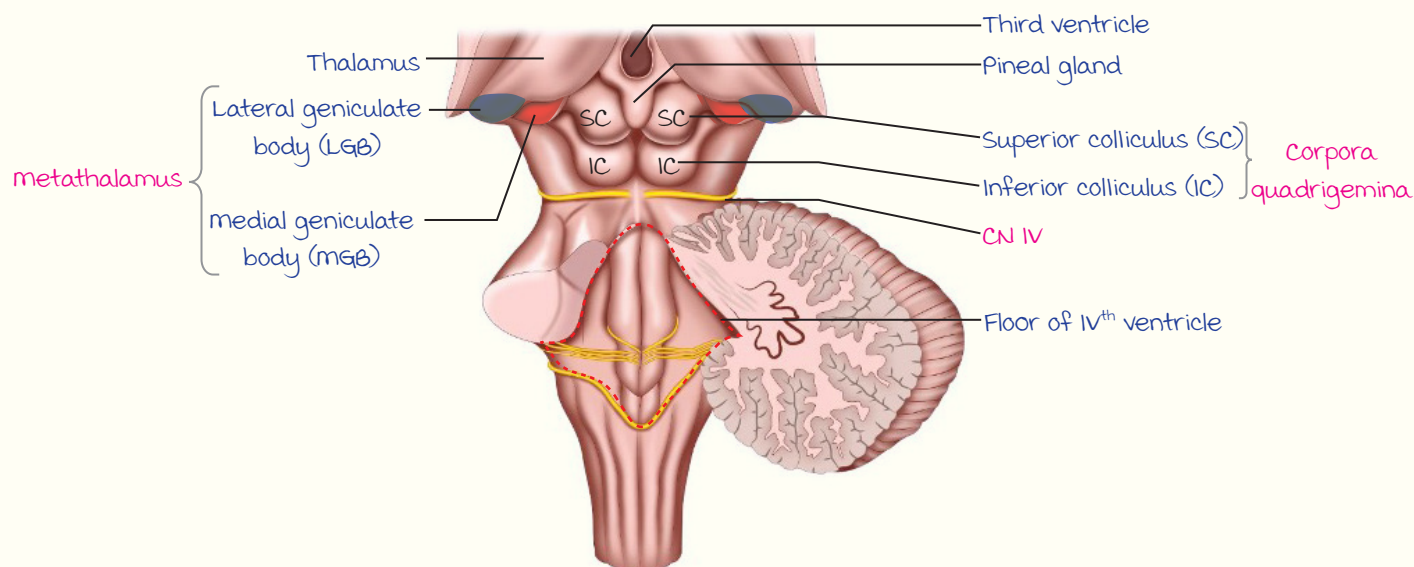
CN III

Posterior perforating substance (gray matter) :

Posterior cerebral artery (PCA) branches perforate

Dorsal Aspect of Brainstem

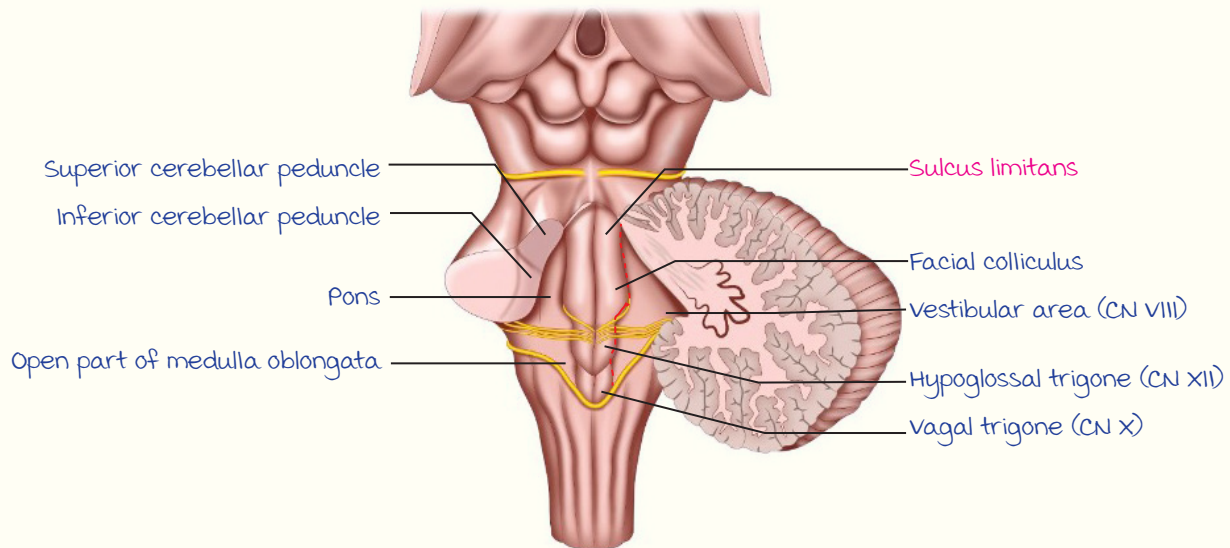
00:44:45



- CN III → Level of SC
- CN IV → Level of IC

Floor of IVth ventricle/Rhomboid Fossa :

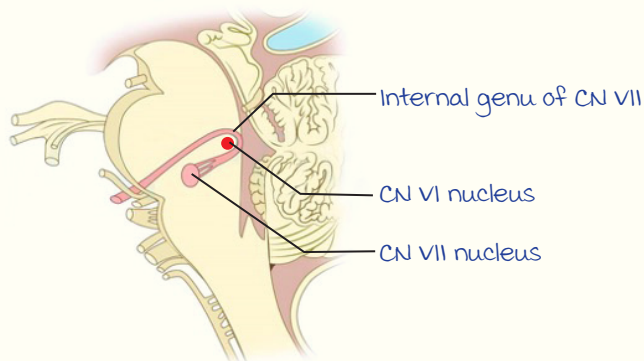
----- Active space -----



Facial colliculus  :

1st genu of **facial nerve** (CN VII) around **abducens nucleus** (CN VI).

Facial colliculus injury
↓
Affects CN VII > CN VI nucleus
↓
LMN lesion of **CN VII**
↓
I/L facial paralysis.



1st genu of CN VII

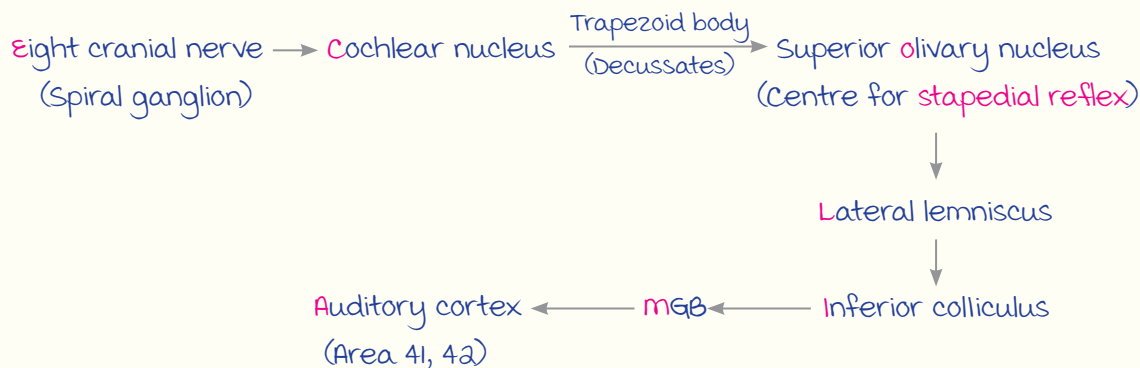
Lemnisci

00:53:27

Lateral Lemniscus :

Auditory pathway :

Mnemonic : **E COLI MA**



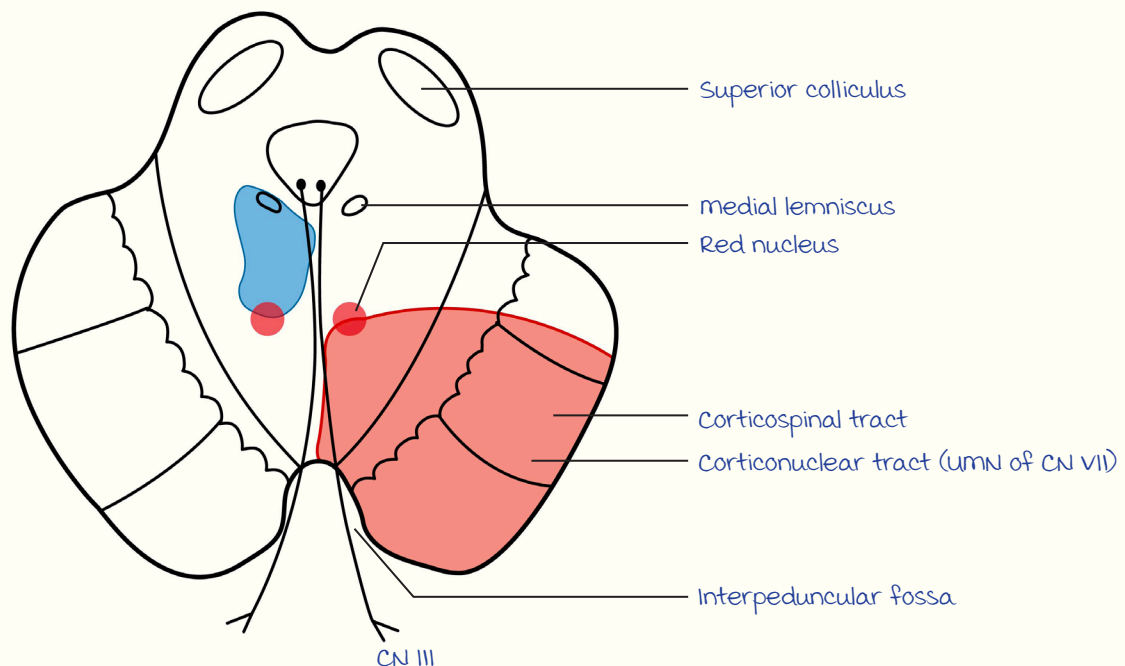
----- Active space -----

Injury :

Lemniscus	Clinical features
medial lemniscus (Anterior spinothalamic tract + Dorsal column tract)	C/L loss of : • Touch • Pressure • Vibration • Stereognosis • Conscious proprioception } Below face
Lateral lemniscus (Auditory pathway)	C/L hearing loss
Spinal lemniscus (Lateral spinothalamic tract)	C/L loss of pain & temperature (Below face)
Trigeminal lemniscus	C/L loss of facial sensations

Brainstem Syndromes

01:04:35

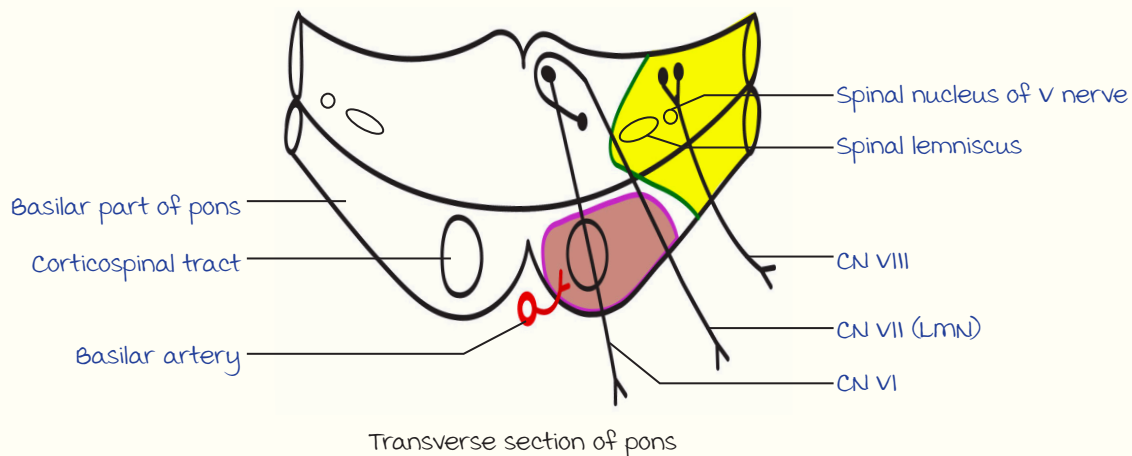
midbrain Syndromes :

Section of midbrain at superior colliculus

----- Active space -----

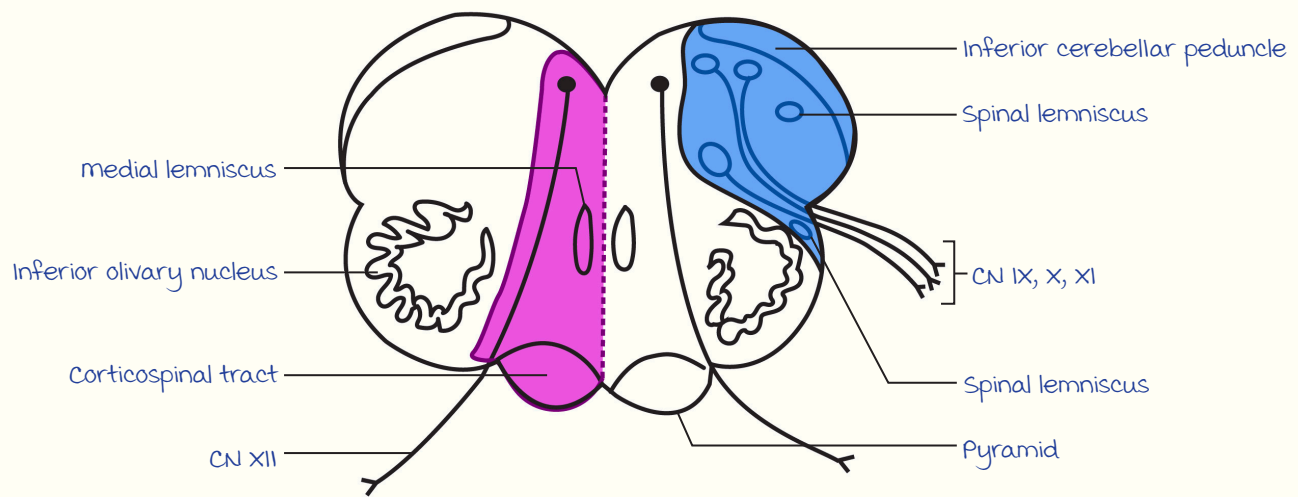
Syndromes	Clinical features
Benedict's syndrome	• I/L CN III palsy : - Ptosis - Divergent squint - Loss of light & accommodation reflex • C/L loss of : - Touch - Vibration - Conscious proprioception - Stereognosis - Pressure
Weber's syndrome	• I/L CN III palsy • C/L hemiplegia • C/L paralysis of lower half of face

Pontine Syndromes :



Syndromes	Clinical features
Millard Gubler syndrome (Basilar artery affected)	• I/L facial paralysis • I/L convergent squint • C/L hemiplegia
Lateral pontine syndrome (D/t acoustic neuroma)	• I/L facial paralysis • I/L progressive deafness • C/L loss of pain & temperature (Below face) • I/L loss of pain & temperature from face

----- Active space ----- medullary Syndromes :



Transverse section of medulla oblongata

Syndromes	Clinical features
medial medullary syndrome	• I/L tongue paralysis • C/L hemiplegia • C/L loss of : - Touch - Pressure - Vibration - Stereognosis - Conscious proprioception
Lateral medullary/ Wallenberg syndrome (Posterior inferior cerebellar artery affected)	• I/L bulbar paralysis (Dysphagia, dysarthria, dyspnea) • I/L loss of gag reflex • C/L loss of pain & temperature (Below face)

NEUROANATOMY : PART 2

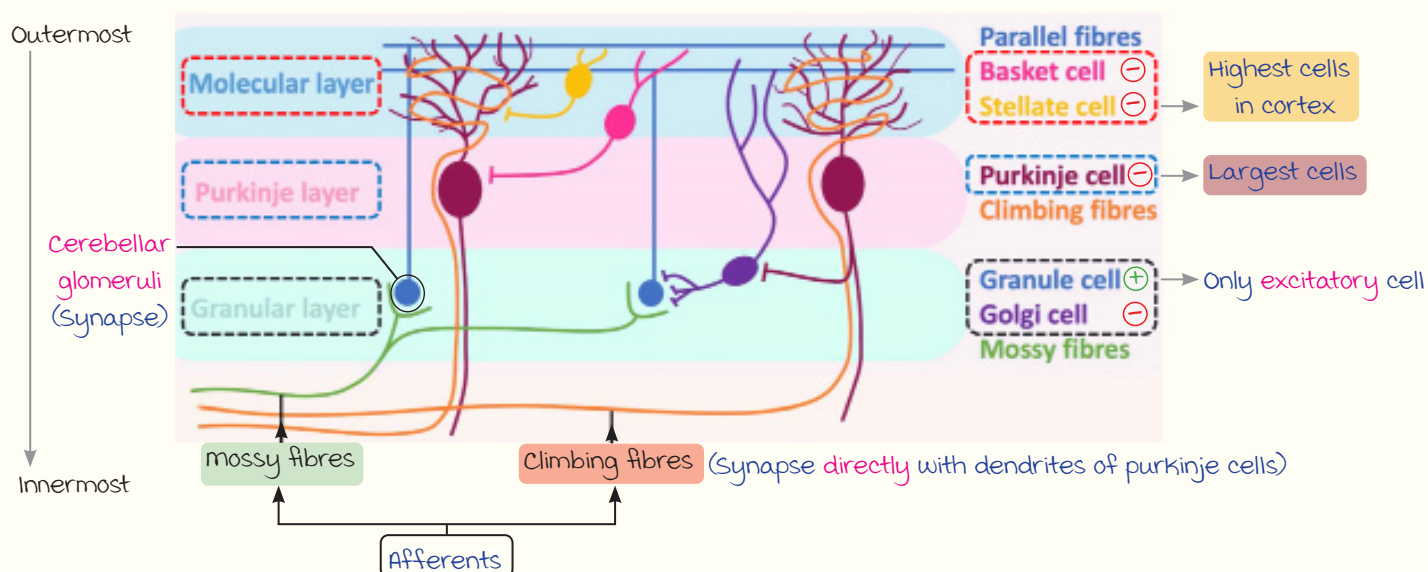
----- Active space -----

Cerebellar Cortex

00:01:12

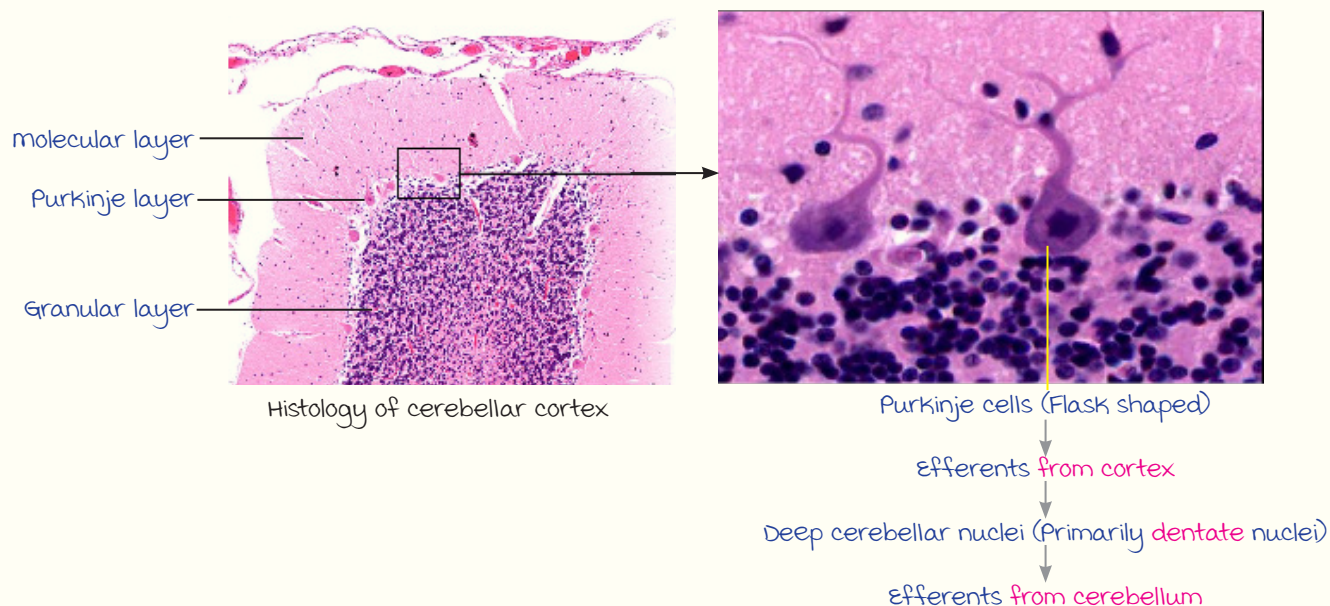
- Outer grey matter.
- Largest collection of $\ominus$ neurons in CNS.

Layers of Cerebellar Cortex :



- Olivocerebellar & Para-olivocerebellar tracts $\rightarrow$ Climbing fibres.
- Other cerebellar tracts $\rightarrow$ mossy fibres.

Purkinje cells : Only efferents from cerebellar cortex.



----- Active space -----

Cerebellar Peduncles

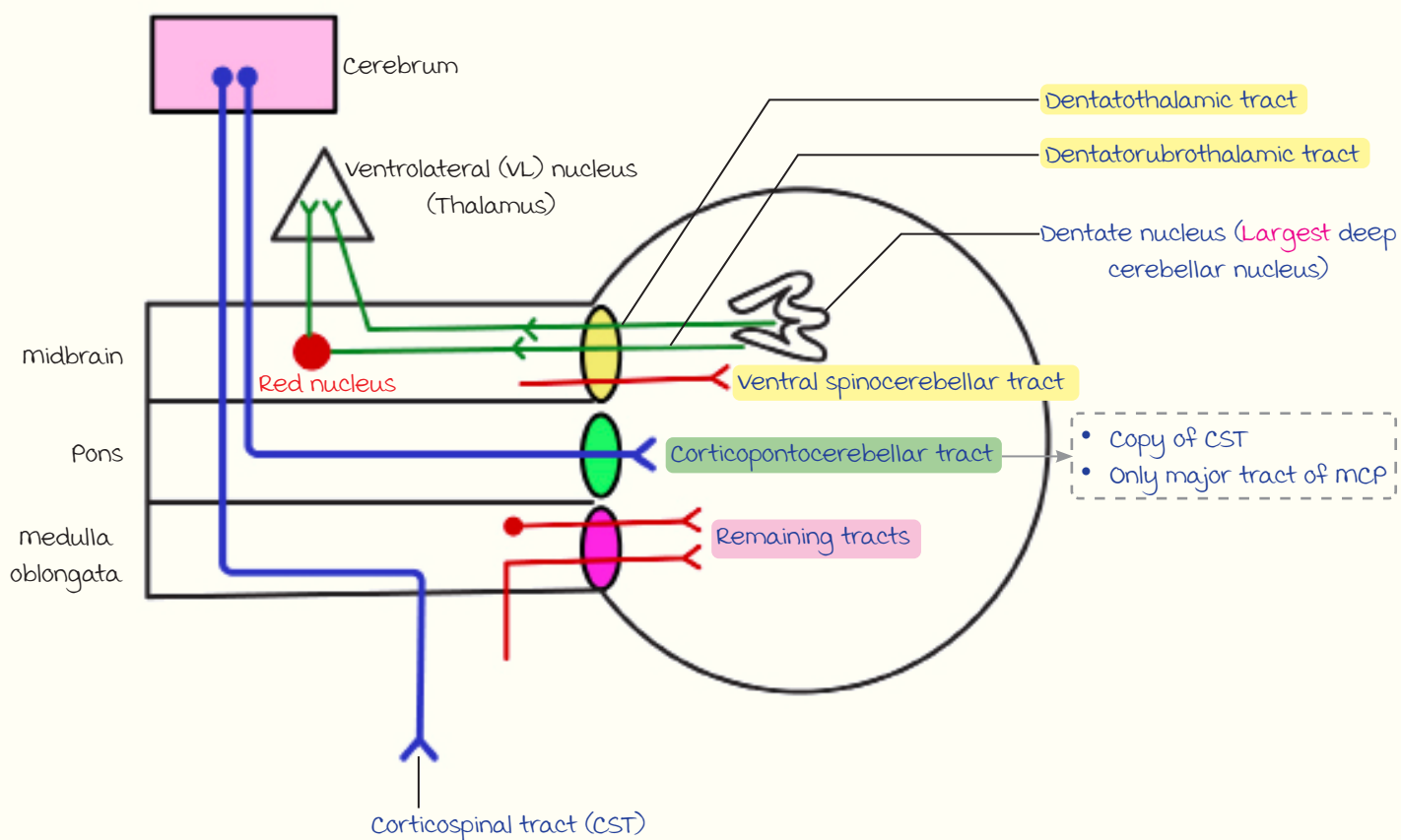
00:12:45

Superior cerebellar peduncle (SCP).

middle cerebellar peduncle (MCP):

- Largest peduncle.
- Purely afferent.

Inferior cerebellar peduncle (ICP).



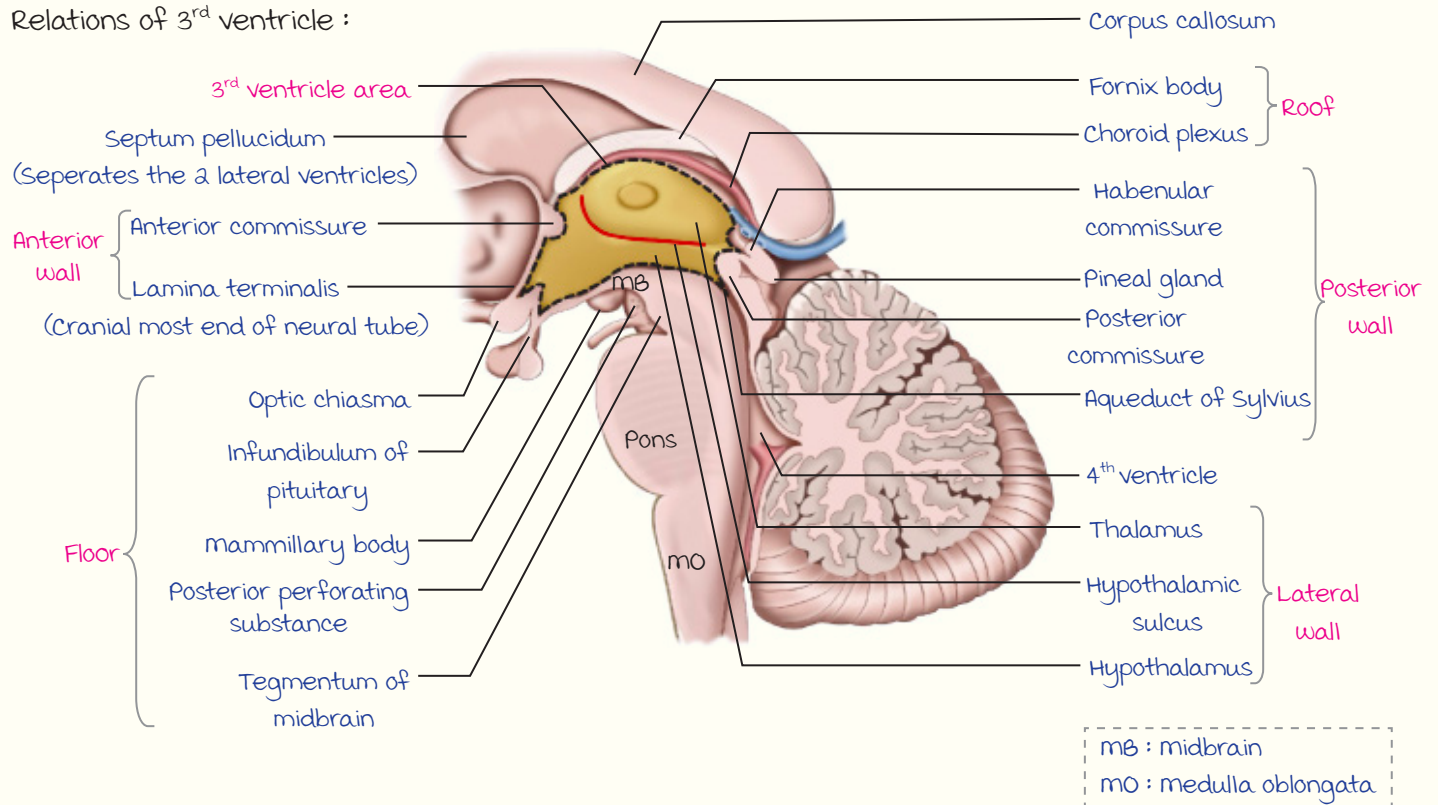
Sections of the Brain

00:18:24

----- Active space -----

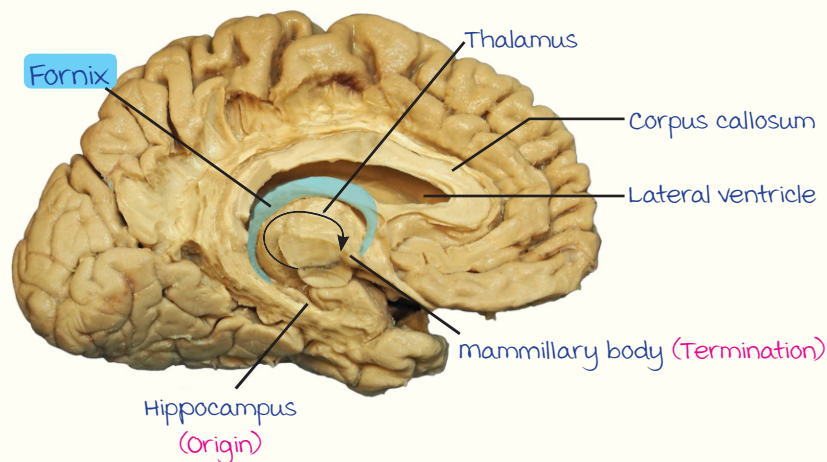
Sagittal Section :

Relations of 3rd ventricle :



Fornix :

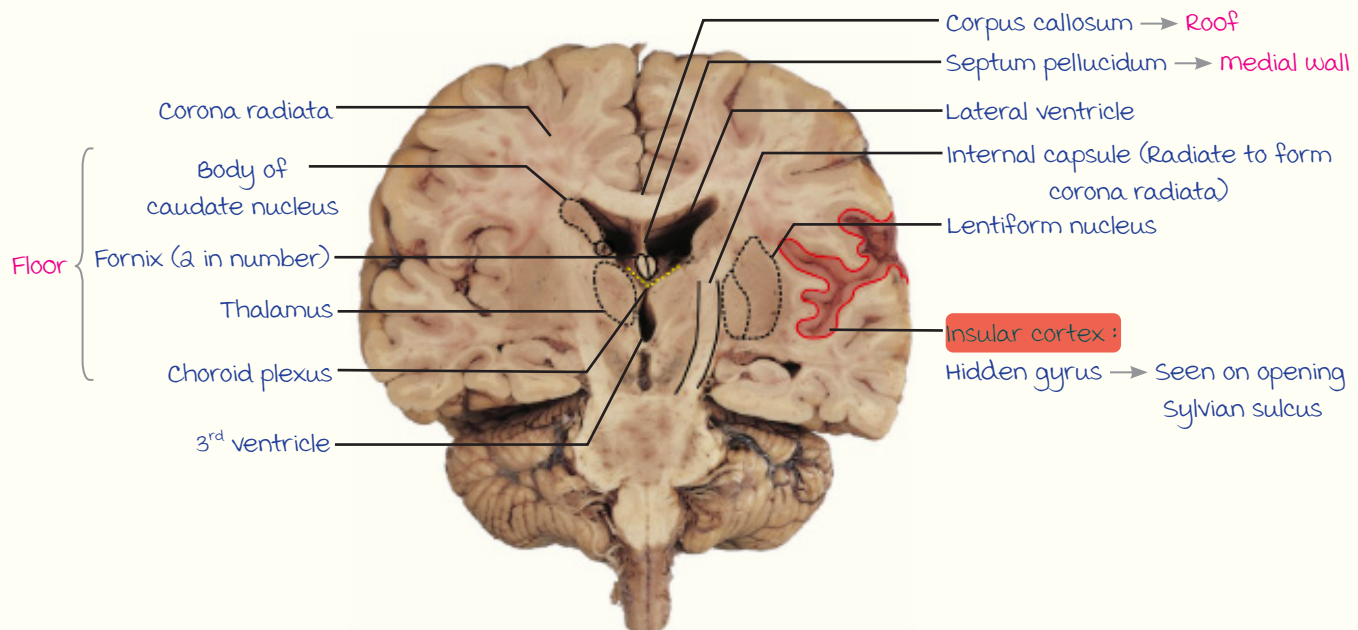
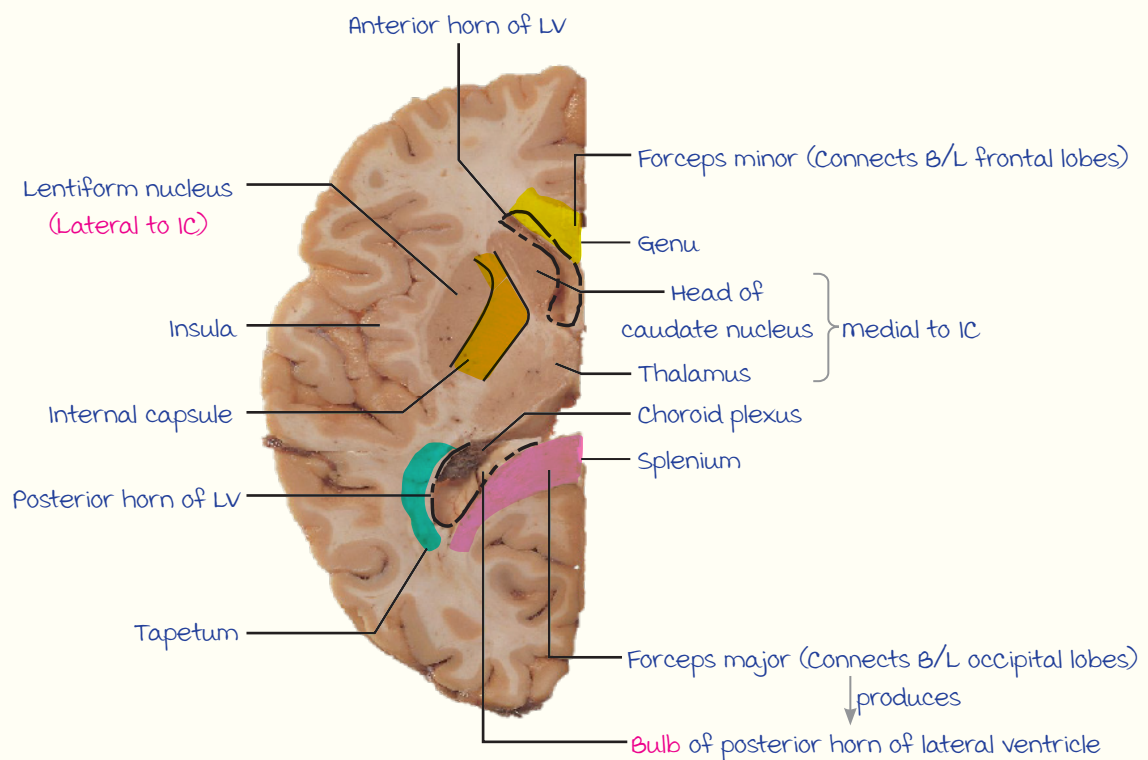
Association > Commissural > Projection fibres.



----- Active space -----

Coronal Section :

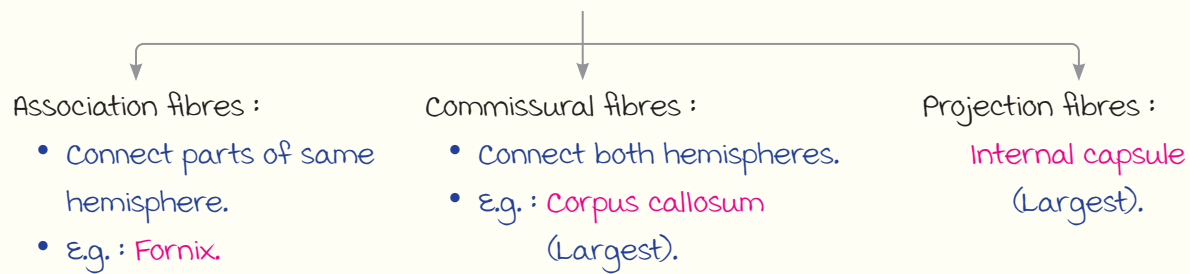
Relations of lateral ventricle :

Transverse Section :

White Matter of Cerebrum

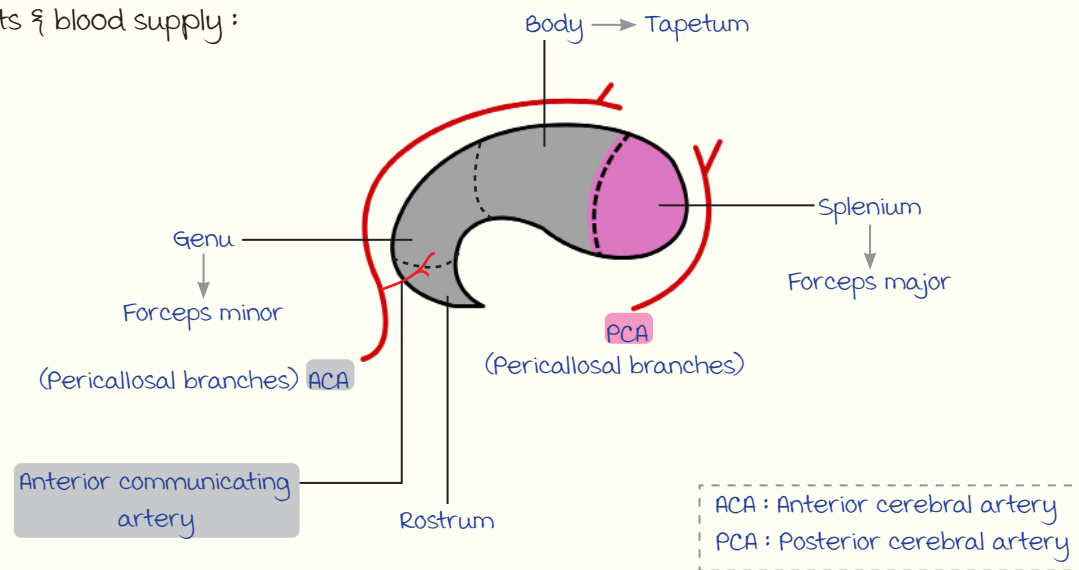
00:42:31

----- Active space -----



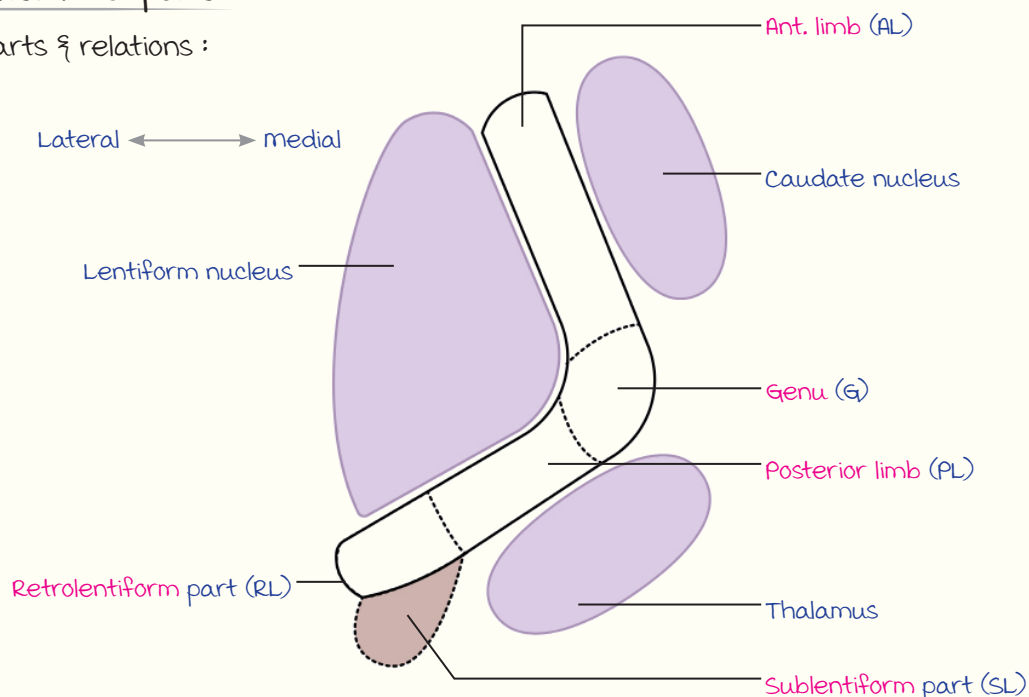
Corpus Callosum :

Parts & blood supply :



Internal Capsule :

Parts & relations :



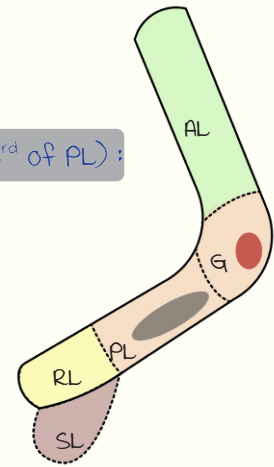
----- Active space ----- Tracts :

Ascending tracts :

- **Ant. thalamic radiation :**
Thalamus → Cerebrum.
- **Superior thalamic radiation :**
Thalamus → Sensory cortex (Area 3, 1, 2).
- **Post. thalamic radiation (Optic radiation) :**
LGB → Visual cortex.
- **Inferior thalamic radiation (Auditory radiation) :**
MGB → Auditory cortex.

Descending tracts :

- **Corticospinal tract :**
UMN of all cranial nerves.
- **Corticospinal tract (Ant. 2/3rd of PL) :**
Supplies limb muscles.

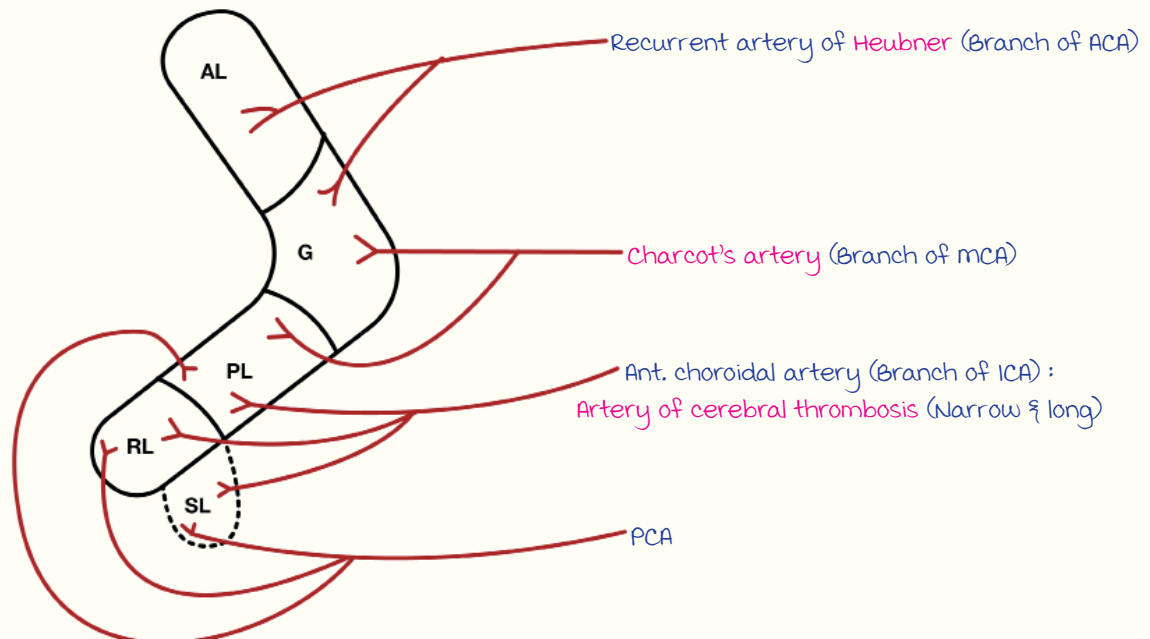


Tracts of IC

Applied anatomy :

Structures affected	Deficit
Genu	C/L facial paralysis
Ant. part of PL	C/L upper limb paralysis
Post. part of PL	C/L lower limb paralysis

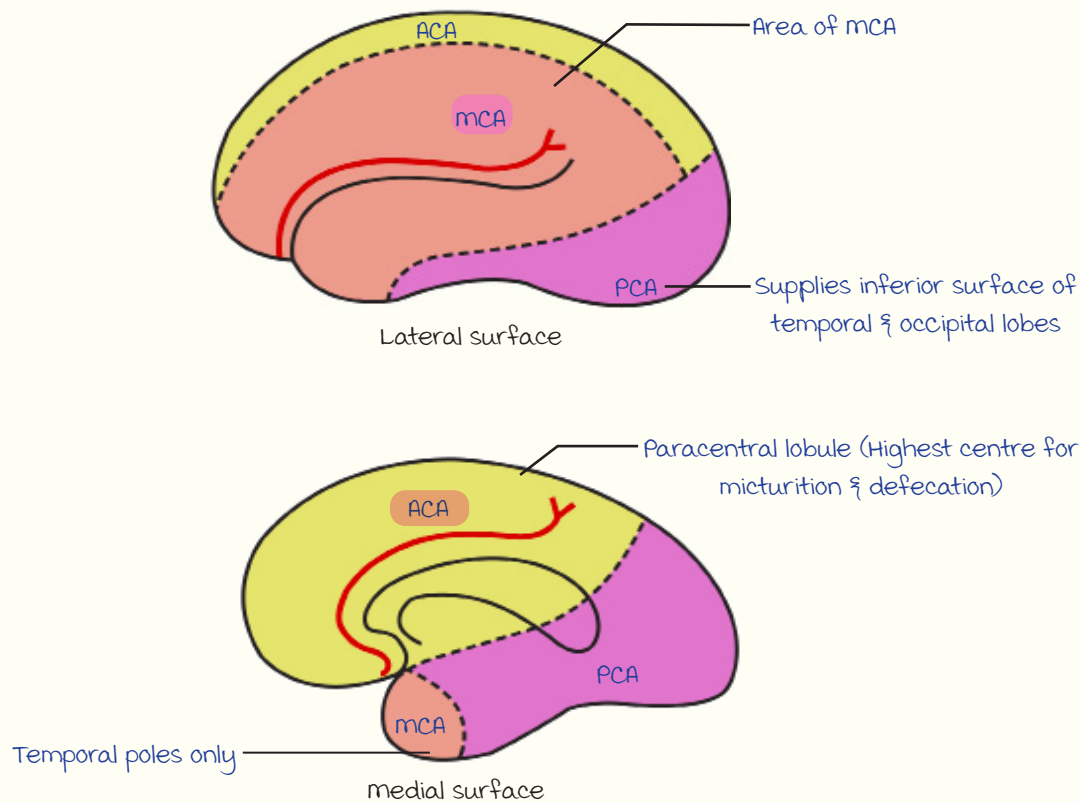
Blood supply :



Blood Supply

01:04:22

----- Active space -----



Applied Anatomy :

vessels affected	Cortical area	Deficit
PCA	Visual cortex	Hemianopia
mCA	- Broca's area (44, 45) - Wernicke's area (22, 39, 40)	Global aphasia
ACA	Paracentral lobule	urinary incontinence

Intracranial Hemorrhages

01:08:32

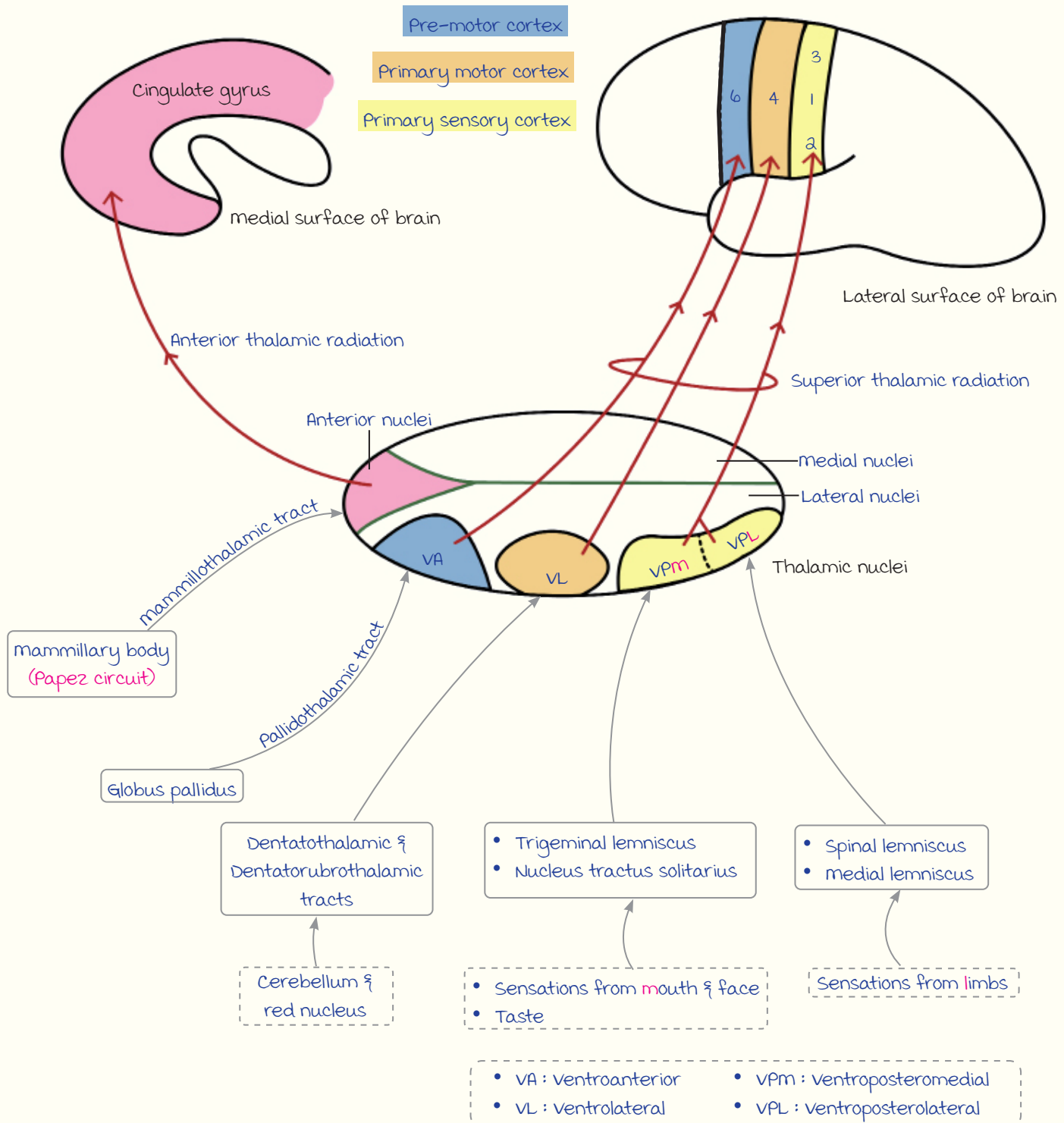
Hemorrhage	vessels involved
Extradural hemorrhage	Ant. division of middle meningeal artery
Subdural hemorrhage	Bridging veins
Subarachnoid hemorrhage	Berry aneurysms of : - Ant. communicating A : Close to ACA affects → CN II - Post. communicating A : Close to PCA affects → CN III
Intracerebral hemorrhage	Charcot's artery

----- Active space -----

Thalamic Connections



01:11:40



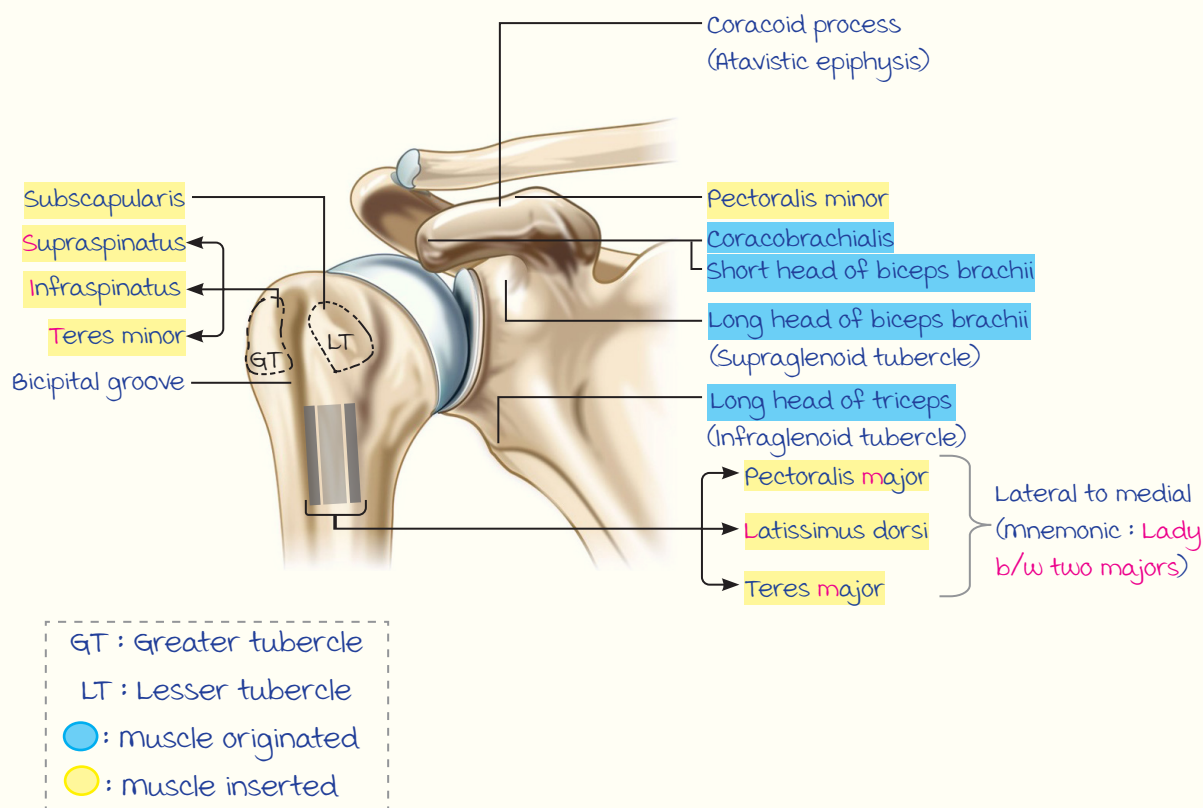
UPPER LIMB

----- Active space -----

Musculoskeletal Anatomy

00:01:22

Scapulohumeral muscles :

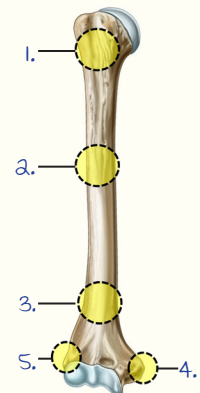


Actions :

muscles	Actions
- Subscapularis - Pectoralis major - Latissimus dorsi - Teres major	Adduction & internal rotation
Supraspinatus	0 - 15° abduction
- Infraspinatus - Teres minor	Adduction & external rotation

----- Active space ----- **Humeral Fractures & Nerve Injuries** :

	Fracture site	Nerve injured
1.	Surgical neck	Axillary nerve
2.	mid shaft (Spiral groove)	Radial nerve
3.	Supracondylar	Anterior interosseus nerve (median nerve)
4.	medial epicondyle	ulnar nerve
5.	Lateral epicondyle	Tardy ulnar nerve palsy (D/t cubitus valgus deformity)



Anterior aspect of humerus

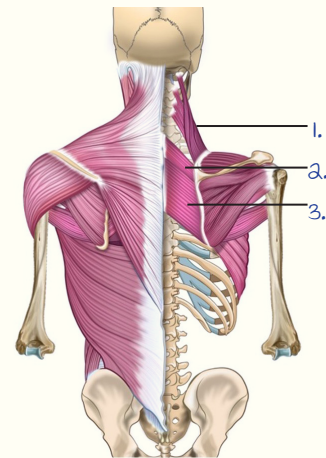
vertebroscapular muscles :

Origin : vertebral column.

Insertion : medial border of scapula.

N. supply : Dorsal scapular N. (C5).

	muscles	Actions
1.	Levator scapulae	Elevates scapula
2.	Rhomboid minor	- Retraction of scapula - Downward rotation of glenoid fossa
3.	Rhomboid major	



vertebroscapular muscles

Clavipectoral Fascia :

Extent : vertically downwards from clavicle to axilla.

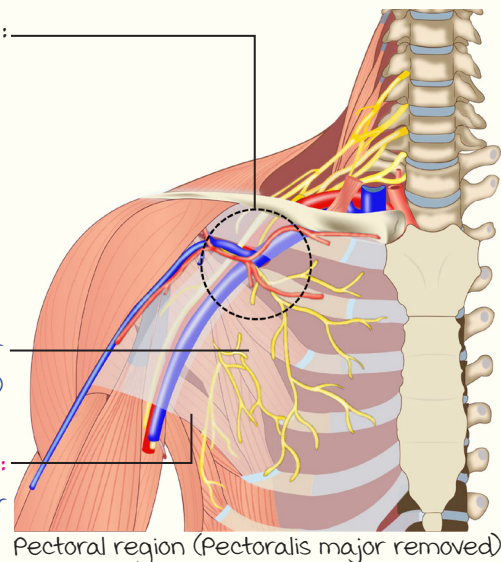
muscles enclosed : Pectoralis minor & subclavius.

Structures piercing the fascia :

- Cephalic V.
- Thoracoacromial A.
- Lateral pectoral N.
- Lymphatics

Pectoralis minor
(Pierced by medial pectoral N.)

Suspensory ligament of axilla :
Extension of CP fascia beyond P. minor

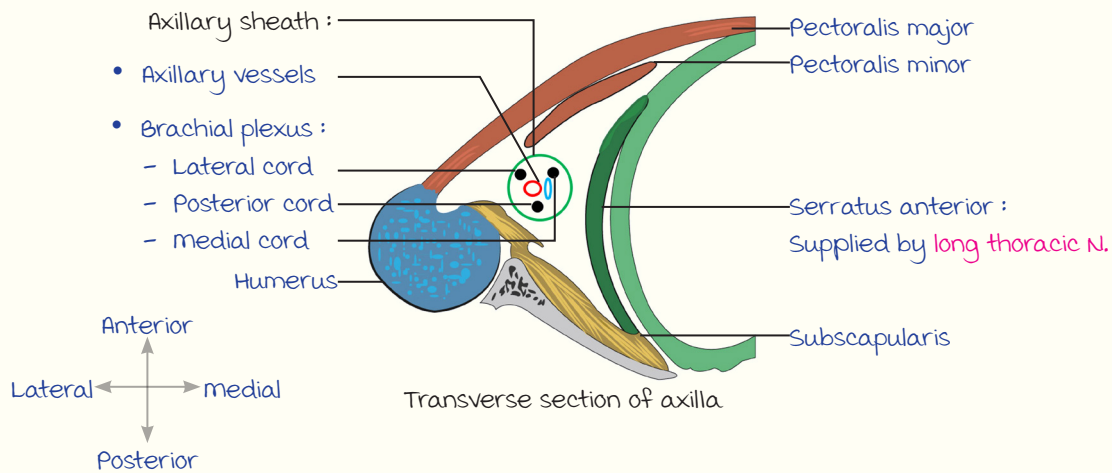


Note :

Suspensory ligament of Cooper : Breast ligament (Lies outside P. major).

Axilla

----- Active space -----



Muscles of the Hand

00:24:43

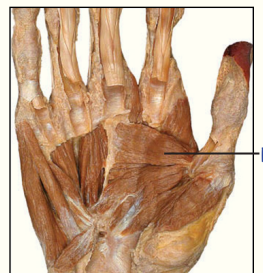
Total of 20 muscles.

- : Supplied by ulnar N.
- : Supplied by median N.

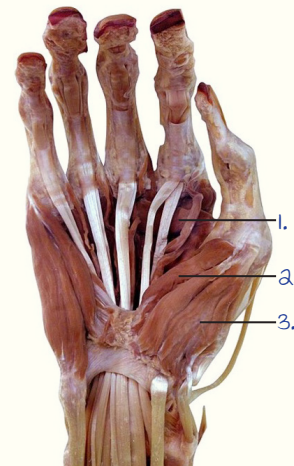
Thenar muscles :

1. Adductor pollicis (AdP) :
 - Origin : Shaft of 3rd metacarpal.
 - Graveyard of ulnar N.
2. Flexor pollicis brevis (FPB).
3. Abductor pollicis brevis (AbPB).
4. Opponens pollicis :
 - Deep to FPB & AbPB.
 - Insertion : Lateral aspect of the shaft of 1st mc.

Insertion : Proximal phalanx of thumb



Adductor pollicis

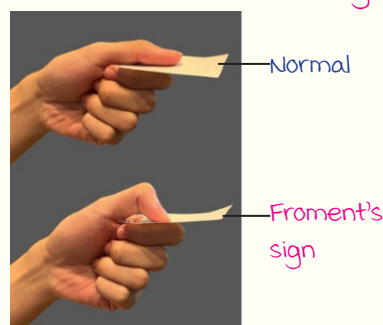


Thenar muscles

Applied aspect

Book test :

Ulnar nerve injury → Action of AdP ⊖ → Weakness compensated by flexion of thumb (Froment's sign).

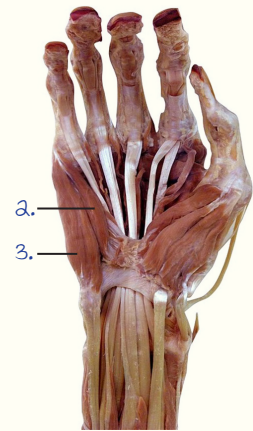


Book test

----- Active space -----

Hypothenar muscles :1. Palmaris brevis :

- Subcutaneous muscle.
- Nerve supply : **Superficial branch of ulnar N.**

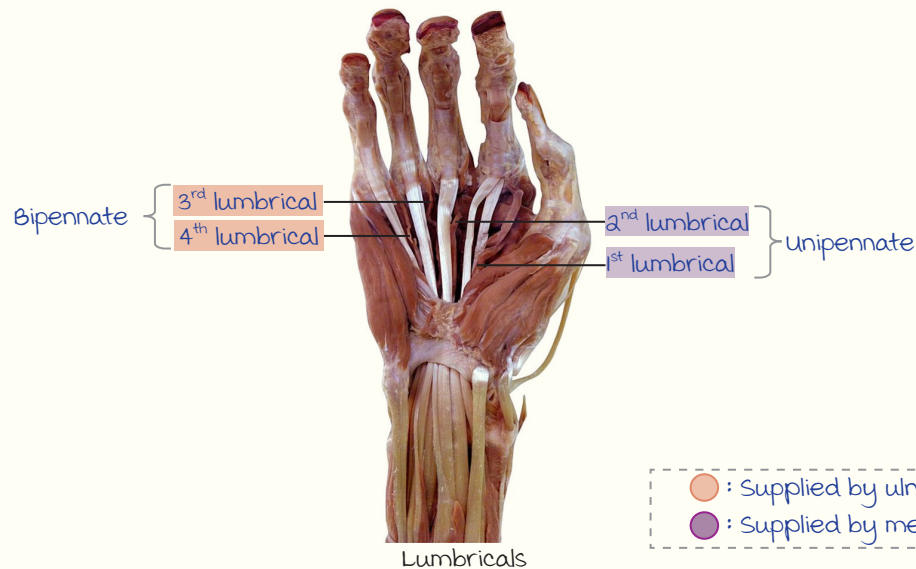
2. Flexor digiti minimi.3. Abductor digiti minimi.4. Opponens digiti minimi
(**Deep** muscle).

Hypothenar muscles

Lumbricals :Origin : **Flexor digitorum profundus (FDP)** tendon.Insertion : **Extensor digitorum** tendon (Dorsal digital expansion).

Action :

- **Flexion** at metacarpophalangeal (**MCP**) joints
 - **Extension** at interphalangeal (**IP**) joints
- } Common to interossei.



Lumbricals

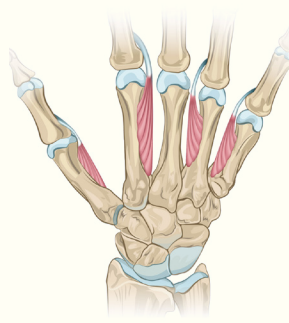
Interossei :

		Palmar interossei	Dorsal interossei
Pennation		Unipennate	Bipennate
Actions		Adduction (PAD)	Abduction (DAB)
Nerve supply		Ulnar nerve	
Relation to	Middle finger	Absent	2nd & 3rd dorsal interossei present
	Other fingers	may be absent on thumb	Absent on thumb & little finger
Tests		Card test	Egawa test (movement of middle finger)

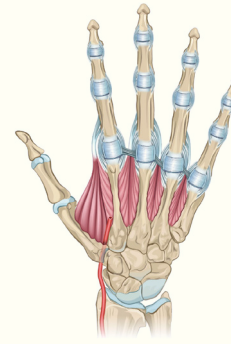
----- Active space -----



Card test



Palmar interossei



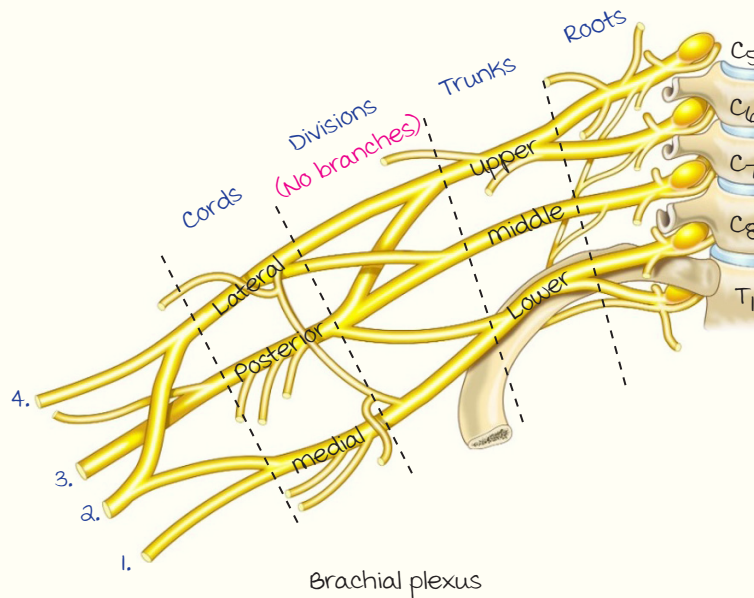
Dorsal interossei

Nerve Supply

00:45:35

Brachial Plexus  :

Ventral rami of C₅-T₁.



Continuation of cords :

1. ulnar nerve.
2. median nerve.
3. Radial nerve.
4. musculocutaneous nerve :

- muscles supplied :
 - Biceps brachii
 - Brachialis
 - Coracobrachialis
 } Anterior compartment of arm.
- Continues as lateral cutaneous nerve of forearm.

----- Active space ----- Brachial plexus injuries :

	Erb's palsy		Klumpke's palsy
Site of injury	upper trunk		Lower trunk
Root value	C ₅ , C ₆		C ₈ , T ₁
Nerves involved	Axillary N.	musculocutaneous N.	ulnar & median N.
muscles affected	Deltoid	Biceps brachii (Coracobrachialis spared : C ₇)	Intrinsic muscles of hand
Position of limb	Adducted shoulder	- Extended elbow - Pronated forearm	- Hyperextension at MCP joint - Flexion at IP joint
Characteristics	Policeman's tip hand		- Claw hand - Horner's syndrome (T₁)



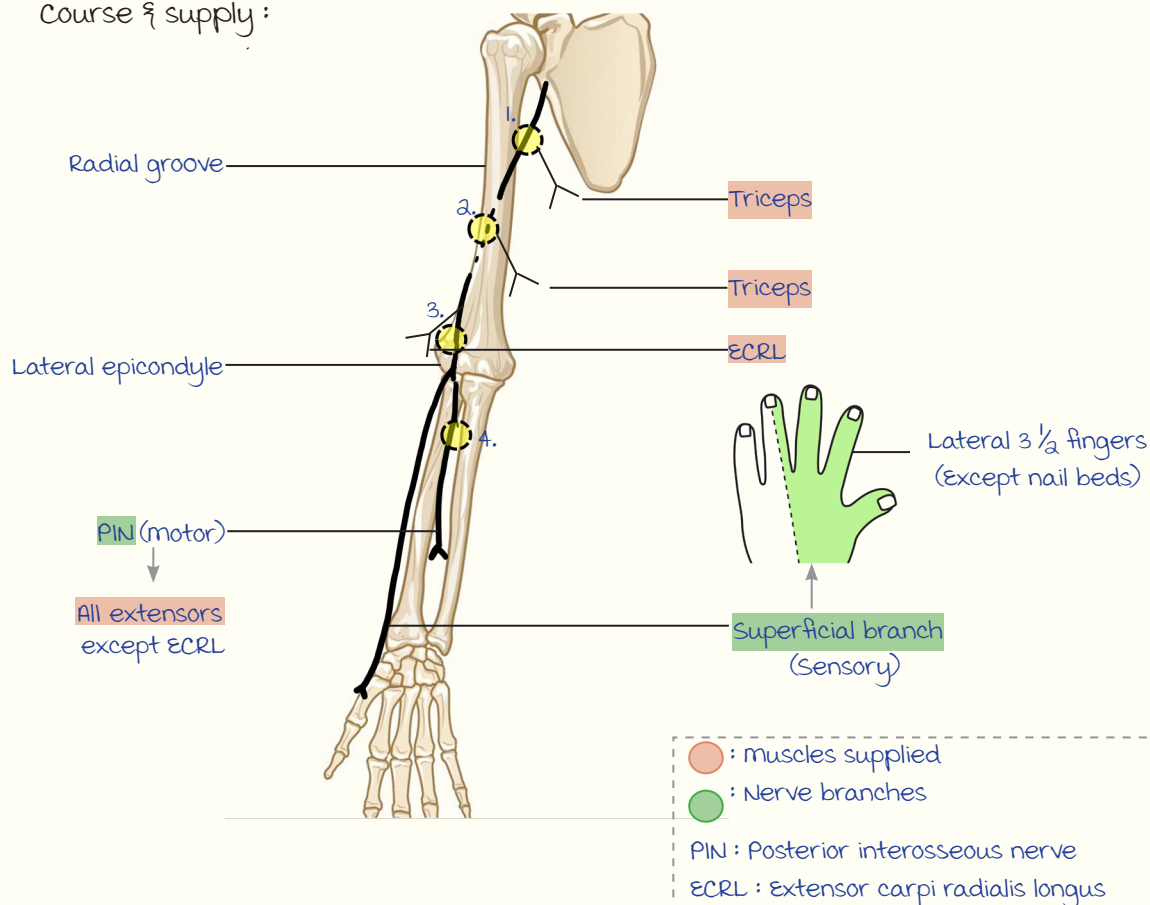
Policeman's tip hand



Claw hand

Radial Nerve (C₅-T₁) :

Course & supply :



Injuries  :

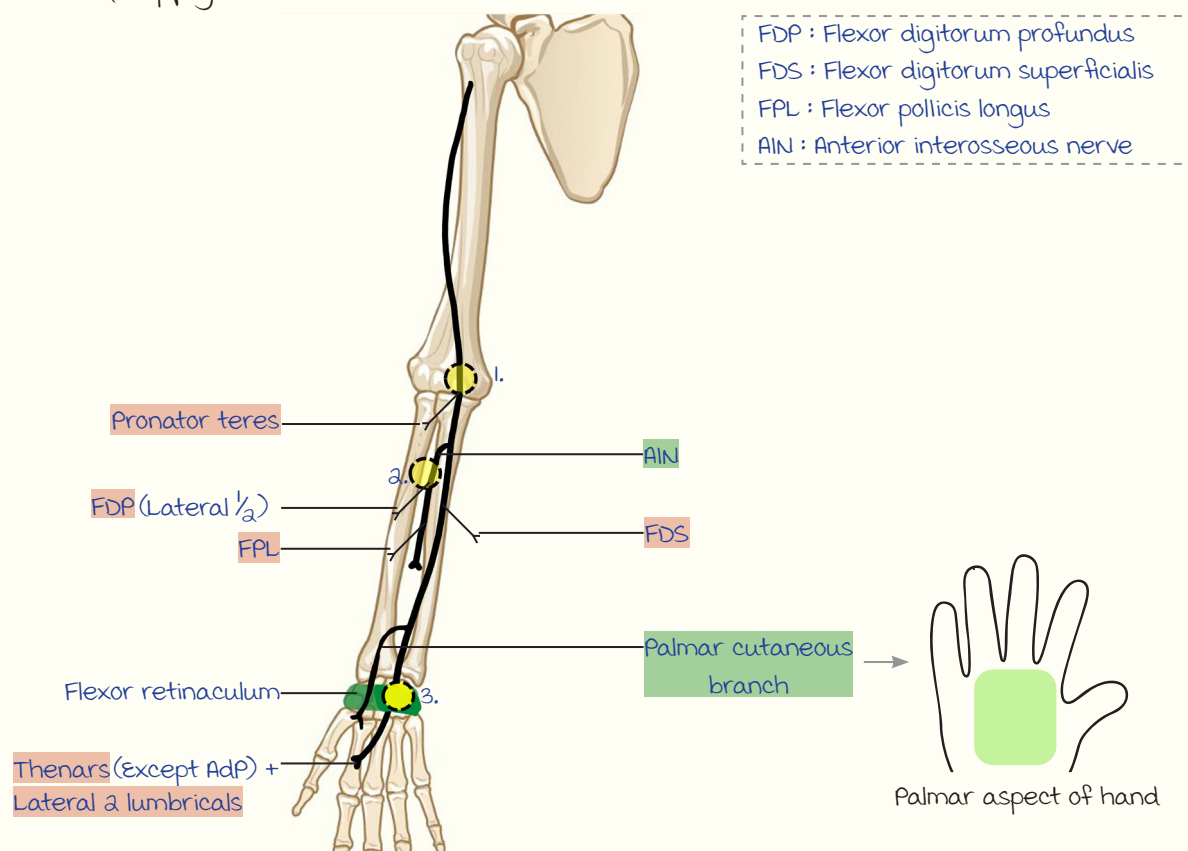
----- Active space -----

Site of injury		1. Axilla (Crutch palsy)	2. Spiral groove (Saturday night palsy)	3. Lateral epicondyle	4. Radial head (Monteggia fracture)
muscles affected	Triceps	⊕	Partial	⊖	⊖
	Wrist extensors	⊕	⊕	⊖	⊖
	Finger extensors	⊕	⊕	⊕	⊕
Features	Elbow flexion	⊕	Weak extension	⊖	⊖
	Wrist drop	⊕	⊕	⊖	⊖
	Finger drop	⊕	⊕	⊕	⊕
	Sensory loss	⊕	⊕	⊕	⊖

Median Nerve (C5-T1) :

AKA labourer's nerve.

Course & supply :



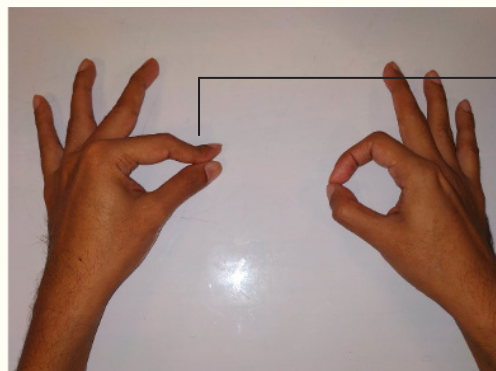
----- Active space -----

Injuries  :

Site of injury		1. High (Above elbow)	2. AIN	3. Low (Carpal tunnel syndrome)
muscles affected	Thenar muscles	⊕	⊖	⊕
	FDS	⊕	⊖	⊖
	FDP (Lateral 1/2)	⊕	⊕	⊖
	FPL	⊕	⊕	⊖
Features	Deformity	- Ape thumb - Hand of benediction/ pointing finger (while making fist)	Weak 'OK' sign	Ape thumb
	Sensory loss over palm	⊕	⊖	⊖



Hand of benediction



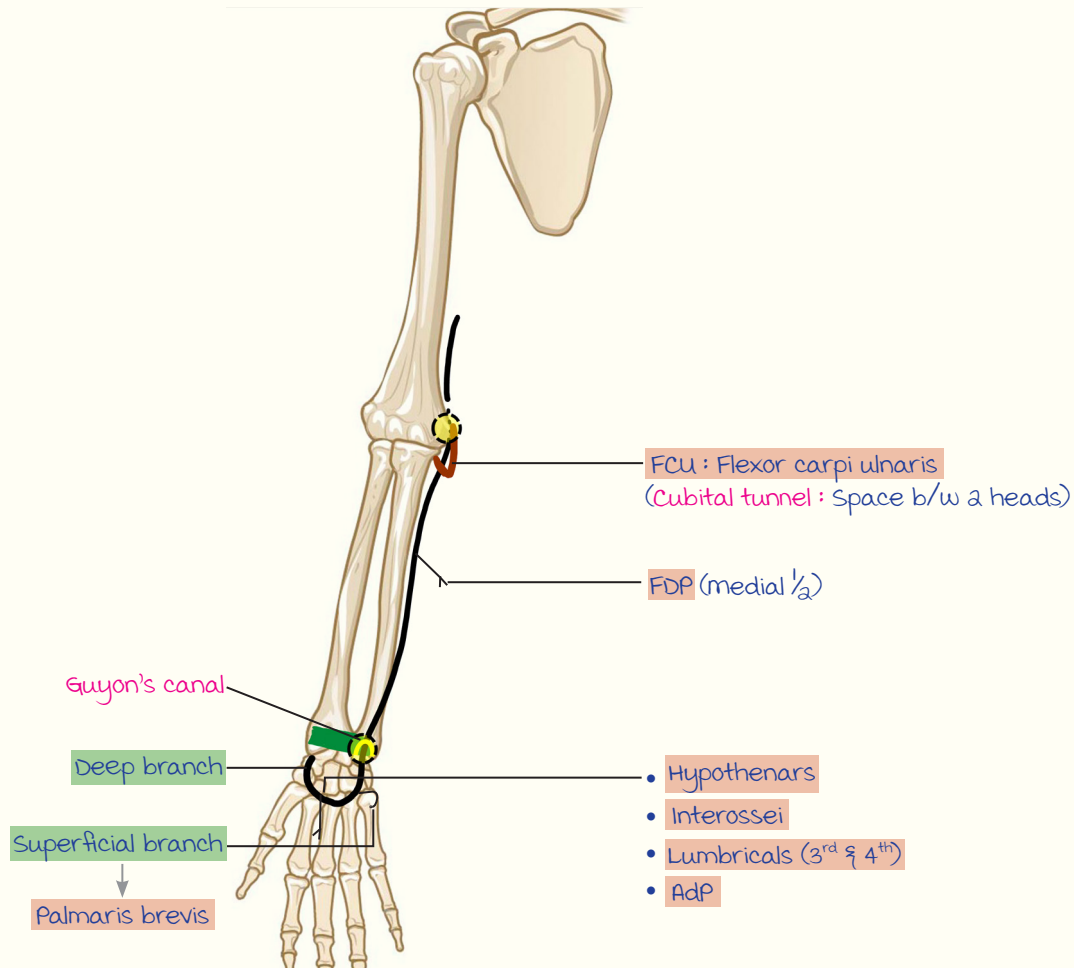
Weak OK sign (AIN injury)

Flexion at DIP joint
of thumb and index
finger affected

Ulnar Nerve (C₇-T₁) :

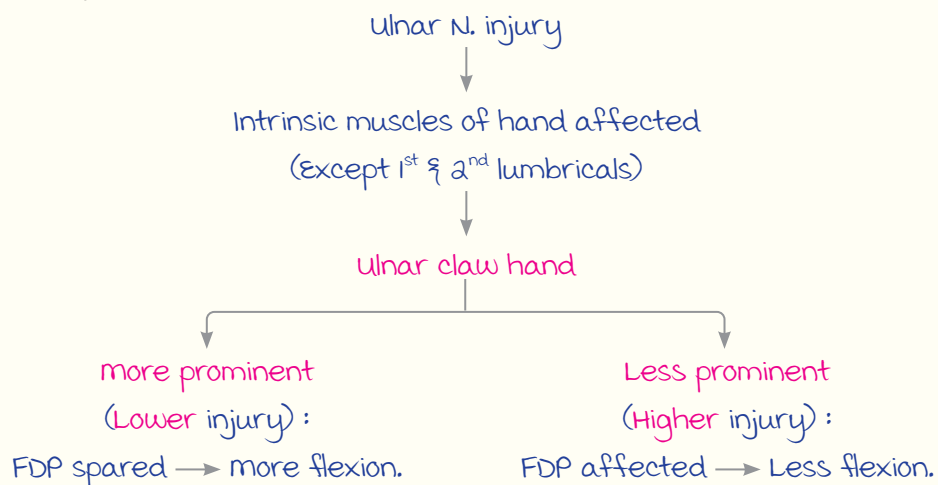
AKA musician's nerve.

Course & supply :



Note :

Cubital fossa injury : median N.

Ulnar N. injury & ulnar paradox .

LOWER LIMB

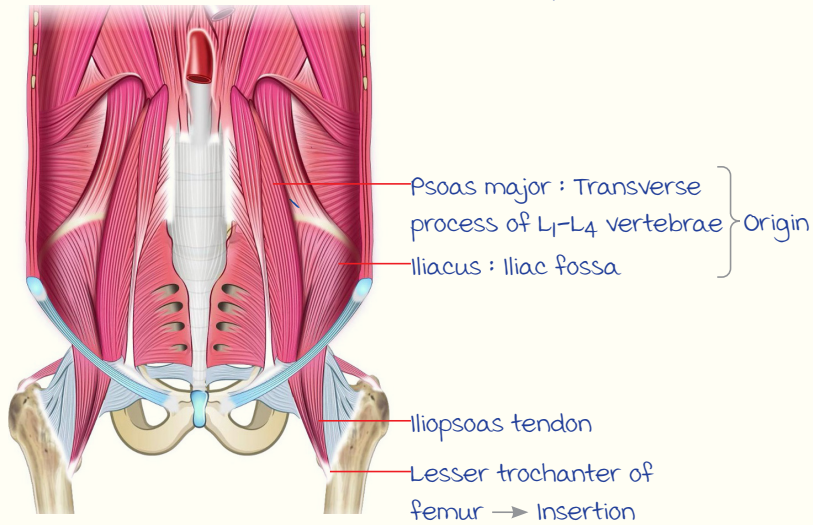
----- Active space -----

Lumbar Plexus

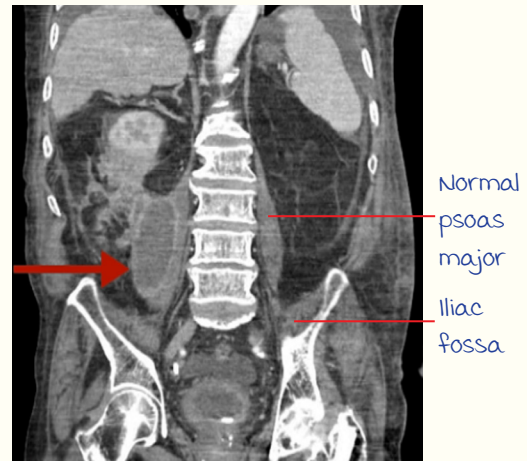
00:02:03

Iliopsoas muscle :

Action : Flexion at hip joint.

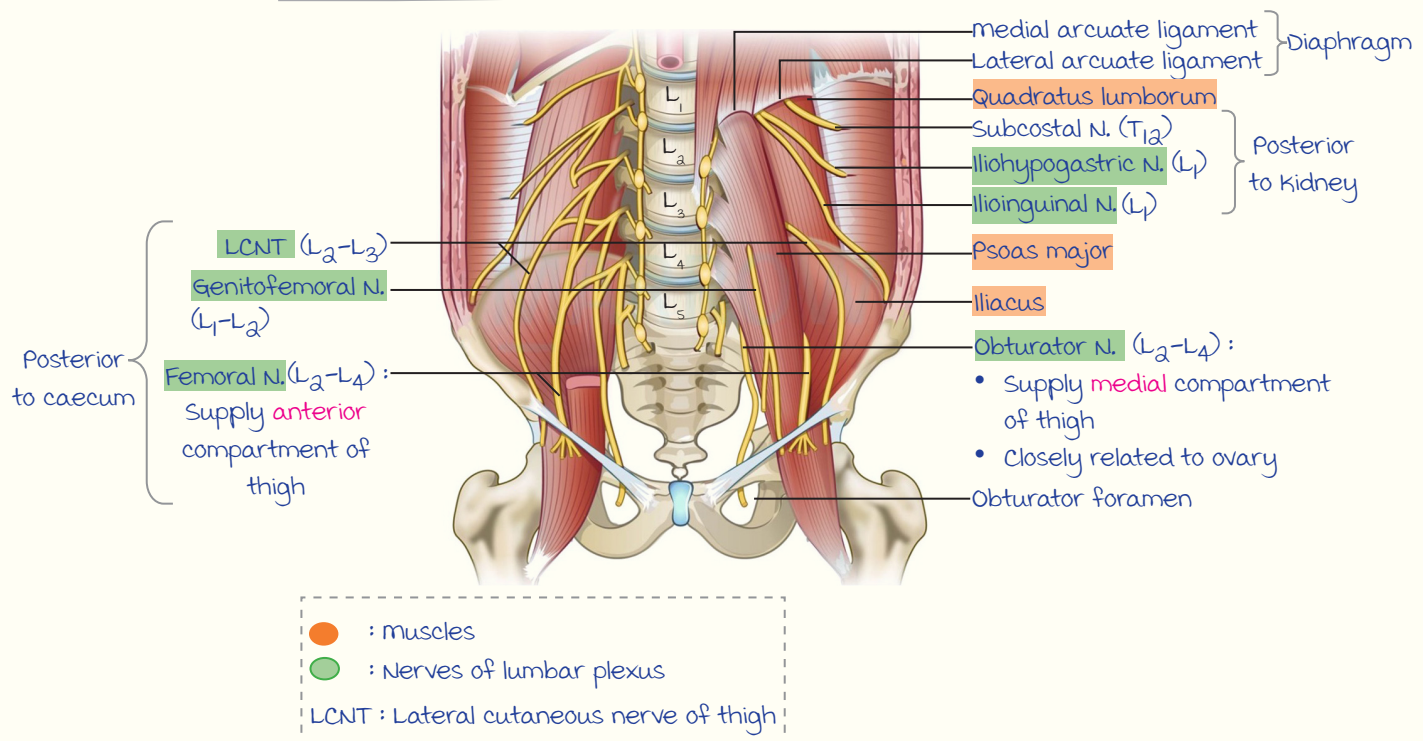


Psoas abscess :



Lumbar plexus (L₁-L₅) lies behind psoas major.

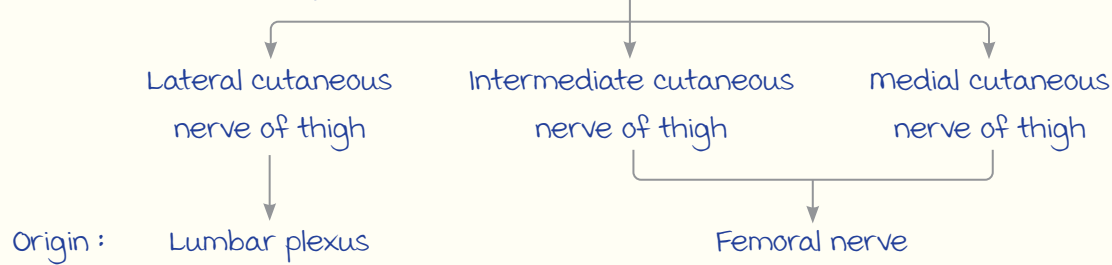
Lumbar Plexus :



Note :

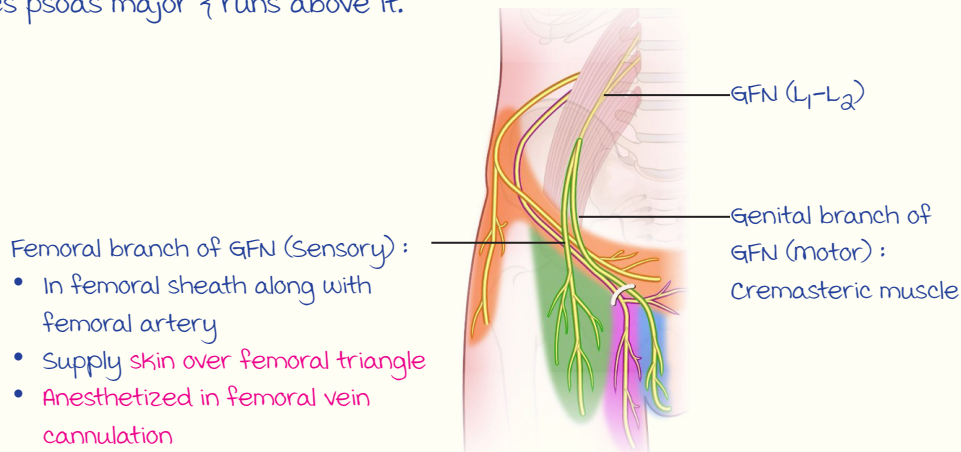
----- Active space -----

Cutaneous nerve supply to anterior compartment of thigh :



Genitofemoral Nerve (GFN) :

Pierces psoas major & runs above it.



Cremasteric reflex :

- Stimulus : Stroke along medial aspect of thigh.
- Afferent : Femoral branch of GFN.
- Efferent : Genital branch of GFN.
- Response : Contraction of cremasteric muscles.

Lateral Cutaneous Nerve of Thigh (LCNT) :

Course :

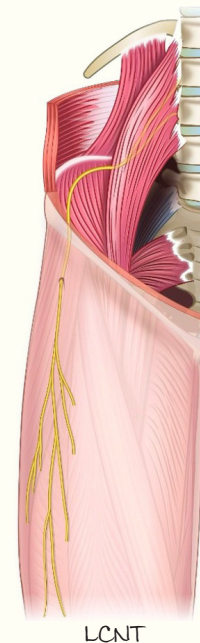
Posterior to Iliac fossa → Passes deep to/ pierces inguinal ligament → Lateral aspect of thigh.

Applied anatomy :

Compression deep to inguinal ligament

Burning/stabbing pain in lateral thigh area

meralgia paresthetica.



Note :

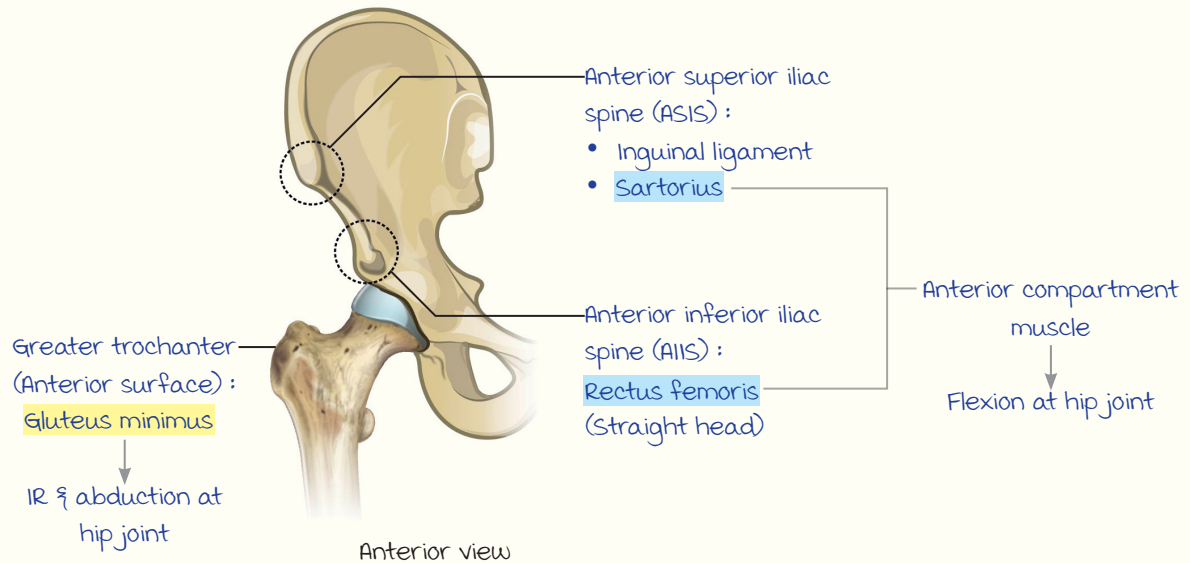
Saphenous N. (Branch of femoral N.) → Pain radiating to : medial leg/foot.

----- Active space -----

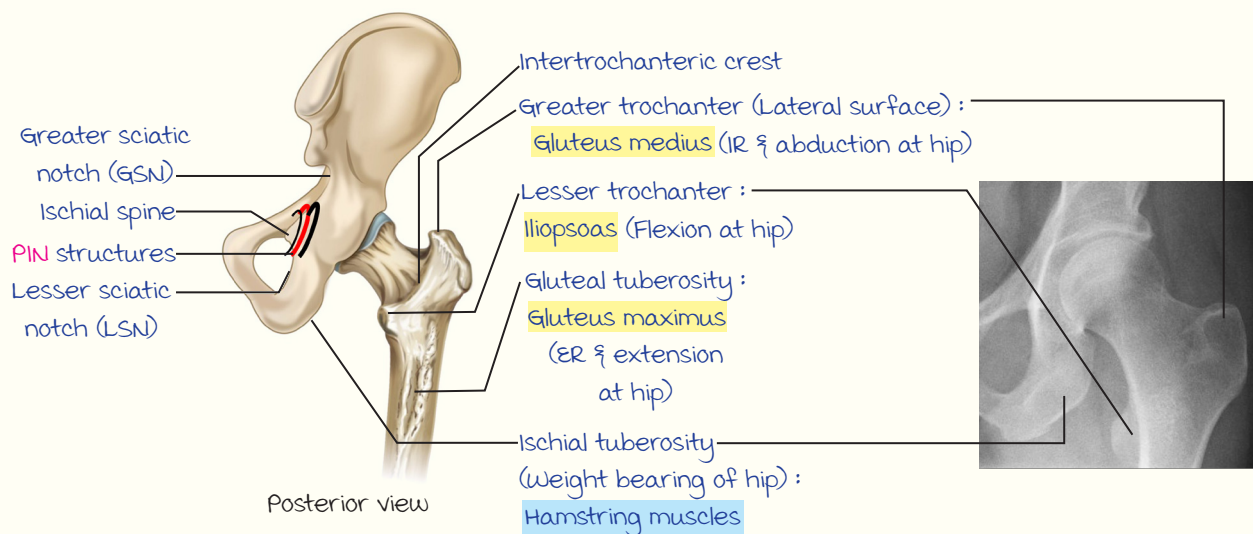
Hip Bone, Femur and Iliotibial Band

00:18:25

Hip Bone of Femur :



IR : Internal rotation
 ER : External rotation
 ● : muscle originated
 ● : muscle inserted

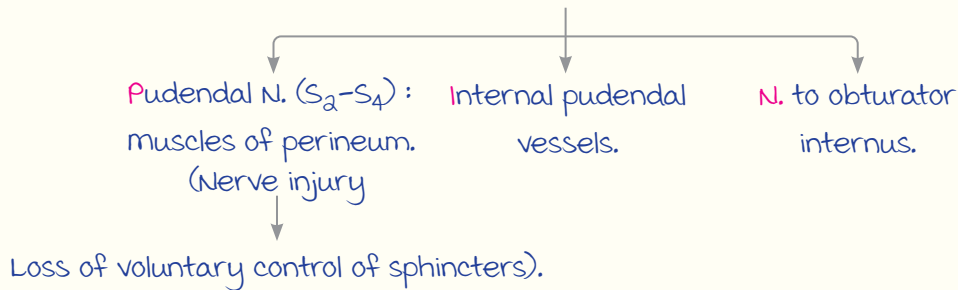


Gluteus maximus :

- Origin**
 - Posterior surface of sacrum.
 - Sacro-tuberous ligament.
- Insertion**
 - Gluteal tuberosity.
 - Iliotibial band.

Ischial spine  :

- At level of **coccygeal vertebrae**.
- Out from GSN into LSN (Pelvis → Perineum) :



----- Active space -----

Iliotibial Band (ITB) :

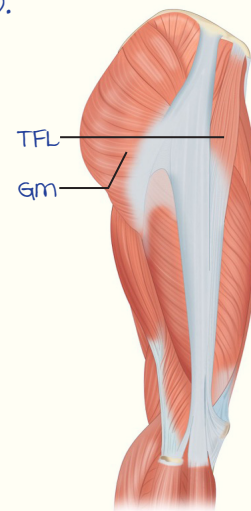
modification of fascia lata (Thickened deep fascia of thigh).

Origin : Iliac bone.

Insertion : Lateral tibial condyle.

Muscles inserted :

- Tensor fascia lata (TFL).
- Gluteus maximus (Gm).



Applied anatomy :

Iliotibial band contracture.

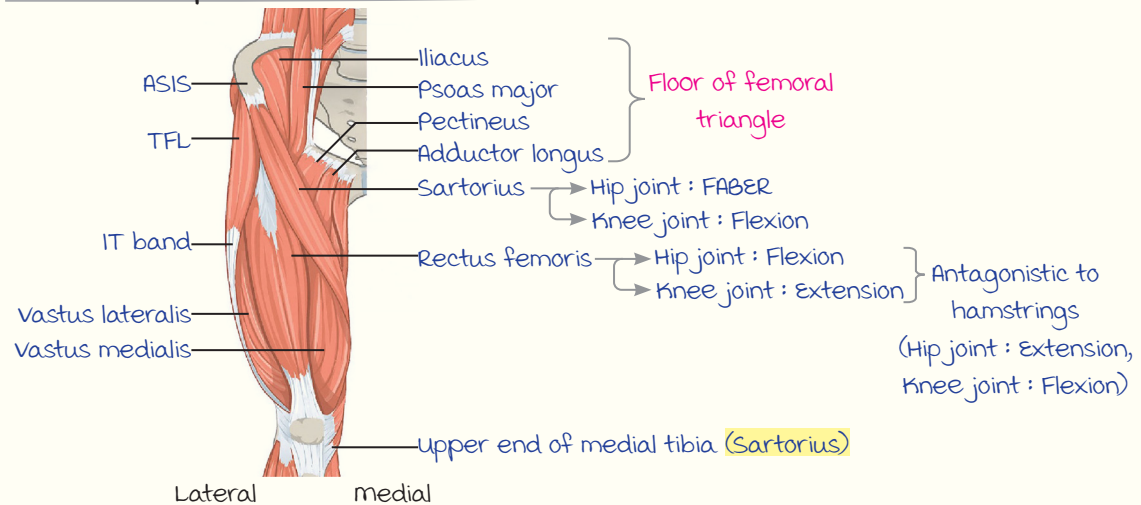


----- Active space -----

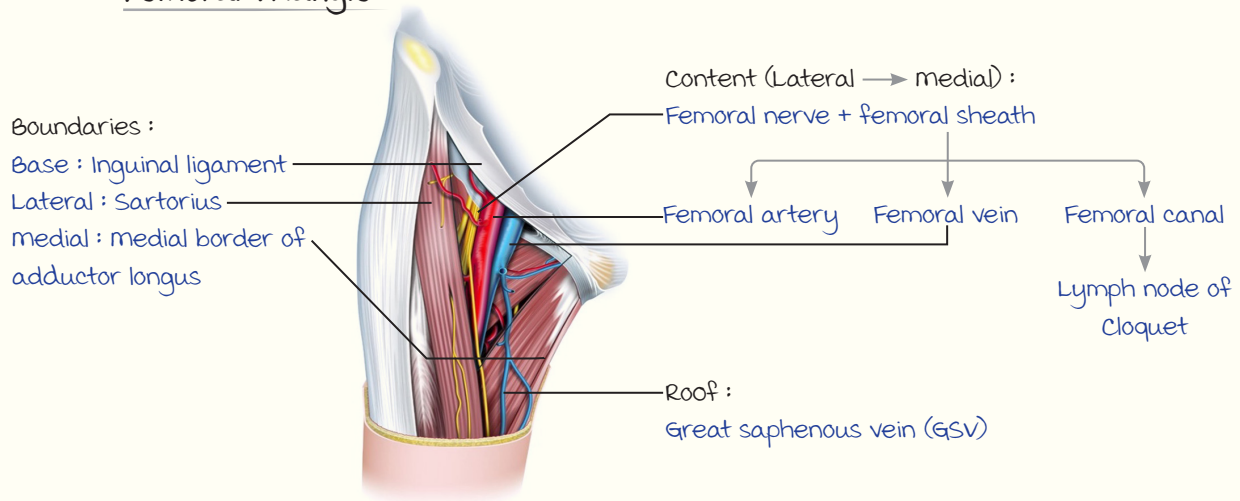
Anterior Compartment of Thigh

00:30:55

Anterior Compartment Muscles :



Femoral Triangle :



Gluteal Region and Saphenous Vein

00:37:48

Gluteal Region :

Inferior gluteal nerve : Gluteus maximus.

Superior gluteal nerve (SGN) :

- muscles → Gluteus medius.
- Gluteus minimus.

- Action :

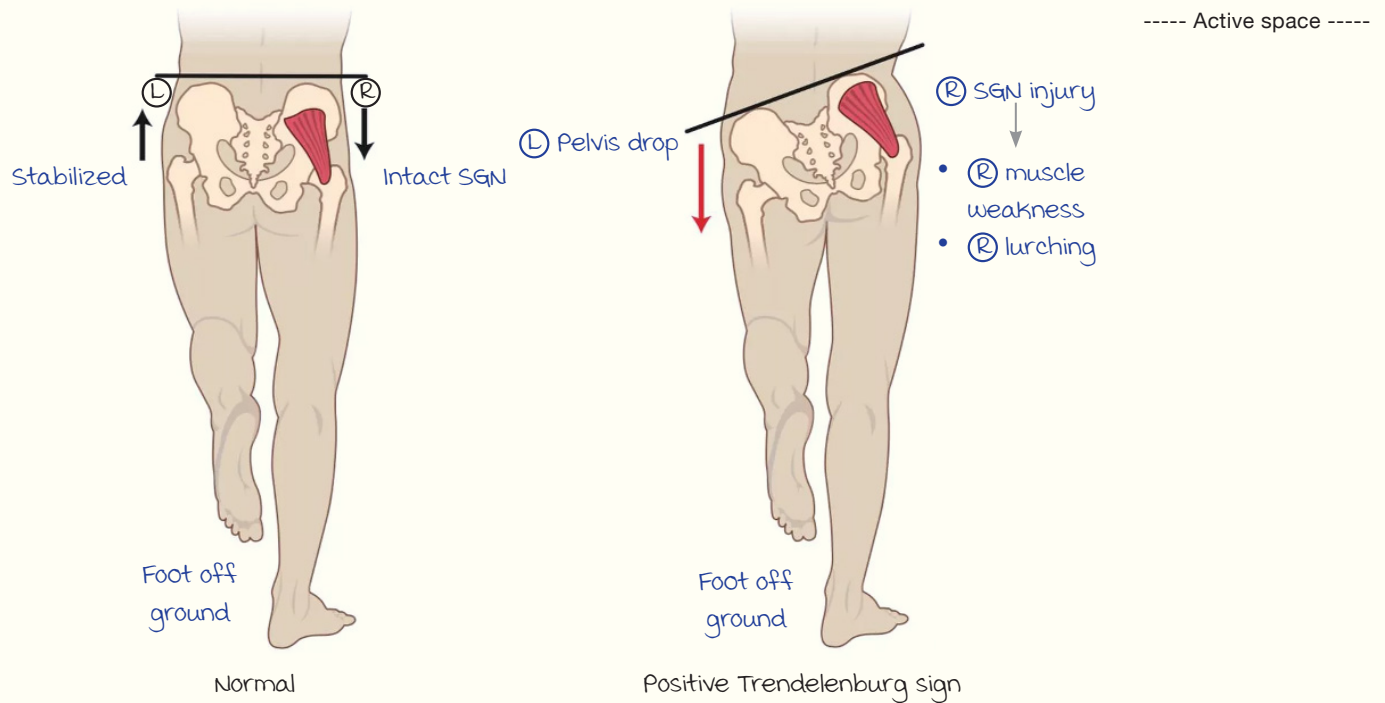
Abduction & internal rotation of hip (On same side)

↓ C/L foot off ground

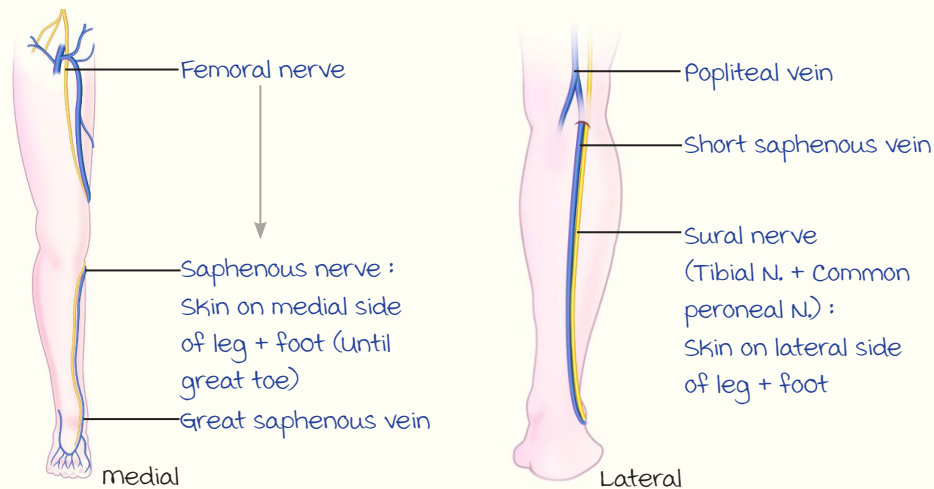
Stabilization of pelvis (Opposite side).

- Injury :

- I/L muscle weakness.
- I/L lurching.
- C/L pelvic drop.



Saphenous vein :



Applied anatomy :

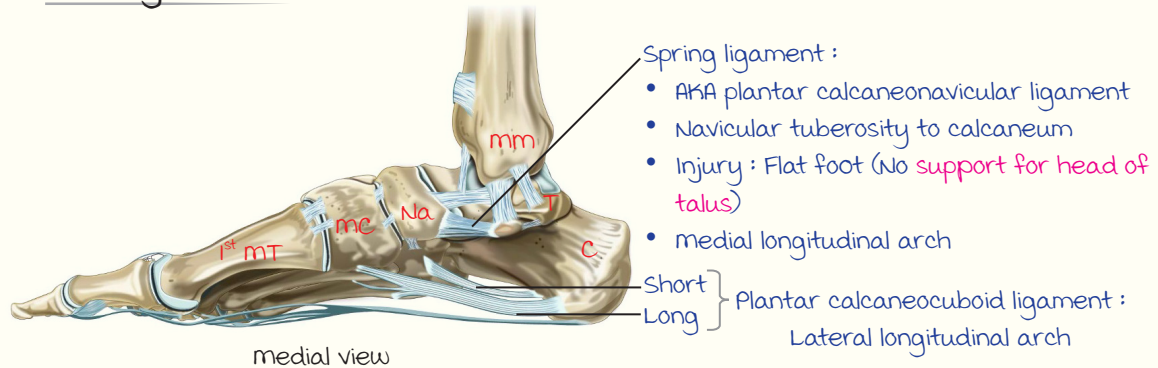
vein grafted	Nerve injured	Sensory loss of leg + foot
GSV	Saphenous N.	medial side
SSV	Sural N.	Lateral side

----- Active space -----

Ligaments of Foot and Ankle

00:43:20

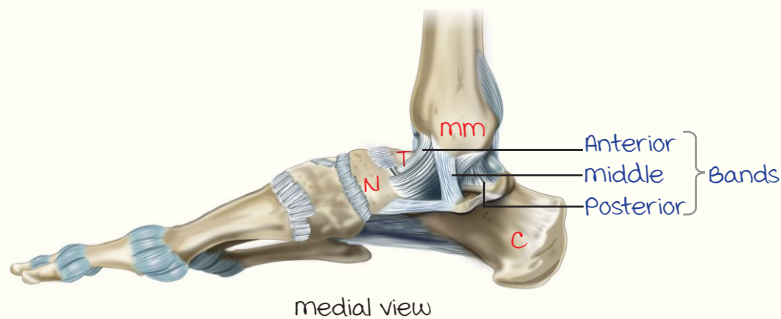
Arch Ligaments :



MT : metatarsal	T : Talus
mc : medial cuneiform	mm : medial malleolus
Na : Navicular	Lm : Lateral malleolus
C : Calcaneus	Cu : Cuboid

Ankle Ligaments :

medial ligaments : **Deltoid ligament**
(Triangular ligament).

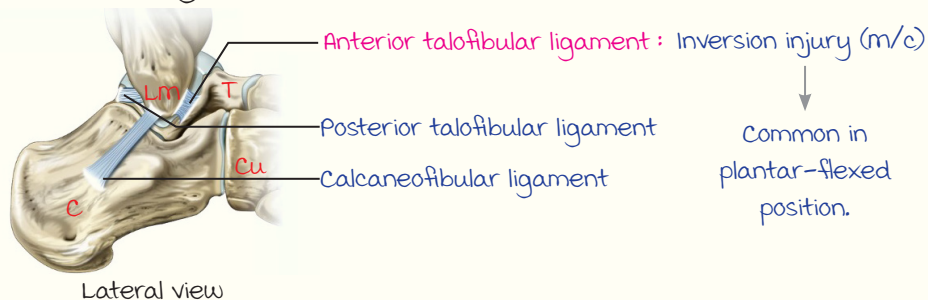


Eversion injury :

- medially : Deltoid injury.
- Laterally : Fracture of Lm.



Lateral ligaments :



Lisfranc Ligament :

Extent : medial cuneiform to base of 2nd metatarsal.

Injury : ↑ Gap between 1st & 2nd metatarsal.



----- Active space -----

Course & Supply:



Applied Anatomy :

Nerve injured	muscles involved	Deformity	Gait
Tibial nerve	- Plantar flexors - Tibialis posterior - Foot muscles	Calcaneovalgus → Dorsiflexed → Everted foot	Calcaneal gait (Walk on heel)
CPN (m/c) (m/c site : Neck of fibula)	- Dorsiflexors - Evertors	Equinovarus → Foot drop → Inverted foot	High steppage gait

.....
Note :

Piriformis nerve supply : Nerve to piriformis/superior gluteal nerve.

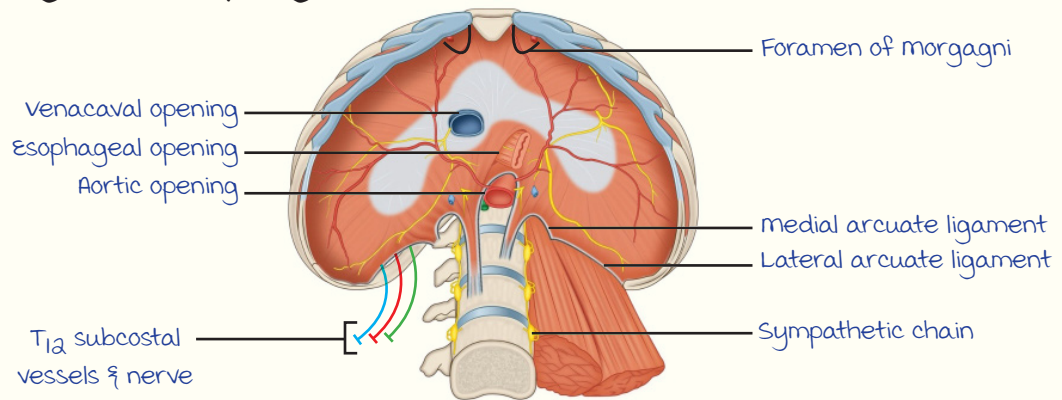
THORAX

----- Active space -----

Diaphragm

00:00:10

Openings in the Diaphragm :



Inferior view of diaphragm

major openings :

	Opening	Level	Location	Traversing structures
1	venacaval opening	T ₈	Central tendon	• IVC • Right phrenic nerve
2	Esophageal opening	T ₁₀	muscular part, surrounded by right crus	• Oesophagus • Right & left X nerve • Esophageal branch of left gastric artery
3	Aortic opening	T ₁₂	B/w right & left crura of diaphragm	• Aorta • Azygos vein • Hemiazygos vein • Thoracic duct

minor openings :

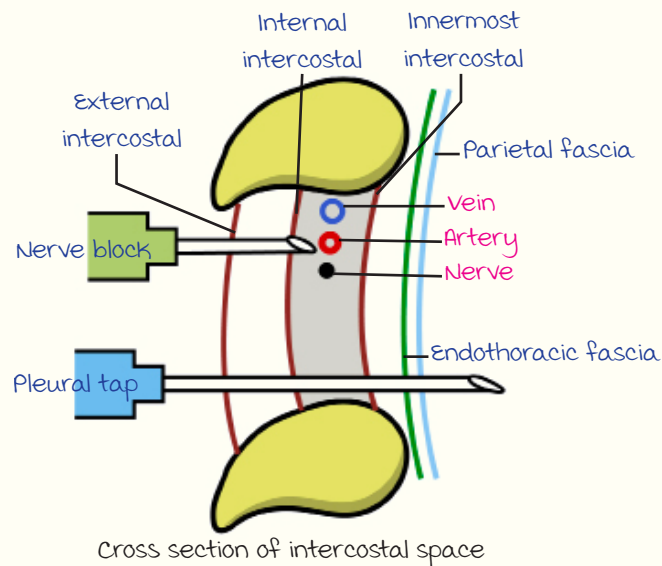
	Opening	Transversing structures
1	Foramen of Morgagni	Superior epigastric vessels
2	medial arcuate ligament	Sympathetic chain
3	Lateral arcuate ligament	T ₁₂ subcostal vein, artery & nerve

Thoracic Wall

00:07:15

----- Active space -----

Neurovascular Arrangement in Intercostal (I/C) Space :



Applied anatomy :

- Pleural tap : Needle inserted at **upper border of lower rib**.
- Intercostal nerve block : Administered at **lower border of upper rib**.

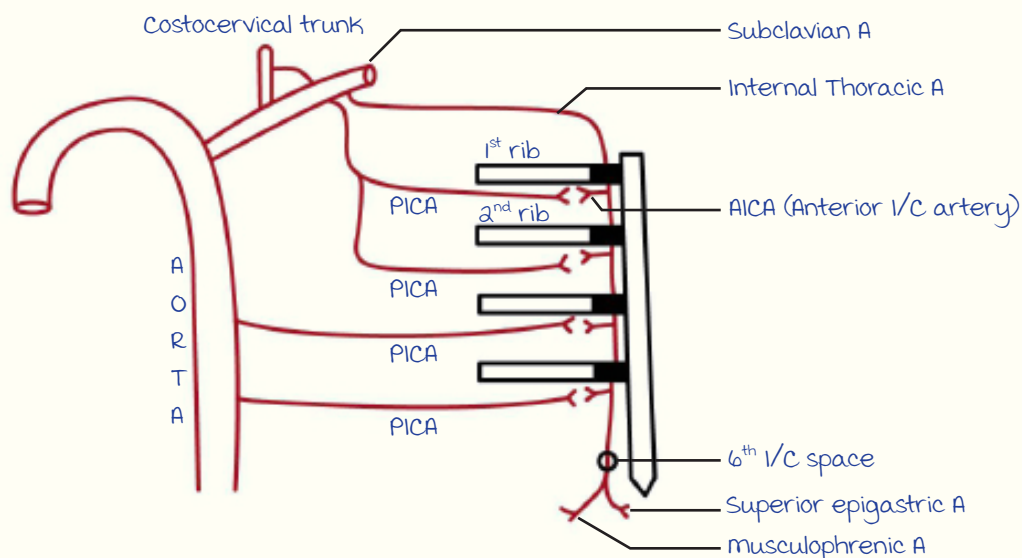
Note :

1st intercostal space arrangement : **Nerve** → **Artery** → **Vein**.

Intercostal nerves :

- T₁, T₂ : Contribute to upper limb.
- T₃ - T₆ : Typical I/C nerve.
- > T₆ : Supply anterior abdominal wall.

Intercostal arteries :



----- Active space -----

1. PICA (Posterior I/C artery) :

- 1st & 2nd I/C spaces : Branch of **costocervical trunk** of subclavian artery.
- Remaining I/C spaces : Branch of aorta.
- Course : Distal part of rib → Costochondral junction.

2. AICA :

- Branch of subclavian artery in all I/C spaces (1st 6 by internal thoracic artery and 7th-9th by musculophrenic artery).
- Course : Costochondral junction → Anastomose with PICA.

- Applied anatomy : **Post-ductal coarctation of aorta**

3rd space onwards : Reverse blood flow from AICA → PICA → Descending aorta

Thick, engorged, tortuous arteries

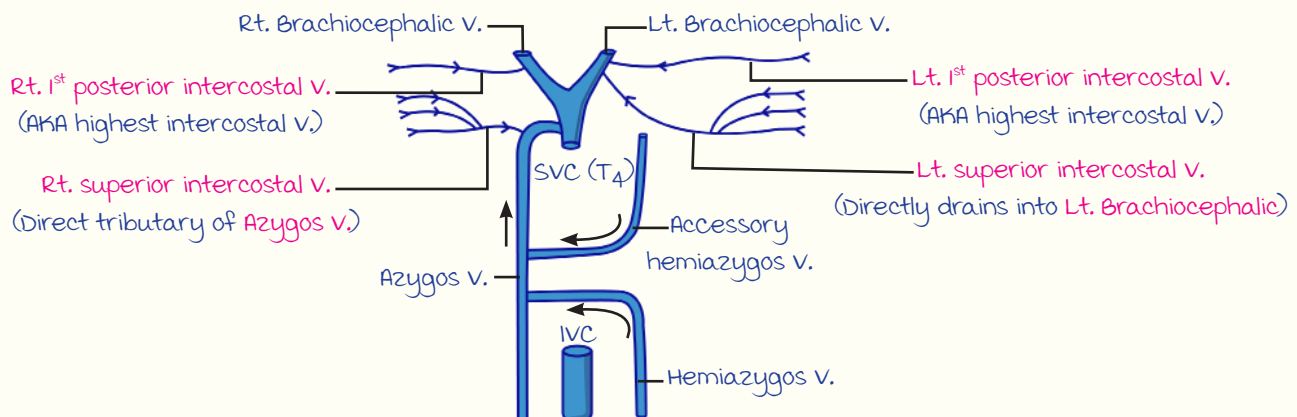
X-ray : **Rib notching.**

- Fate of internal thoracic artery :

At 6th I/C space



Intercostal venous drainage :



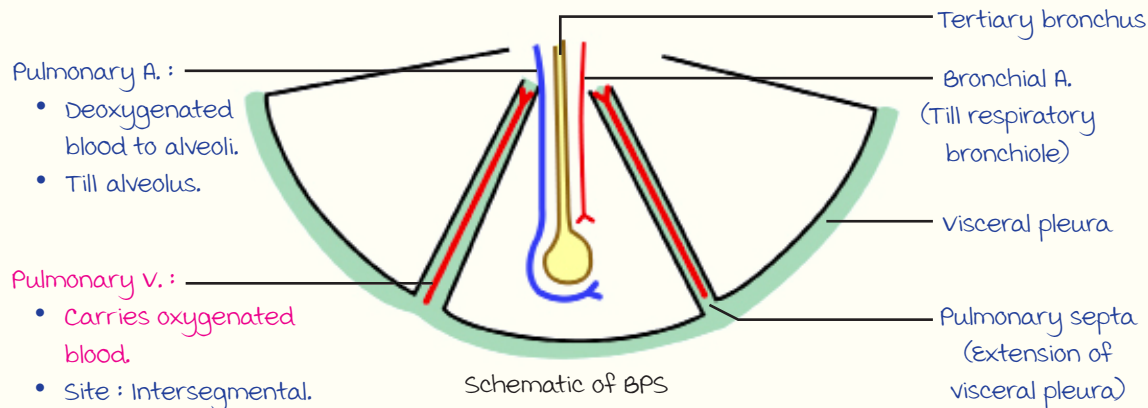
- 1st I/C vein drains $\xrightarrow{\text{Drains}}$ Brachiocephalic vein.
- Superior I/C vein (2nd to 4th) :
 - Right → Azygos vein.
 - Left → Lt. Brachiocephalic vein.

Bronchopulmonary Segment (BPS)

00:24:00

----- Active space -----

- Portion of lung aerated by 1 tertiary bronchus.
- Right lung : 10 BPS, Left lung : 9 BPS.



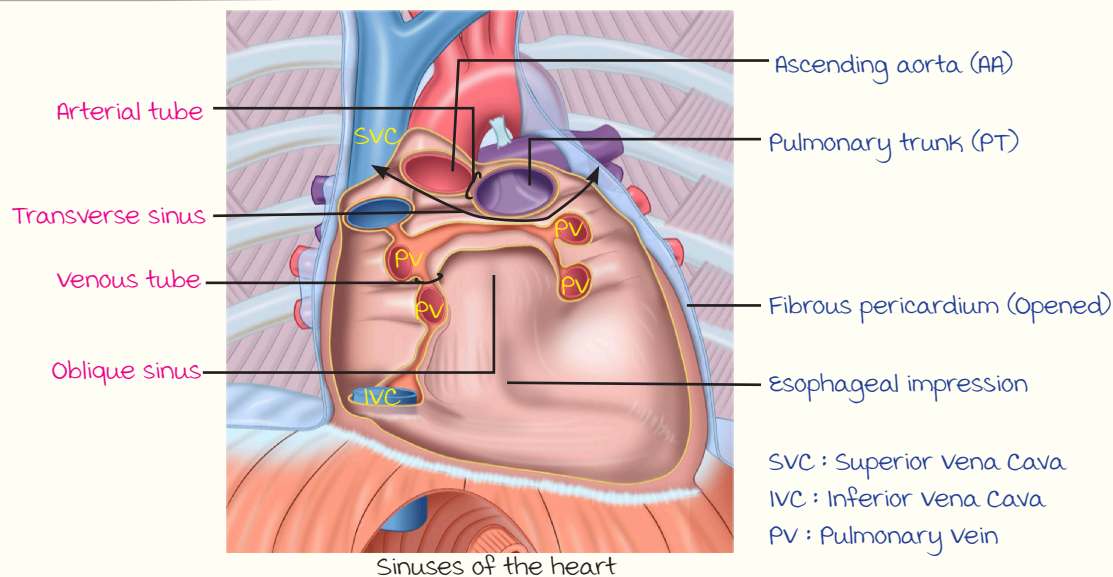
Right Lung :

- Right bronchus : Shorter, wider, vertically oriented → ↑ Chance of infection. (Aspiration)
- Apical segment of upper lobe : **Least vascular** (Site for TB).
- Right superior basal of lower lobe : **most dependent BPS** in **supine** position.
- Right posterior basal segment of lower lobe : **most dependent BPS** in **sitting/standing**.

Pericardial Sinuses & Interior of Heart

00:32:28

Pericardial Sinuses :



Transverse sinus :

- Continuous space b/w the arterial & venous tubes.
- Surgical importance : Site for **ligation of ascending aorta & pulmonary trunk**.

----- Active space -----

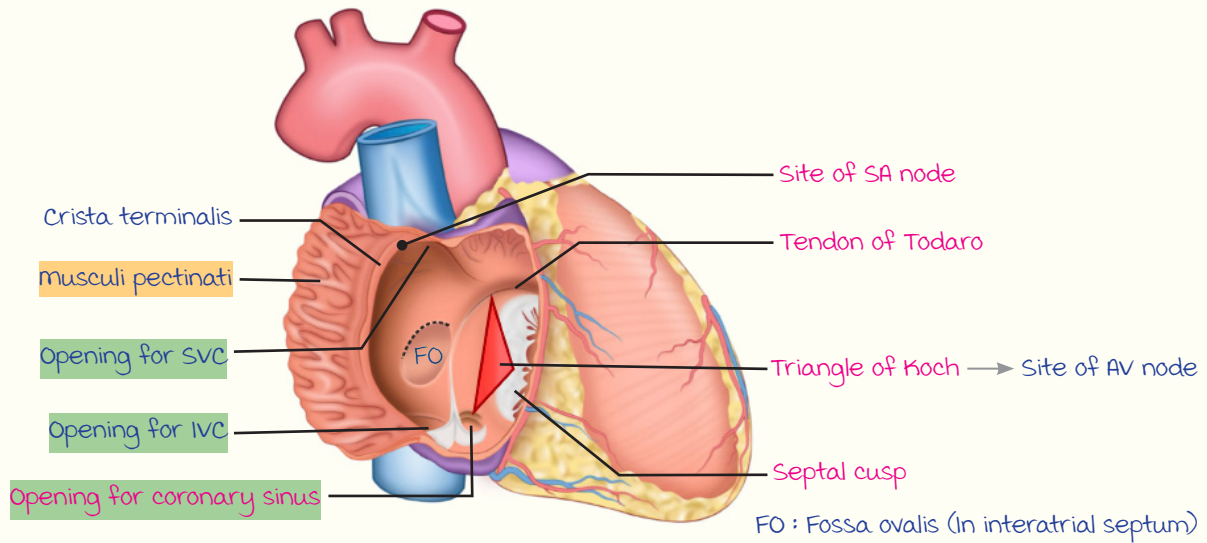
Oblique sinus :

- Cul-de-sac : Closed above.
- Space b/w the 4 pulmonary veins, behind **left atrium**.
- Allows expansion of left atrium **without compressing esophagus**.

Interior of the Heart :

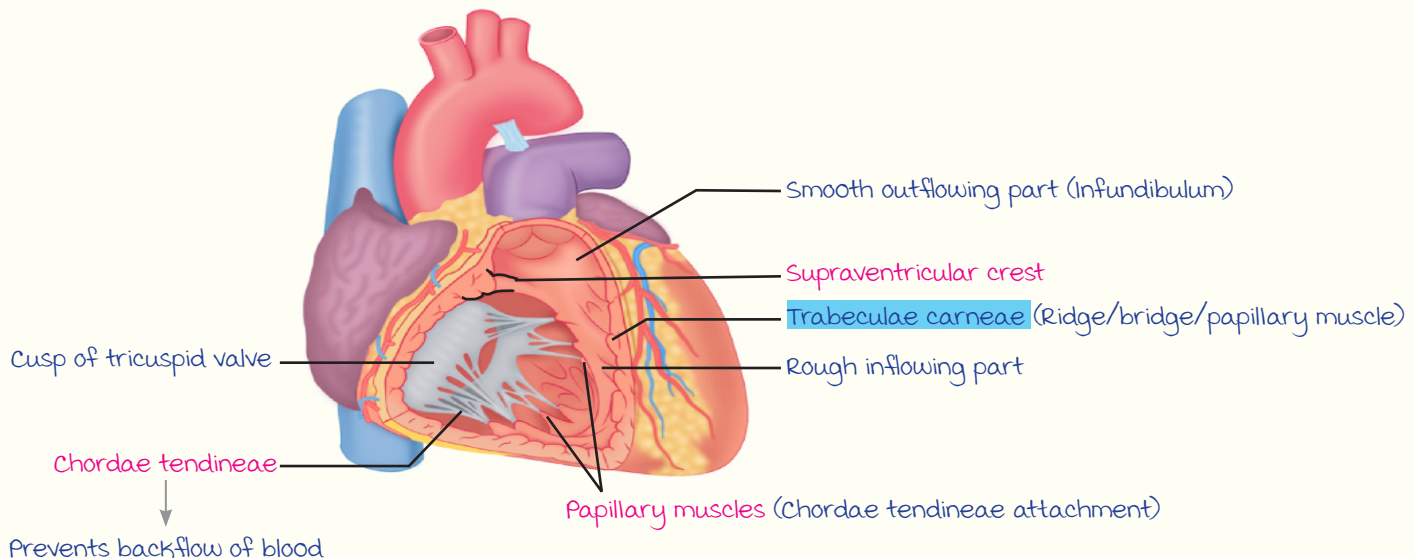
Interior of right atrium :

Parts → Rough anterior wall } Separated by crista terminalis.
Smooth posterior wall }



Interior of right ventricle :

Parts → Rough inflowing part } Separated by supraventricular crest.
Smooth outflowing part }

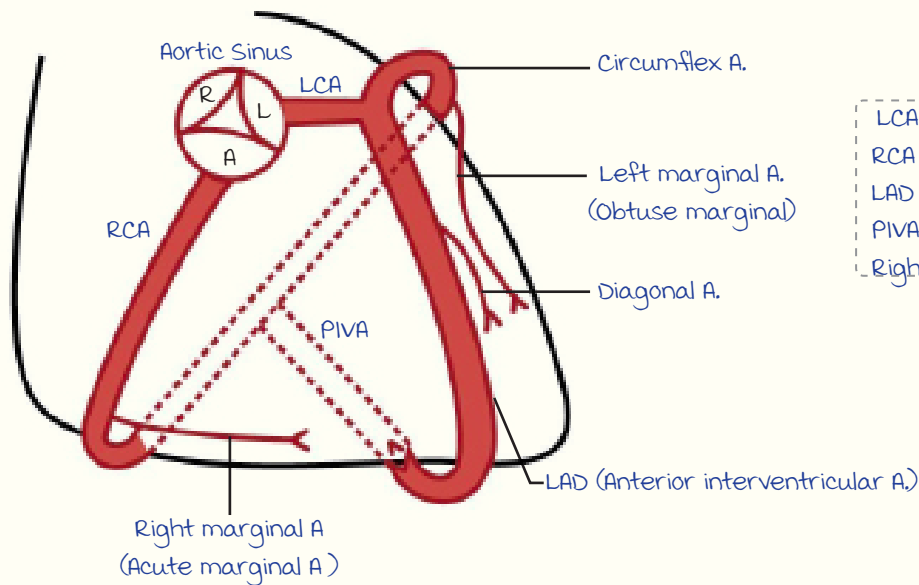


Coronary Circulation

00:45:20

----- Active space -----

Arterial Supply :



LCA : Left coronary artery
 RCA : Right coronary artery
 LAD : Left anterior descending artery
 PIVA : Posterior interventricular artery
 Right (R), Left (L), Anterior (A) : Aortic sinus

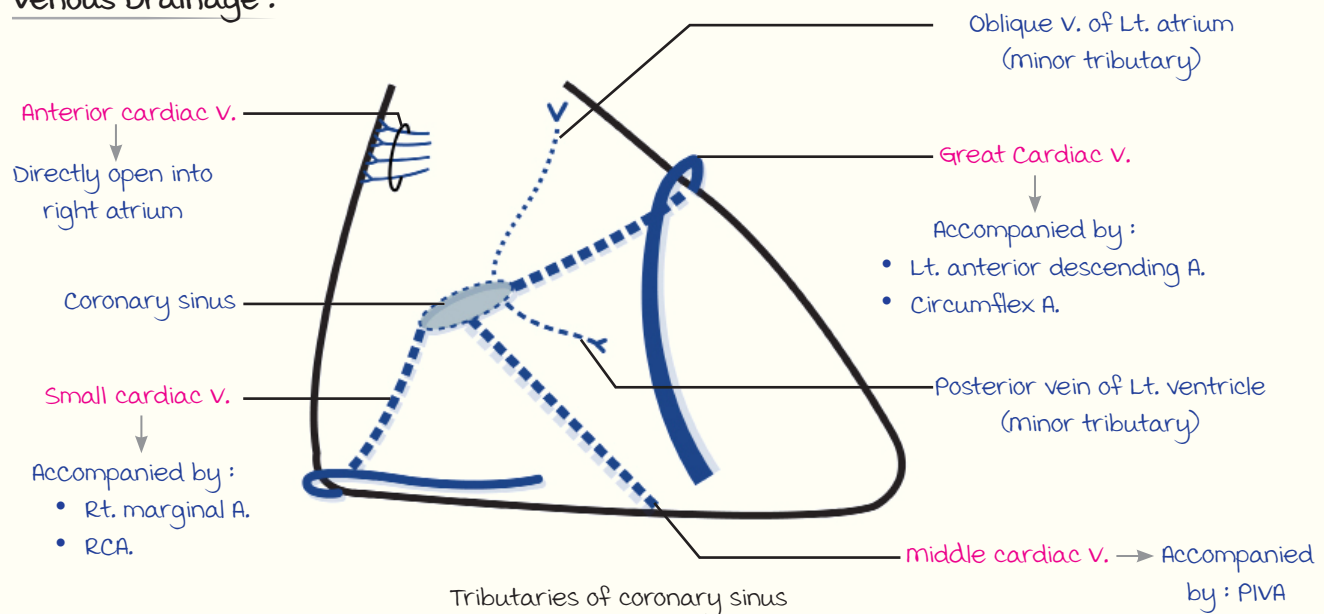
Cardiac dominance :

Based on origin of PIVA —
 RCA → Right dominance.
 Circumflex artery → Left dominance.

Blood supply to the conduction system :

- RCA : SA node, AV node, Bundle of His.
- LCA : RBB & LBB.

Venous Drainage :



Note :

Anterior cardiac veins are not tributaries of coronary sinus.

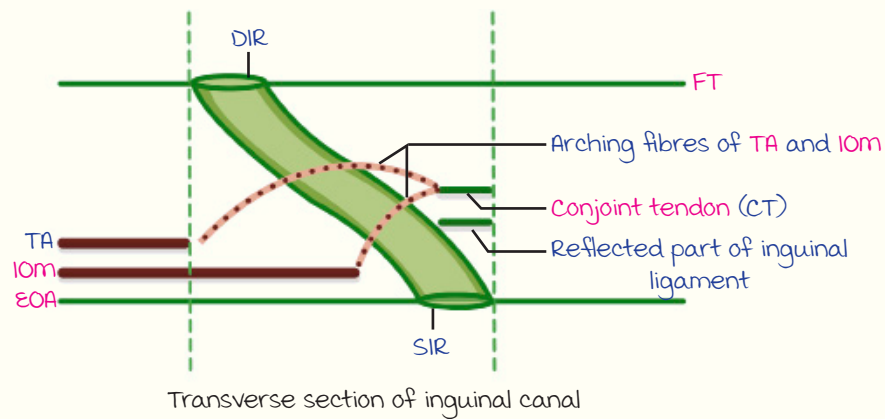
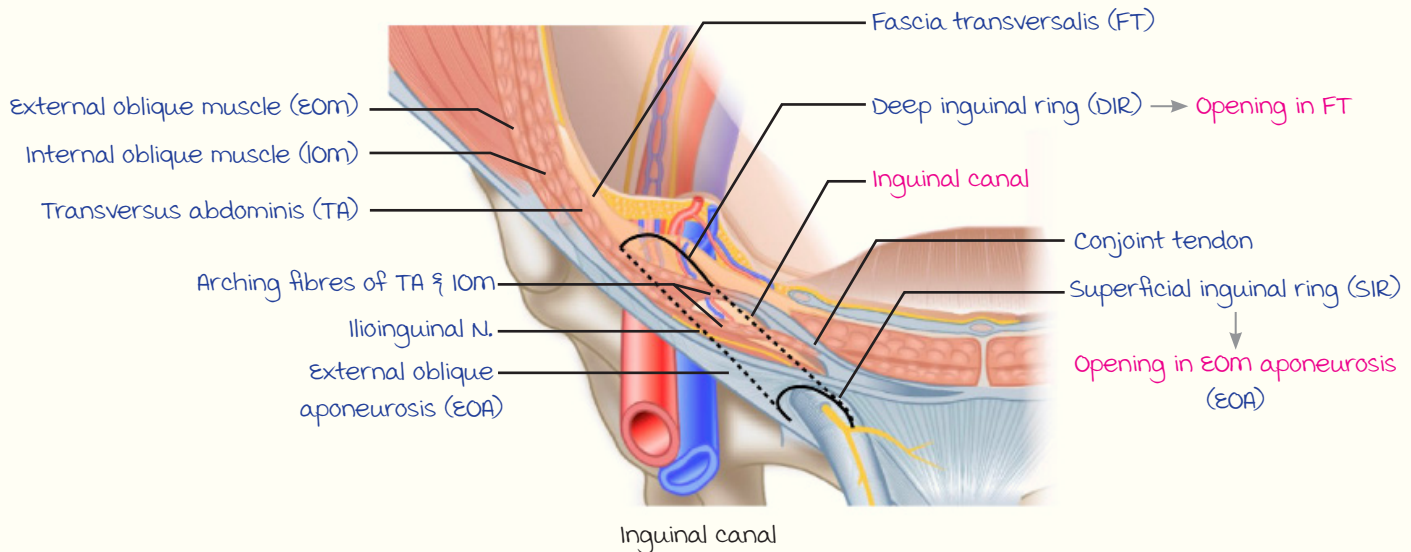
ABDOMEN

----- Active space -----

Inguinal Canal

00:00:15

Relations of Inguinal Canal :



Anterior wall :

- External oblique aponeurosis.
- Internal oblique muscle (IOM).

Posterior wall :

- Fascia transversalis (FT).
- Conjoint tendon.

Roof : Arched fibers of internal oblique & transversus abdominis (TA).

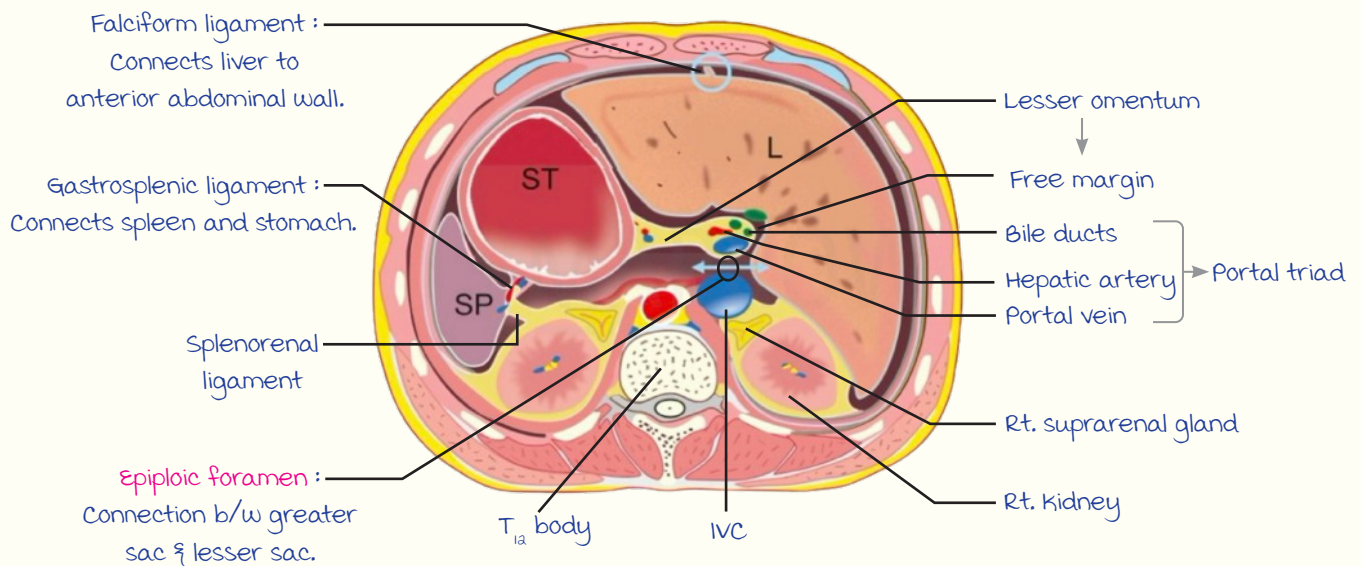
Floor :

- Inguinal ligament.
- Lacunar ligament.

Peritoneal Folds

00:06:30

----- Active space -----



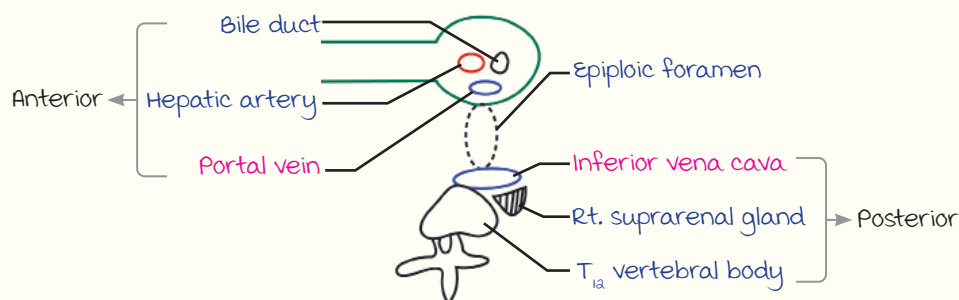
SP : Spleen
ST : Stomach

L : Liver
IVC : Inferior vena cava

Contents :

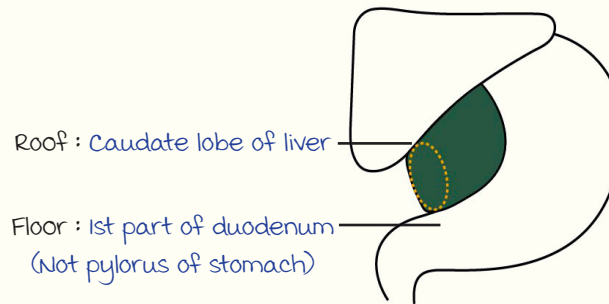
Peritoneal fold	Contents
Falciform ligament	Ligamentum teres (Remnant of left umbilical vein)
Splenorenal/lienorenal ligament	- Splenic artery - Tail of pancreas
Gastrosplenic ligament	- Short gastric artery - Left gastroepiploic artery
Lesser Omentum	Free margin : - Hepatic artery - Bile duct - Portal vein Attached margin : - Left gastric artery - Right gastric artery

Epiploic Foramen/Foramen of Winslow :



Transverse section of epiploic foramen

----- Active space -----



most dependent site of peritoneal cavity :

- Supine : Hepatorenal pouch of Morrison
- Sitting/standing → Females : Rectouterine pouch
- Males : Rectovesical pouch

Phrenicocolic Ligament :

- Connects diaphragm to splenic flexure of colon.
- Responsible for spleen being directed toward right iliac fossa in splenomegaly.

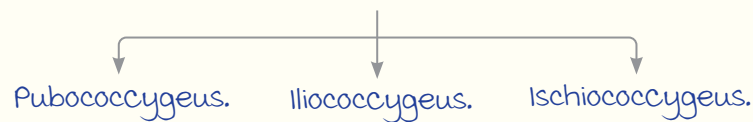
Perineum

00:17:50

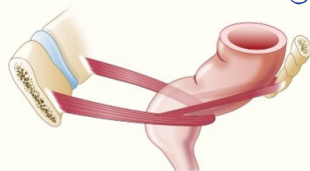
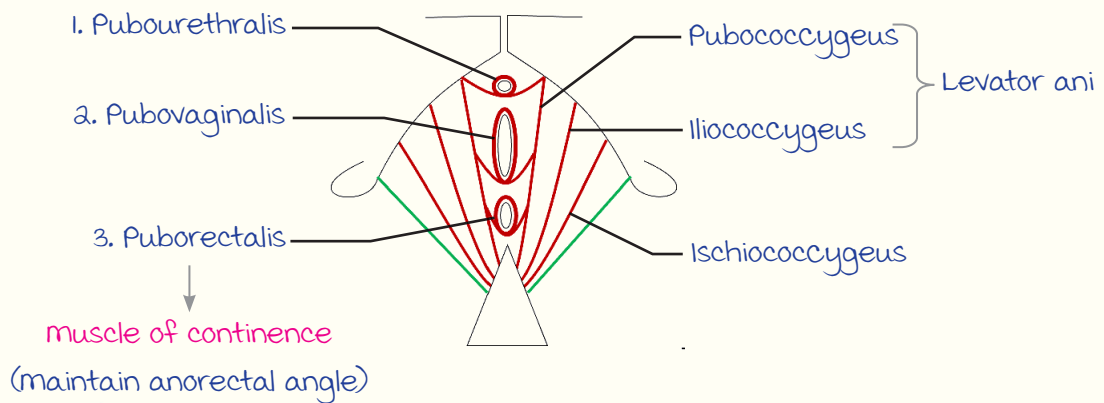
Pelvic Diaphragm :

Superiormost layer of perineum, that separates the pelvis from perineum.

Components :

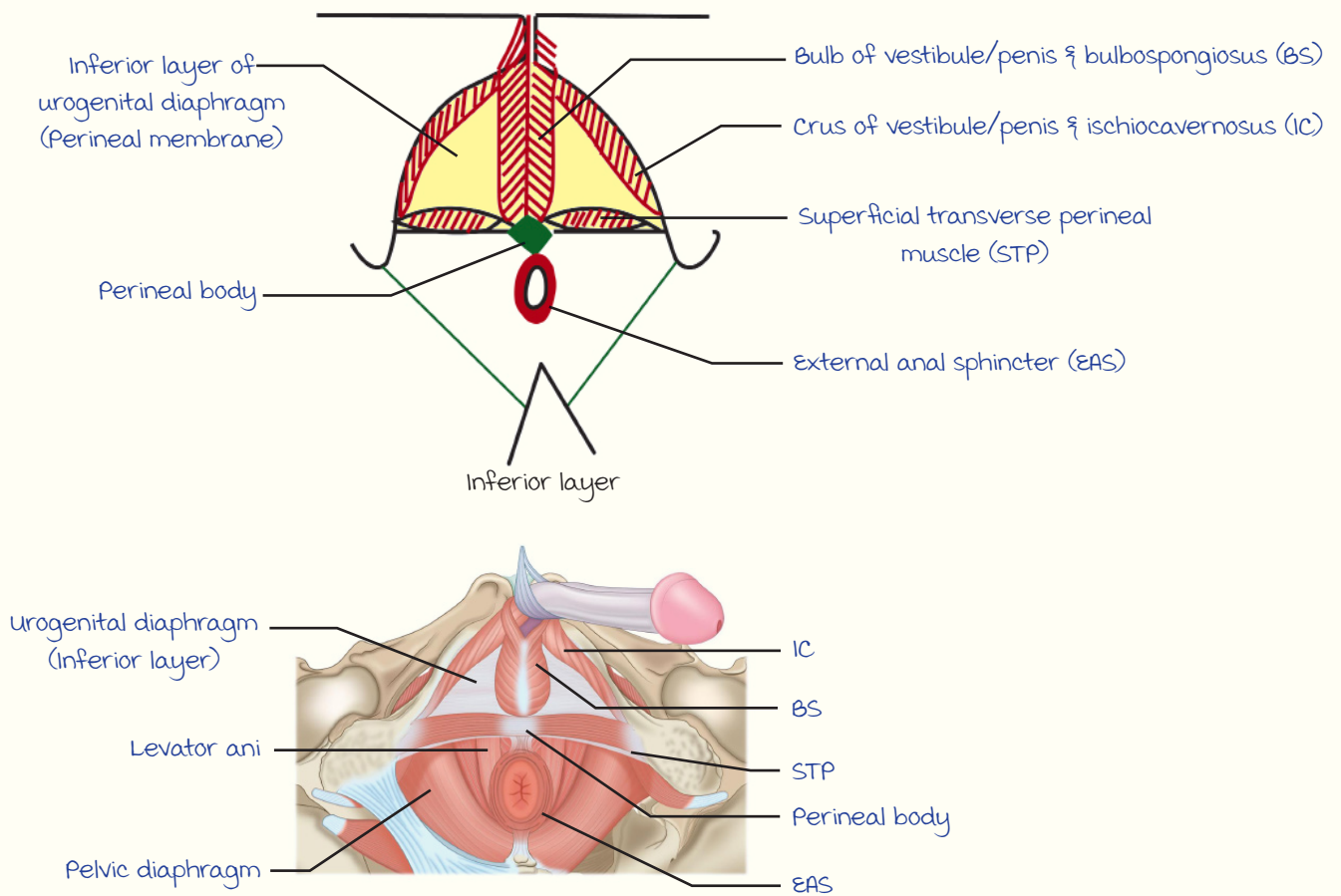


modifications of pubococcygeus :



Urogenital Diaphragm :

----- Active space -----



Perineal Body :

Paired (4) :

1. superficial transverse perineal muscle.
2. Deep transverse perineal muscle.
3. Bulbospongiosus.
4. Levator ani.

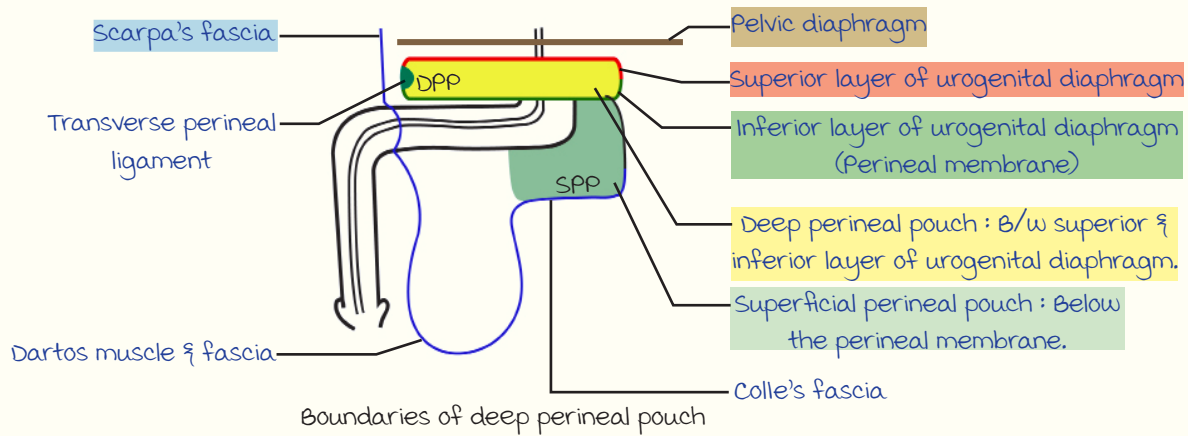
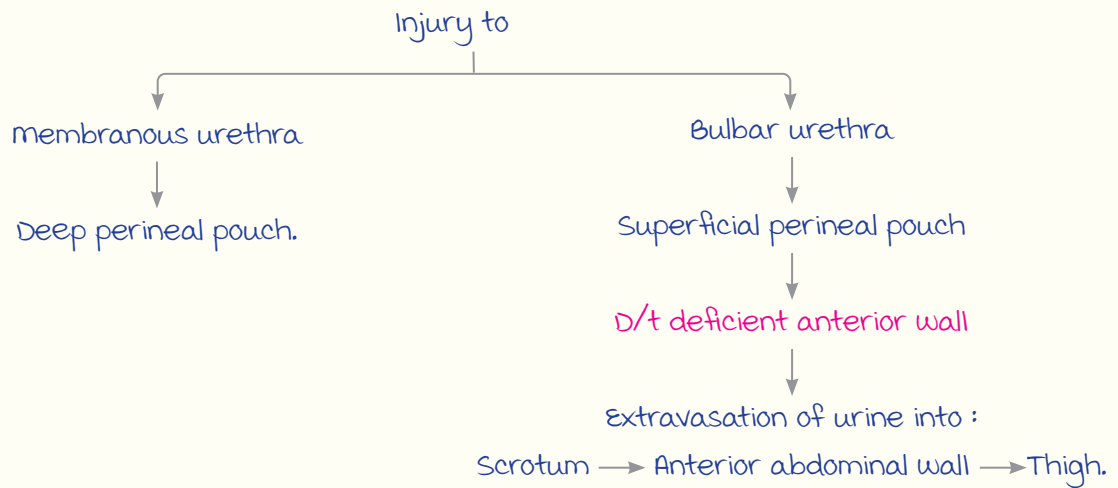
Unpaired (2) :

1. External anal sphincter.
2. Longitudinal muscle of anal canal.

Note :

Ischiocavernosus is not a part of perineal body.

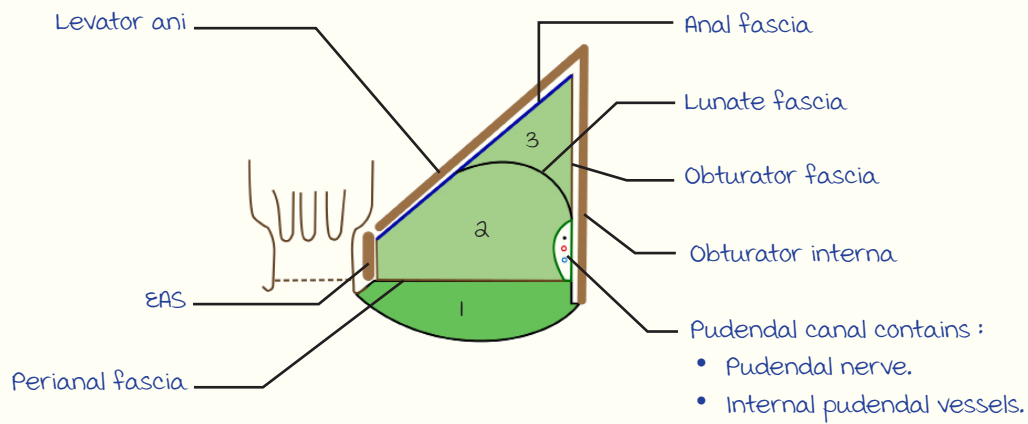
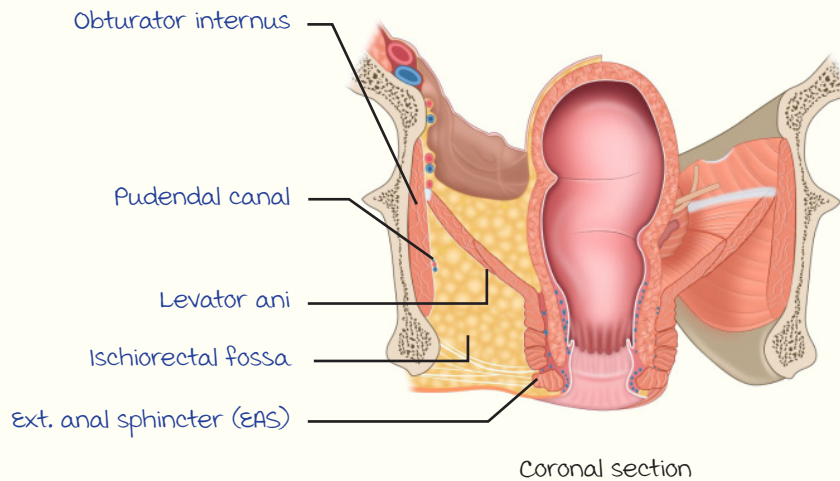
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Perineal Pouch :Extravasation of urine :Contents :

- Deep perineal pouch :
 - membranous urethra.
 - External urethral sphincter.
 - Bulbourethral (Cowper's) gland (male).
- Superficial perineal pouch :
 - Bulb of penis & bulbospongiosus.
 - Crus of penis & ischiocavernosus.
 - Superficial transverse perinei & other muscles below perineal membrane.
 - Bartholin's gland (Females)

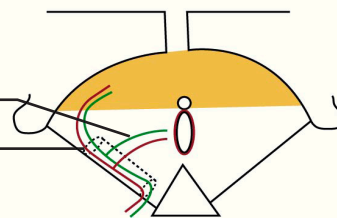
Ischioanal Fossa/Ischioanal Fossa :

----- Active space -----



Contents :

1. Fat.
2. Inferior rectal nerves & vessels.
3. Pudendal canal & its contents.



Applied anatomy :

Drainage of Ischioanal fossa abscess :

- Risk of injury to inferior rectal nerves & vessels (D/t horizontal course).
- Horseshoe recess :
Communication of both IRF → Spread of IRF abscess to the C/L side.

